



**Generation Interconnection
Impact Study Report
for
Queue Project AE1-024/AE1-025/AE1-026
CASHIE 230 KV
189.9 MW Capacity / 240 MW Energy**

June, 2020

Table of Contents

1	Introduction.....	4
2	Preface.....	4
3	General	5
3.1	Point of Interconnection	5
3.2	Cost Summary.....	5
4	Transmission Owner Scope of Work	6
5	Attachment Facilities	7
6	Direct Connection Cost Estimate.....	7
7	Non-Direct Connection Cost Estimate.....	8
8	Schedule.....	8
9	Transmission Owner Analysis.....	8
9.1	Power Flow Analysis	8
9.2	Short Circuit Analysis	8
9.3	Stability Analysis	8
10	Interconnection Customer Requirements.....	9
10.1	Compliance Issues and Interconnection Customer Requirements	9
10.2	Power Factor Requirements.....	9
11	Revenue Metering and SCADA Requirements	10
11.1	PJM Requirements	10
11.1.1	Meteorological Data Reporting Requirement.....	10
11.2	Dominion Requirements.....	10
12	Network Impacts.....	10
12.1	Generation Deliverability	12
12.2	Multiple Facility Contingency	12
12.3	Contribution to Previously Identified Overloads.....	13
12.4	Potential Congestion due to Local Energy Deliverability.....	14
12.5	System Reinforcements.....	18
12.6	Flow Gate Details.....	33
12.6.1	Contingency Descriptions	33
12.6.2	Index 1	40
12.6.3	Index 2	44

12.6.4	Index 3	48
12.6.5	Index 4	50
12.6.6	Index 5	53
12.6.7	Index 6	55
12.6.8	Index 7	58
12.6.9	Index 8	61
12.6.10	Index 9.....	66
12.6.11	Index 10.....	68
12.6.12	Index 11.....	70
12.6.13	Index 12.....	73
12.6.14	Index 13.....	76
12.6.15	Index 14.....	79
12.6.16	Index 15.....	82
12.6.17	Index 16.....	85
12.6.18	Index 17.....	88
12.6.19	Index 18.....	90
12.6.20	Index 19.....	94
12.6.21	Index 20.....	96
12.6.22	Index 21.....	98
12.6.23	Index 22.....	100
12.6.24	Index 23.....	103
12.6.25	Index 24.....	105
12.6.26	Index 25.....	107
12.6.27	Index 26.....	110
12.6.28	Index 27.....	113
12.6.29	Index 28.....	119
13	Affected Systems	122
13.1	Duke Energy Progress.....	122
14	Short Circuit.....	124
15	Stability Analysis.....	126
15.1	Steady-State Voltage Requirements.....	126
15.2	Stability and Reactive Power Requirement for Low Voltage Ride Through	126
	Attachment 1	127

1 Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the Feasibility Study Agreement the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (Dominion).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 General

The Interconnection Customer (IC), has proposed a hybrid solar/storage generating facility located in Bertie County, North Carolina. Each individual request will have a capability of 80 MW, with 63.3 MW of this output being recognized by PJM as Capacity. The total installed facilities of AE1-024/AE1-025/AE1-026 will have a total capability of 240 MW with 189.9 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 15, 2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AE1-024/AE1-025/AE1-026
Project Name	CASHIE 230 KV
State	North Carolina
County	Bertie
Transmission Owner	Dominion
MFO	240
MWE	240
MWC	189.9
Fuel	Solar / Storage
Basecase Study Year	2022

3.1 Point of Interconnection

AE1-024/AE1-025/AE1-026 will interconnect with the Dominion transmission system at the Cashie 230kV substation.

3.2 Cost Summary

The AE1-024/AE1-025/AE1-026 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 1,800,000
Direct Connection Network Upgrade	\$ 6,500,000
Non Direct Connection Network Upgrades	\$ 0 – TBD in the Facilities Study
Total Costs	\$ 8,300,000

In addition, the AE1-024/AE1-025/AE1-026 project may be responsible for a contribution to the following costs

Description	Total Cost
System Upgrades	\$ 160,687,353

Note: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not

closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

4 Transmission Owner Scope of Work

Dominion assessed the impact of the proposed Queue Projects AE1-024 / AE1-025 / AE1-026 were evaluated as a 189.9 MW Capacity (240 MW energy) injection at the Cashie 230 kV substation in the Dominion Transmission System, for compliance with NERC Reliability Criteria on Dominion Transmission System. The system was assessed using the summer 2022 AE1 case provided to Dominion by PJM. When performing a generation analysis, Dominion's main analysis will be load flow study results under single contingency (both normal and stressed system conditions). Dominion Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of Dominion's Planning Criteria and interconnection requirements can be found in the Company's Facility Connection Requirements which are publicly available at: <http://www.dominionenergy.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically in Planning Studies NERC Category C Contingency Conditions (Bus Fault, Tower Line, N-1-1, and Stuck Breaker scenarios) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For Dominion Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

The required Attachment Facilities, Direct Connection and Non-Direct Connection work for the interconnection of the AE1-024 / AE1-025 / AE1-026 generation project to the Dominion Transmission System is detailed in the following sections. The associated one-line with the generation project attachment facilities and primary direct and non-direct connection are shown in Attachment 1.

Note that the ITO findings were made from a conceptual review of this project. A more detailed review of the connection facilities and their cost will be identified in a future study phases. Further note that the cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. ITO herein reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

5 Attachment Facilities

Generation Substation: Install metering and associated protection equipment. Estimated Cost \$600,000.

Transmission: Construct approximately one span of 230 kV Attachment line between the generation substation and Cashie 230kV substation. The estimated cost for this work is \$1,200,000.

The estimated total cost of the Attachment Facilities is \$1,800,000. It is estimated to take 18-24 months to complete this work upon execution of an Interconnection Construction Service Agreement (ICSA). These preliminary cost estimates are based on typical engineering costs. A more detailed engineering cost estimates are normally done when the IC provides an exact site plan location for the generation substation during the Facility Study phase. See Attachment 1.

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Substation Work	\$ 600,000
Transmission Work	\$1,200,000
Total Attachment Facility Costs	\$1,800,000

6 Direct Connection Cost Estimate

Substation: The proposed interconnection substation will consist of a three 230 kV breakers added to an existing station, which will require a site expansion. The arrangement in the substation will be as shown below on Attachment 1. The estimated cost of this work scope is \$6,500,000.

Transmission: None

The total estimated cost to complete the Direct Connection Network Facilities is \$6,500,000. It is estimated to take 24-36 months to complete this work upon execution of an Interconnection Construction Service Agreement.

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Substation Work	\$6,500,000
Transmission Work	\$0
Total Direct Connection Facility Costs	\$6,500,000

7 Non-Direct Connection Cost Estimate

During the Facilities Study, ITO's System Protection Engineering Department will review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system. The review is based on maintaining system reliability by reviewing ITO's protection requirements with the known transmission system configuration which includes generating facilities in the area. This review may determine that transmission line protection and communication upgrades are required at remote substations.

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Substation Work	\$ TBD in Facilities Study phase
Transmission Work	\$ 0
Total Non-Direct Connection Facility Costs	\$ TBD in Facilities Study phase

8 Schedule

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimate elapsed time to complete each of the required reinforcements is identified in the "System Reinforcements" section of the report.

9 Transmission Owner Analysis

9.1 Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2022 summer peak load flow model and the results were verified by Dominion. Additionally, Dominion performed an analysis of its transmission system and no further deficiencies were identified.

9.2 Short Circuit Analysis

PJM performed a short circuit analysis and the results were verified by Dominion. The connection of the AE1-024 / AE1-025 / AE1-026 projects to the system does not result in any newly overdutied circuit breakers on the Dominion transmission system and does not have a significant fault current contribution to existing overdutied circuit breakers

9.3 Stability Analysis

PJM will complete a dynamic stability analysis as part of the Facilities Study. The results of this analysis will be reviewed by Dominion. Should stability concerns be identified in PJM's study, Dominion will develop appropriate system reinforcement(s) and included the estimated cost of any reinforcement(s) in Dominion's System Impact Study report.

10 Interconnection Customer Requirements

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion's "Dominion Energy Electric Transmission Generator Interconnection Requirements" documented in Dominion's Facility Interconnection Requirements "Exhibit C" located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

10.1 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion's "Dominion's Facility Interconnection Requirements" document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC's GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion's "Dominion's Facility Interconnection Requirements" document located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

10.2 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

11 Revenue Metering and SCADA Requirements

11.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

11.1.1 Meteorological Data Reporting Requirement

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

11.2 Dominion Requirements

See Section 3.4.6 "Metering and Telecommunications" of Dominion's "Dominion's Facility Interconnection Requirements" document located at: <https://www.dominionenergy.com/company/moving-energy/electrictransmission-access>.

12 Network Impacts

The Queue Project AE1-024/AE1-025/AE1-026 was evaluated as a 240.0 MW (Capacity 189.9 MW) injection at the Cashie 230 kV substation in the Dominion area. Project AE1-024/AE1-025/AE1-026 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-024/AE1-025/AE1-026 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

12.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308856	314569	6EARLEYS	DVP	314575	6NUCO TP	DVP	1	DVP_P1-2: LN 2092	single	571.52	99.99	111.6	AC	67.55
3308857	314569	6EARLEYS	DVP	314575	6NUCO TP	DVP	1	DVP_P1-2: LN 2131	single	571.52	99.15	110.77	AC	67.57
3309004	314574	6EVERETS	DVP	936530	AD2-068 TAP	DVP	1	DVP_P1-2: LN 246	single	449.32	98.52	107.17	AC	40.88
3309005	314574	6EVERETS	DVP	936530	AD2-068 TAP	DVP	1	DVP_P1-2: LN 2092	single	449.32	98.21	106.94	AC	41.71
3308906	314575	6NUCO TP	DVP	919140	AA1-138 TAP	DVP	1	DVP_P1-2: LN 2092	single	571.52	93.67	105.24	AC	67.55
3308907	314575	6NUCO TP	DVP	919140	AA1-138 TAP	DVP	1	DVP_P1-2: LN 2131	single	571.52	92.83	104.41	AC	67.57
3309047	314583	6LAKEVEW	DVP	924510	AB2-100 TAP	DVP	1	DVP_P1-2: LN 246	single	375.06	97.77	105.35	AC	29.33
3309048	314583	6LAKEVEW	DVP	924510	AB2-100 TAP	DVP	1	DVP_P1-2: LN 6005_FSA	single	375.06	96.14	103.7	AC	29.33
3309133	314591	6NASH	DVP	304226	6PA-RMOUNT#4	CPL	1	DVP_P1-2: LN 2058	single	470.0	98.55	102.57	AC	19.21
3308916	919140	AA1-138 TAP	DVP	314537	6SUFFOLK	DVP	1	DVP_P1-2: LN 2092	single	571.52	93.54	105.09	AC	67.55
3308917	919140	AA1-138 TAP	DVP	314537	6SUFFOLK	DVP	1	DVP_P1-2: LN 2131	single	571.52	92.7	104.26	AC	67.57
3309232	934330	AD1-057 TAP	DVP	313845	6HATHAWAY	DVP	1	DVP_P1-2: LN 238	single	441.8	97.15	100.18	AC	12.95
3309030	936530	AD2-068 TAP	DVP	304451	6GREENVILLE	CPL	1	DVP_P1-2: LN 246	single	478.0	96.81	104.9	AC	40.88
3309031	936530	AD2-068 TAP	DVP	304451	6GREENVILLE	CPL	1	DVP_P1-2: LN 2092	single	478.0	96.36	104.51	AC	41.71

12.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308592	313714	6PERQUIMANS	DVP	314662	6S HERTFORD	DVP	1	DVP_P4-2: 24642	breaker	897.0	98.76	107.01	AC	82.05
3308614	314662	6S HERTFORD	DVP	314651	6WINFALL	DVP	1	DVP_P4-2: 24642	breaker	897.0	96.92	105.16	AC	82.05
3308520	314905	8CHANCE	DVP	314900	8BRISTER	DVP	1	DVP_P4-2: H1T594	breaker	3351.0	99.19	100.27	AC	41.63
3308521	314905	8CHANCE	DVP	314900	8BRISTER	DVP	1	DVP_P4-2: 573T594	breaker	3351.0	98.8	99.87	AC	41.5
3308493	314911	8LADYSMITH	DVP	314905	8CHANCE	DVP	1	DVP_P4-2: 57302	breaker	3351.0	99.71	100.81	AC	42.35

12.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/DC	MW IMPACT
3308862	313845	6HATHAWAY	DVP	304222	6ROCKYMT230T	CPL	1	DVP_P1-2: LN 2181	single	374.0	120.44	125.39	AC	18.84
3309021	313845	6HATHAWAY	DVP	314591	6NASH	DVP	1	DVP_P1-2: LN 2058	single	449.32	108.77	112.97	AC	19.21
3309010	314209	6SKIFF CREEK	DVP	314386	6KINGS M	DVP	1	DVP_P1-2: LN 557	single	441.8	136.07	138.02	AC	10.03
3308370	314282	6CARSON	DVP	314285	6CHRL249	DVP	1	DVP_P4-2: 562T563	breaker	684.0	112.62	114.47	AC	14.76
3308390	314285	6CHRL249	DVP	314316	6LOCKS	DVP	1	DVP_P4-2: 562T563	breaker	684.0	109.9	111.75	AC	14.76
3308283	314287	6CHESTF B	DVP	314276	6BASIN	DVP	1	DVP_P4-2: 562T563	breaker	549.0	136.7	138.99	AC	14.69
3308808	314287	6CHESTF B	DVP	314276	6BASIN	DVP	1	DVP_P1-2: LN 563	single	449.32	130.25	132.53	AC	11.78
3309067	314296	6PENNIMAN	DVP	314415	6WALR209	DVP	1	DVP_P1-2: LN 557	single	441.8	124.85	126.8	AC	10.03
3309039	314386	6KINGS M	DVP	314296	6PENNIMAN	DVP	1	DVP_P1-2: LN 557	single	441.8	128.46	130.4	AC	10.03
3309242	314415	6WALR209	DVP	314391	6LIGH209	DVP	1	DVP_P1-2: LN 557	single	441.8	110.47	112.42	AC	10.03
3309494	314554	3BTLEBRO	DVP	304223	3ROCKYMT115T	CPL	1	DVP_P7-1: LN 2058-2181	tower	164.0	294.06	302.25	AC	15.3
3308310	314568	3EARLEYS	DVP	314569	6EARLEYS	DVP	2	DVP_P4-2: 201262	breaker	208.0	144.76	150.48	AC	10.96
3308311	314568	3EARLEYS	DVP	314569	6EARLEYS	DVP	2	DVP_P4-2: WT2171	breaker	208.0	125.43	129.22	AC	7.57
3308426	314568	3EARLEYS	DVP	314569	6EARLEYS	DVP	1	DVP_P4-2: 217162	breaker	208.0	128.41	132.2	AC	7.57
3308365	314574	6EVERETS	DVP	936530	AD2-068 TAP	DVP	1	DVP_P4-2: 2020T2144	breaker	485.0	129.67	140.07	AC	52.36
3309544	314574	6EVERETS	DVP	936530	AD2-068 TAP	DVP	1	DVP_P7-1: LN 2058-2181	tower	485.0	148.13	157.18	AC	46.54
3308400	314583	6LAKEVEW	DVP	924510	AB2-100 TAP	DVP	1	DVP_P4-2: 246T247	breaker	427.0	127.66	135.54	AC	39.55
3308346	314638	6ELIZ CT	DVP	314647	6SHAWBRO	DVP	1	DVP_P4-2: 246T247	breaker	658.0	132.18	143.81	AC	71.46
3308395	314903	8CHCKAHM	DVP	314908	8ELMONT	DVP	1	DVP_P4-2: 563T576	breaker	3144.0	114.47	116.19	AC	62.85
3308396	314903	8CHCKAHM	DVP	314908	8ELMONT	DVP	1	DVP_P4-2: 56372	breaker	3144.0	110.5	112.2	AC	59.51
3308887	314903	8CHCKAHM	DVP	314908	8ELMONT	DVP	1	DVP_P1-2: LN 563	single	2442.12	112.42	114.1	AC	47.45
3308888	314903	8CHCKAHM	DVP	314908	8ELMONT	DVP	1	DVP_P1-2: LN 576	single	2442.12	112.14	113.83	AC	47.61
3308951	314905	8CHANCE	DVP	314900	8BRISTER	DVP	1	DVP_P1-2: LN 594	single	2442.12	114.02	115.2	AC	33.11
3308952	314905	8CHANCE	DVP	314900	8BRISTER	DVP	1	DVP_P1-2: LN 573	single	2442.12	109.02	110.18	AC	32.49
3308329	314908	8ELMONT	DVP	314911	8LADYSMITH	DVP	1	DVP_P4-2: 57602	breaker	3351.0	123.2	124.95	AC	68.1
3308330	314908	8ELMONT	DVP	314911	8LADYSMITH	DVP	1	DVP_P4-2: WT576	breaker	3351.0	123.2	124.95	AC	68.1
3308782	314908	8ELMONT	DVP	314911	8LADYSMITH	DVP	1	DVP_P1-2: LN 576	single	2442.12	138.87	140.79	AC	54.31
3308783	314908	8ELMONT	DVP	314911	8LADYSMITH	DVP	1	DVP_P1-2: LN 563	single	2442.12	128.32	129.97	AC	46.23

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308492	314911	8LADYSMITH	DVP	314905	8CHANCE	DVP	1	DVP_P4-2: SPOTS H1T594	breaker	3351.0	103.18	104.27	AC	41.92
3309061	314911	8LADYSMITH	DVP	314905	8CHANCE	DVP	1	DVP_P1-2: LN 573	single	2738.22	103.9	104.97	AC	33.6
3309062	314911	8LADYSMITH	DVP	314905	8CHANCE	DVP	1	DVP_P1-2: LN 594	single	2738.22	102.4	103.43	AC	32.36
3308436	314914	8MDLTHAN	DVP	314918	8NO ANNA	DVP	1	DVP_P4-2: 557T574	breaker	3637.0	111.97	113.4	AC	67.94
3308845	314914	8MDLTHAN	DVP	314918	8NO ANNA	DVP	1	DVP_P1-2: LN 574	single	2442.12	110.19	111.96	AC	49.94
3308847	314914	8MDLTHAN	DVP	314918	8NO ANNA	DVP	1	DVP_P1-2: LN 557	single	2442.12	103.46	105.17	AC	47.58
3308927	924510	AB2-100 TAP	DVP	314563	6CLUBHSE	DVP	1	DVP_P1-2: LN 246	single	375.06	113.03	120.39	AC	29.33
3308928	924510	AB2-100 TAP	DVP	314563	6CLUBHSE	DVP	1	DVP_P1-2: LN 6005_FSA	single	375.06	111.44	118.8	AC	29.33
3308929	924510	AB2-100 TAP	DVP	314563	6CLUBHSE	DVP	1	DVP_P1-2: AA1-138_FSA	single	375.06	111.43	118.79	AC	29.33
3308932	924510	AB2-100 TAP	DVP	314563	6CLUBHSE	DVP	1	Base Case	single	346.86	100.36	105.99	AC	20.58
3308324	936530	AD2-068 TAP	DVP	304451	6GREENVILE T	CPL	1	DVP_P4-2: 2020T2144	breaker	478.0	136.47	146.94	AC	52.36
3309535	936530	AD2-068 TAP	DVP	304451	6GREENVILE T	CPL	1	DVP_P7-1: LN 2058-2181	tower	478.0	156.39	165.41	AC	46.54

12.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308956	313713	6SWAMP	DVP	313721	6ACORN HILL	DVP	1	DVP_P1-2: LN 2020	operation	449.32	147.05	159.38	AC	52.98
3309221	313714	6PERQUIMANS	DVP	314662	6S HERTFORD	DVP	1	DVP_P1-2: LN 246	operation	733.2	120.58	130.71	AC	82.35
3309226	313714	6PERQUIMANS	DVP	314662	6S HERTFORD	DVP	1	Base Case	operation	733.2	95.45	104.66	AC	66.73
3308958	313721	6ACORN HILL	DVP	314648	6SUNBURY	DVP	1	DVP_P1-2: LN 2020	operation	449.32	147.01	159.34	AC	52.98
3308861	313845	6HATHAWAY	DVP	304222	6ROCKYMT230 T	CPL	1	DVP_P1-2: LN 2181	operation	374.0	159.35	164.85	AC	23.62
3309018	313845	6HATHAWAY	DVP	314591	6NASH	DVP	1	DVP_P1-2: LN 2058	operation	449.32	141.82	146.49	AC	24.09

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPAC T
3309447	313885	6WALBMRL	DVP	313714	6PERQUIMANS	DVP	1	DVP_P1-2: LN 246	operation	733.2	91.99	101.94	AC	82.4
3309396	314203	6MACKEYS	DVP	314637	6EDENTON	DVP	1	DVP_P1-2: LN 246	operation	731.32	96.09	106.09	AC	82.4
3309009	314209	6SKIFF CREEK	DVP	314386	6KINGS M	DVP	1	DVP_P1-2: LN 557	operation	441.8	139.04	141.67	AC	12.57
3309421	314225	6CHARCTY	DVP	314227	6LAKESIDE	DVP	1	DVP_P1-2: LN 557	operation	984.18	86.35	87.67	AC	15.35
3309238	314282	6CARSON	DVP	314285	6CHRL249	DVP	1	DVP_P1-2: LN 563	operation	559.3	103.37	105.7	AC	14.97
3309280	314285	6CHRL249	DVP	314316	6LOCKS	DVP	1	DVP_P1-2: LN 563	operation	559.3	100.39	102.71	AC	14.97
3308806	314287	6CHESTF B	DVP	314276	6BASIN	DVP	1	DVP_P1-2: LN 563	operation	449.32	161.68	164.49	AC	14.78
3309394	314287	6CHESTF B	DVP	314225	6CHARCTY	DVP	1	DVP_P1-2: LN 557	operation	984.18	89.89	91.21	AC	15.35
3309066	314296	6PENNIMAN	DVP	314415	6WALR209	DVP	1	DVP_P1-2: LN 557	operation	441.8	127.88	130.51	AC	12.57
3309392	314299	6HARROWG	DVP	314263	6TYLER1	DVP	1	DVP_P1-2: LN 563	operation	441.8	71.5	73.7	AC	12.5
3309393	314331	6POE	DVP	314299	6HARROWG	DVP	1	DVP_P1-2: LN 563	operation	441.8	71.15	73.36	AC	12.5
3309038	314386	6KINGS M	DVP	314296	6PENNIMAN	DVP	1	DVP_P1-2: LN 557	operation	441.8	131.49	134.12	AC	12.57
3309241	314415	6WALR209	DVP	314391	6LIGH209	DVP	1	DVP_P1-2: LN 557	operation	441.8	114.05	116.68	AC	12.57
3309171	314435	6SAPONY	DVP	314282	6CARSON	DVP	1	DVP_P1-2: LN 246	operation	678.68	113.64	119.18	AC	38.47
3309176	314435	6SAPONY	DVP	314282	6CARSON	DVP	1	Base Case	operation	678.68	96.31	100.27	AC	26.7
3308972	314563	6CLUBHSE	DVP	314435	6SAPONY	DVP	1	DVP_P1-2: LN 246	operation	598.78	130.58	136.82	AC	38.49
3308977	314563	6CLUBHSE	DVP	314435	6SAPONY	DVP	1	Base Case	operation	598.78	111.01	115.46	AC	26.72
3308986	314568	3EARLEYS	DVP	314569	6EARLEYS	DVP	1	DVP_P1-3: 6EARLEYS -TX#4	operation	175.78	150.15	154.87	AC	7.57
3309013	314568	3EARLEYS	DVP	314569	6EARLEYS	DVP	2	DVP_P1-3: 6EARLEYS -TX#3	operation	175.78	146.47	151.19	AC	7.57
3308851	314569	6EARLEYS	DVP	314575	6NUCO TP	DVP	1	DVP_P1-2: LN 2092	operation	571.52	156.73	169.1	AC	84.71
3308859	314569	6EARLEYS	DVP	314575	6NUCO TP	DVP	1	Base Case	operation	571.52	98.0	105.79	AC	55.72
3308999	314574	6EVERETS	DVP	936530	AD2-068 TAP	DVP	1	DVP_P1-2: LN 2092	operation	449.32	139.57	150.84	AC	52.3
3309003	314574	6EVERETS	DVP	936530	AD2-068 TAP	DVP	1	Base Case	operation	449.32	118.36	126.68	AC	39.76
3308901	314575	6NUCO TP	DVP	919140	AA1-138 TAP	DVP	1	DVP_P1-2: LN 2092	operation	571.52	150.26	162.58	AC	84.71
3308909	314575	6NUCO TP	DVP	919140	AA1-138 TAP	DVP	1	Base Case	operation	571.52	91.75	99.52	AC	55.72
3309345	314579	6HORNRTN	DVP	934330	AD1-057 TAP	DVP	1	DVP_P1-2: LN 238	operation	441.8	103.56	106.68	AC	16.24
3308875	314583	6LAKEVEW	DVP	314561	6CAROLNA	DVP	1	DVP_P1-2: LN 2056-A	operation	375.06	138.48	142.21	AC	14.38
3309041	314583	6LAKEVEW	DVP	924510	AB2-100 TAP	DVP	1	DVP_P1-2: LN 246	operation	375.06	137.41	145.73	AC	36.78
3309046	314583	6LAKEVEW	DVP	924510	AB2-100 TAP	DVP	1	Base Case	operation	346.86	114.42	122.06	AC	25.8

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPAC T
3309130	314591	6NASH	DVP	304226	6PA-RMOUNT#4	CPL	1	DVP_P1-2: LN 2058	operation	470.0	130.13	134.6	AC	24.09
3309293	314616	6TRWBRDG	DVP	933990	AD1-023 TAP	DVP	1	DVP_P1-2: LN 2092	operation	571.52	121.34	124.43	AC	18.65
3308867	314620	6CASHIE	DVP	314569	6EARLEYS	DVP	1	DVP_P1-2: LN 2092	operation	571.52	136.03	170.82	AC	202.69
3309444	314637	6EDENTON	DVP	313885	6WALBMRL	DVP	1	DVP_P1-2: LN 246	operation	733.2	91.99	101.95	AC	82.4
3309209	314638	6ELIZ CT	DVP	314647	6SHAWBRO	DVP	1	DVP_P1-2: LN 6001	operation	571.52	121.47	131.98	AC	57.37
3309214	314638	6ELIZ CT	DVP	314647	6SHAWBRO	DVP	1	Base Case	operation	530.16	96.67	104.28	AC	38.73
3308970	314648	6SUNBURY	DVP	314537	6SUFFOLK	DVP	1	DVP_P1-2: LN 2020	operation	449.32	144.84	157.16	AC	52.98
3309336	314651	6WINFALL	DVP	313713	6SWAMP	DVP	1	DVP_P1-2: LN 2020	operation	449.32	107.71	119.7	AC	53.03
3309265	314662	6S HERTFORD	DVP	314651	6WINFALL	DVP	1	DVP_P1-2: LN 246	operation	733.2	118.33	128.45	AC	82.35
3309270	314662	6S HERTFORD	DVP	314651	6WINFALL	DVP	1	Base Case	operation	733.2	93.24	102.43	AC	66.73
3309185	314902	8CARSON	DVP	314914	8MDLTHAN	DVP	1	DVP_P1-2: LN 557	operation	3218.56	107.21	108.88	AC	61.11
3308881	314903	8CHCKAHM	DVP	314908	8ELMONT	DVP	1	DVP_P1-2: LN 563	operation	2442.12	141.73	143.82	AC	59.5
3308886	314903	8CHCKAHM	DVP	314908	8ELMONT	DVP	1	Base Case	operation	2442.12	114.55	116.17	AC	45.83
3308948	314905	8CHANCE	DVP	314900	8BRISTER	DVP	1	DVP_P1-2: LN 594	operation	2442.12	135.46	136.93	AC	41.52
3308779	314908	8ELMONT	DVP	314911	8LADYSMITH	DVP	1	DVP_P1-2: LN 576	operation	2442.12	168.92	171.32	AC	68.1
3309058	314911	8LADYSMITH	DVP	314905	8CHANCE	DVP	1	DVP_P1-2: LN 573	operation	2738.22	121.72	123.06	AC	42.13
3308843	314914	8MDLTHAN	DVP	314918	8NO ANNA	DVP	1	DVP_P1-2: LN 574	operation	2442.12	145.35	147.57	AC	62.62
3308850	314914	8MDLTHAN	DVP	314918	8NO ANNA	DVP	1	Base Case	operation	2442.12	87.57	89.23	AC	46.9
3309199	314918	8NO ANNA	DVP	314934	8SPOTSYL	DVP	1	DVP_P1-2: LN 581	operation	3218.56	111.12	112.37	AC	46.68
3309207	314918	8NO ANNA	DVP	314934	8SPOTSYL	DVP	1	Base Case	operation	3218.56	87.35	88.3	AC	35.2
3309404	314918	8NO ANNA	DVP	314911	8LADYSMITH	DVP	1	DVP_P1-2: LN 573	operation	3218.56	91.4	92.3	AC	32.84
3309151	314934	8SPOTSYL	DVP	314916	8MORRSVL	DVP	1	DVP_P1-2: LN 552	operation	3218.56	114.74	115.99	AC	46.23
3309159	314934	8SPOTSYL	DVP	314916	8MORRSVL	DVP	1	Base Case	operation	3218.56	90.53	91.48	AC	34.93
3308911	919140	AA1-138 TAP	DVP	314537	6SUFFOLK	DVP	1	DVP_P1-2: LN 2092	operation	571.52	150.07	162.39	AC	84.71
3308919	919140	AA1-138 TAP	DVP	314537	6SUFFOLK	DVP	1	Base Case	operation	571.52	91.66	99.43	AC	55.72
3308921	924510	AB2-100 TAP	DVP	314563	6CLUBHSE	DVP	1	DVP_P1-2: LN 246	operation	375.06	156.63	164.95	AC	36.78
3308926	924510	AB2-100 TAP	DVP	314563	6CLUBHSE	DVP	1	Base Case	operation	346.86	135.27	142.88	AC	25.8
3309107	933990	AD1-023 TAP	DVP	314620	6CASHIE	DVP	1	DVP_P1-2: LN 2092	operation	571.52	137.91	140.94	AC	18.65
3309025	936530	AD2-068 TAP	DVP	304451	6GREENVILLE T	CPL	1	DVP_P1-2: LN 2092	operation	478.0	136.09	146.56	AC	52.3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPAC T
3309029	936530	AD2-068 TAP	DVP	304451	6GREENVILE T	CPL	1	Base Case	operation	478.0	116.12	123.83	AC	39.76

12.5 System Reinforcements

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN
3308856, 3308857	3	GEARLEYS - 6NUCO TP 230 kV	Rebuild 13.7 miles 2-636 ACSR Project Type : FAC Cost : \$20.55 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 AE1-026 is the driver for this upgrade.	\$20,550,000	\$20,550,000	N6138
3308906, 3308907	5	6NUCO TP - AA1-138 TAP 230 kV	Rebuild 13.35 miles 2-636 ACSR Project Type : FAC Cost : \$20.025 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 AE1-026 is the driver for this upgrade.	\$20,025,000	\$20,025,000	N6139
3308916, 3308917	9	AA1-138 TAP - 6SUFFOLK 230 kV	Rebuild 17.3 miles 2-636 ACSR Project Type : FAC Cost : \$25.95 million Time Estimate : 36-40 Months New Ratings: 1047/1047/1204 AE1-026 is the driver for this upgrade.	\$25,950,000	\$25,950,000	N6140

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN
3309047, 3309048, 3308927, 3308928, 3308929, 3308932	6, 28	6LAKEVEW - AB2-100 TAP - 6CLUBHSE 230 kV	Rebuild 2.0 miles 2-636 ACSR & Rebuild 16.0 miles 2-636 ACSR Project Type : FAC Cost : \$27 million Projected in-service date: 12/31/2024 New Ratings: 1047/1047/1204 Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study.	\$27,000,000	\$0	B3121
3309232	10	AD1-057 TAP - 6HATHAWAY 230 kV	Rebuild 12.8 miles 2-636 ACSR Project Type : FAC Cost : \$ 19.2 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 AE1-026 is the driver for this upgrade	\$19,200,000	\$19,200,000	N6223

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN
3309133	7	6NASH - 6PA-RMOUNT#4 230 kV	DOMINION: Rebuild DEV portion 3.3 miles 2-636 ACSR Project Type : FAC Cost : \$ 8.25 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 AE1-026 is the driver for this upgrade. Duke Energy/Progress (CPLE): Replace wave trap Project Type : FAC Cost : \$ 0.5 million Time Estimate : 18 Months New Ratings: 542/542/542 AE1-026 is the driver for this upgrade.	\$8,750,000	\$8,750,000	N6444

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN																																																												
3309004, 3309005, 3309030, 3309031, 3308365, 3309544, 3308324, 3309535	4, 11	6EVERETS - AD2-068 TAP 230 kV & AD2-068 TAP - 6GREENVILLE T 230 kV	DOMINION:	\$39,250,000	\$6,670,696	N6164																																																												
			Rebuild 20.32 miles of 230 kV Line 218 from Everetts to Greenville with 2-636 ACSR																																																															
			Project Type : FAC																																																															
			Cost : \$ 30.75 million																																																															
			Time Estimate : 30-36 Months																																																															
			New Ratings: 1047/1047/1204																																																															
			The cost allocation table is below:																																																															
			<table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost (\$30,750,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD1-023</td><td>18.24</td><td>8.50%</td><td>\$2,614,947</td><td>'DVP_P7-1: LN 2058-2181'</td><td>tower</td></tr><tr><td>AD1-047</td><td>7.1</td><td>3.31%</td><td>\$1,017,880</td><td>'DVP_P7-1: LN 2058-2181'</td><td>tower</td></tr><tr><td>AD1-057</td><td>13.49</td><td>6.29%</td><td>\$1,933,971</td><td>'DVP_P7-1: LN 2058-2181'</td><td>tower</td></tr><tr><td>AD1-076</td><td>82.59</td><td>38.51%</td><td>\$11,840,377</td><td>'DVP_P7-1: LN 2058-2181'</td><td>tower</td></tr><tr><td>AD2-051</td><td>13.6</td><td>6.34%</td><td>\$1,949,741</td><td>'DVP_P7-1: LN 2058-2181'</td><td>tower</td></tr><tr><td>AD2-089</td><td>17.17</td><td>8.01%</td><td>\$2,461,548</td><td>'DVP_P7-1: LN 2058-2181'</td><td>tower</td></tr><tr><td>AD2-090</td><td>6.55</td><td>3.05%</td><td>\$939,030</td><td>'DVP_P7-1: LN 2058-2181'</td><td>tower</td></tr><tr><td>AD2-169</td><td>9.22</td><td>4.30%</td><td>\$1,321,810</td><td>'DVP_P7-1: LN 2058-2181'</td><td>tower</td></tr><tr><td>AE1-026</td><td>46.53</td><td>21.69%</td><td>\$6,670,696</td><td>'DVP_P7-1: LN 2058-2181'</td><td>tower</td></tr></table>				Queue	MW contribution	Percentage of Cost	Cost (\$30,750,000)	Contingency Name	Contingency Type	AD1-023	18.24	8.50%	\$2,614,947	'DVP_P7-1: LN 2058-2181'	tower	AD1-047	7.1	3.31%	\$1,017,880	'DVP_P7-1: LN 2058-2181'	tower	AD1-057	13.49	6.29%	\$1,933,971	'DVP_P7-1: LN 2058-2181'	tower	AD1-076	82.59	38.51%	\$11,840,377	'DVP_P7-1: LN 2058-2181'	tower	AD2-051	13.6	6.34%	\$1,949,741	'DVP_P7-1: LN 2058-2181'	tower	AD2-089	17.17	8.01%	\$2,461,548	'DVP_P7-1: LN 2058-2181'	tower	AD2-090	6.55	3.05%	\$939,030	'DVP_P7-1: LN 2058-2181'	tower	AD2-169	9.22	4.30%	\$1,321,810	'DVP_P7-1: LN 2058-2181'	tower	AE1-026	46.53	21.69%	\$6,670,696	'DVP_P7-1: LN 2058-2181'	tower
			Queue				MW contribution	Percentage of Cost	Cost (\$30,750,000)	Contingency Name	Contingency Type																																																							
			AD1-023				18.24	8.50%	\$2,614,947	'DVP_P7-1: LN 2058-2181'	tower																																																							
			AD1-047				7.1	3.31%	\$1,017,880	'DVP_P7-1: LN 2058-2181'	tower																																																							
			AD1-057				13.49	6.29%	\$1,933,971	'DVP_P7-1: LN 2058-2181'	tower																																																							
			AD1-076				82.59	38.51%	\$11,840,377	'DVP_P7-1: LN 2058-2181'	tower																																																							
AD2-051	13.6	6.34%	\$1,949,741	'DVP_P7-1: LN 2058-2181'	tower																																																													
AD2-089	17.17	8.01%	\$2,461,548	'DVP_P7-1: LN 2058-2181'	tower																																																													
AD2-090	6.55	3.05%	\$939,030	'DVP_P7-1: LN 2058-2181'	tower																																																													
AD2-169	9.22	4.30%	\$1,321,810	'DVP_P7-1: LN 2058-2181'	tower																																																													
AE1-026	46.53	21.69%	\$6,670,696	'DVP_P7-1: LN 2058-2181'	tower																																																													
Duke Energy/Progress (CPLE):																																																																		
Reconductor 1.87 miles to bundled 1590 ACSR, plus terminal equipment																																																																		
Project Type : FAC																																																																		
Cost : \$ 8.5 million																																																																		
Time Estimate : 36 Months																																																																		
New Ratings: 1195/1195/1195																																																																		

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN
3308592	12	6PERQUIMANS - 6S HERTFORD 230 kV	Rebuild 4.65 miles of 230 kV Line 2131 from Perquimans to South Hertford with 2-636 ACSR. Project Type : FAC Cost : \$ 6.975 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 AE1-026 is the driver for this upgrade.	\$6,975,000	\$6,975,000	N6438
3308520, 3308521, 3308951, 3308952	13	8CHANCE - 8BRISTER 500 kV	Rebuild 500kV Line #552 Bristers to Chancellor – 21.6 miles long Project Type : FAC Cost : \$62.15 million Projected in-service date : 12/31/2022 New Ratings: 4330/4330/4979 Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study.	\$62,150,000	\$0	B3019
3308492, 3309061, 3309062, 3308493	14	8LADYSMITH - 8CHANCE 500 kV	Rebuild 500kV Line #581 Ladysmith to Chancellor - 15.2 miles long Project Type : FAC Cost : \$45.6 million Projected in-service date : 12/30/2023 New Ratings: 4330/4330/4979 Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study.	\$45,600,000	\$0	B3021

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN
3308614	13	6S HERTFORD - 6WINFALL 230 kV	Rebuild 3.3 miles of 230 kV Line 2131 from South Hertford to Winfall with 2-636 ACSR. Project Type : FAC Cost : \$ 4.95 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 AE1-026 is the driver for this upgrade.	\$4,950,000	\$4,950,000	N6439
3308862	15	6HATHAWAY - 6ROCKYMT230 T 230 kV	DOMINION: Rebuild Hathaway-Rocky Mount (Duke Energy Progress) 230 kV Line #2181 and Line #2058 with double circuit steel structures using double circuit conductor at current 230 kV standards with a minimum rating of 1047 MVA. Project Type : FAC Cost : \$ 13 million Projected in-service date: 12/31/2024 New Ratings: 1047/1047/1204 Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study. Duke Energy/Progress (CPL): No upgrade on CPL&E. The line is rated at 542/542/542 MVA.	\$13,000,000	\$0	B3122

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN
3309021	16	6HATHAWAY - 6NASH 230 kV	Rebuild 0.76 miles of 230 kV Line 2181 from Hathaway to Nash with 2-636 ACSR. Project Type : FAC Cost : \$ 1.9 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AE1-026 could receive cost allocation. Note 2: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study.	\$1,900,000	\$0	N6442
3309010, 330906, 3309039	17, 20, 19	6SKIFF CREEK - 6KINGS M 230 kV & 6KINGS M - 6PENNIMAN 230 kV & 6PENNIMAN - 6WALR209 230 kV	Rebuild 6.1 miles of Waller-Skiffes Creek 230 kV Line (#2154) between Waller and Kings Mill to current standards with a minimum summer emergency rating of 1047 MVA utilizing single circuit steel structures. Remove this 6.1 mile section of Line #58 between Waller and Kings Mill. Rebuild the 1.6 miles of Line #2154 and #19 between Kings Mill and Skiffe Creek to current standards with a minimum summer emergency rating of 1047 MVA at 230 kV for Line #2154 and 261 MVA at 115 kV for Line #19, utilizing double circuit steel structures. Project Type : CON Cost : \$10 million Projected in-service date : 12/30/2024 New Ratings: 1047/1047/1204 Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study.	\$10,000,000	\$0	B3057

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN																																				
3308370	1	6CARSON - 6CHRL249 230 kV	Line #249 Carson - Chaparral: wreck and rebuild the line of 7.5 miles from Carson to Chaparral to increase its line rating to 1047 MVA (normal), 1047 MVA (emergency), and 1204 MVA (load dump). Project Type : FAC Cost : \$ 18.675 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 The cost allocation table is below: <table><tr><th>Queue</th><th>MW contribution</th><th>Percent age of Cost</th><th>Cost (\$18,675,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD1-088</td><td>0.62</td><td>1.35%</td><td>\$252,255</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-033</td><td>8.56</td><td>18.65%</td><td>\$3,482,745</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-063</td><td>10.63</td><td>23.16%</td><td>\$4,324,951</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-169</td><td>11.33</td><td>24.68%</td><td>\$4,609,755</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-026</td><td>14.76</td><td>32.16%</td><td>\$6,005,294</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr></table>	Queue	MW contribution	Percent age of Cost	Cost (\$18,675,000)	Contingency Name	Contingency Type	AD1-088	0.62	1.35%	\$252,255	'DVP_P4-2: 562T563'	breaker	AD2-033	8.56	18.65%	\$3,482,745	'DVP_P4-2: 562T563'	breaker	AD2-063	10.63	23.16%	\$4,324,951	'DVP_P4-2: 562T563'	breaker	AD2-169	11.33	24.68%	\$4,609,755	'DVP_P4-2: 562T563'	breaker	AE1-026	14.76	32.16%	\$6,005,294	'DVP_P4-2: 562T563'	breaker	\$18,675,000	\$6,005,294	N6112
Queue	MW contribution	Percent age of Cost	Cost (\$18,675,000)	Contingency Name	Contingency Type																																					
AD1-088	0.62	1.35%	\$252,255	'DVP_P4-2: 562T563'	breaker																																					
AD2-033	8.56	18.65%	\$3,482,745	'DVP_P4-2: 562T563'	breaker																																					
AD2-063	10.63	23.16%	\$4,324,951	'DVP_P4-2: 562T563'	breaker																																					
AD2-169	11.33	24.68%	\$4,609,755	'DVP_P4-2: 562T563'	breaker																																					
AE1-026	14.76	32.16%	\$6,005,294	'DVP_P4-2: 562T563'	breaker																																					

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN																														
3308390	2	6CHRL249 - 6LOCKS 230 kV	Line #249 Chaparra - Locks: Wreck and rebuild the line of 3 miles from Carson to Chaparral to increase its line rating to 1047 MVA (normal), 1047 MVA (emergency), and 1204 MVA (load dump). A Va CPCN is required. Project Type : FAC Cost : \$ 7.15 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 The cost allocation table is below: <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost (\$7,150,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD2-033</td><td>5.79</td><td>13.61%</td><td>\$973,127</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-063</td><td>10.63</td><td>25.01%</td><td>\$1,788,131</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-169</td><td>11.33</td><td>26.66%</td><td>\$1,905,882</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-026</td><td>14.76</td><td>34.73%</td><td>\$2,482,861</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost (\$7,150,000)	Contingency Name	Contingency Type	AD2-033	5.79	13.61%	\$973,127	'DVP_P4-2: 562T563'	breaker	AD2-063	10.63	25.01%	\$1,788,131	'DVP_P4-2: 562T563'	breaker	AD2-169	11.33	26.66%	\$1,905,882	'DVP_P4-2: 562T563'	breaker	AE1-026	14.76	34.73%	\$2,482,861	'DVP_P4-2: 562T563'	breaker	\$7,150,000	\$2,482,861	N6116
Queue	MW contribution	Percentage of Cost	Cost (\$7,150,000)	Contingency Name	Contingency Type																															
AD2-033	5.79	13.61%	\$973,127	'DVP_P4-2: 562T563'	breaker																															
AD2-063	10.63	25.01%	\$1,788,131	'DVP_P4-2: 562T563'	breaker																															
AD2-169	11.33	26.66%	\$1,905,882	'DVP_P4-2: 562T563'	breaker																															
AE1-026	14.76	34.73%	\$2,482,861	'DVP_P4-2: 562T563'	breaker																															
3308283, 3308808	18	6CHESTF B - 6BASIN 230 kV	PJM baseline upgrade b2990: Chesterfield to Basin 230 kV line - Replace 0.14 miles of 1109 ACAR with a conductor which will increase the line rating to approximately 706 MVA. The baseline project is in-service. Project Type : FAC Cost : \$0.35 million In-service date : 04/27/2018 New Ratings: 706/706/812 This baseline is in-service and no further upgrade is required.	\$350,000	\$0	B2990																														

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN																														
3309242	21	6WALR209 - 6LIGH209 230 kV	PJM Baseline Upgrade b3056. Partial Rebuild 230 kV Line #2113 Waller to Lightfoot. The baseline project has a projected in-service date of 12/30/2024. Project Type : FAC Cost : \$4 million Projected In-service date : 12/30/2024 New Ratings: 1047/1047/1204 Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study.	\$4,000,000	\$0	B3056																														
3308310, 3308311, 3308426	23, 24	3EARLEYS - 6EARLEYS 230/115 kV Transformer 1 & 2	Earleys 230 – 115 kV Transformer: add additional 230-115 kV transformer. Project Type : CON Cost : \$ 5.5 million Time Estimate : 24-30 Months New Ratings: 276/292/328 The cost allocation table is below: <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost (\$5,500,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD1-047</td><td>1.71</td><td>5.10%</td><td>\$280,697</td><td>'DVP_P4-2: 2012TH4'</td><td>breaker</td></tr><tr><td>AD2-090</td><td>10.34</td><td>30.91%</td><td>\$1,700,301</td><td>'DVP_P4-2: 2012TH4'</td><td>breaker</td></tr><tr><td>AD2-169</td><td>10.44</td><td>31.21%</td><td>\$1,716,745</td><td>'DVP_P4-2: 2012TH4'</td><td>breaker</td></tr><tr><td>AE1-026</td><td>10.96</td><td>32.77%</td><td>\$1,802,254</td><td>DVP_P4-2: 201262'</td><td>breaker</td></tr></table> The upgrade will need to be tested during the Facilities Study phase to ensure the upgrade mitigates the identified violations and also does not cause any additional violations.	Queue	MW contribution	Percentage of Cost	Cost (\$5,500,000)	Contingency Name	Contingency Type	AD1-047	1.71	5.10%	\$280,697	'DVP_P4-2: 2012TH4'	breaker	AD2-090	10.34	30.91%	\$1,700,301	'DVP_P4-2: 2012TH4'	breaker	AD2-169	10.44	31.21%	\$1,716,745	'DVP_P4-2: 2012TH4'	breaker	AE1-026	10.96	32.77%	\$1,802,254	DVP_P4-2: 201262'	breaker	\$5,500,000	\$1,802,254	N6658
Queue	MW contribution	Percentage of Cost	Cost (\$5,500,000)	Contingency Name	Contingency Type																															
AD1-047	1.71	5.10%	\$280,697	'DVP_P4-2: 2012TH4'	breaker																															
AD2-090	10.34	30.91%	\$1,700,301	'DVP_P4-2: 2012TH4'	breaker																															
AD2-169	10.44	31.21%	\$1,716,745	'DVP_P4-2: 2012TH4'	breaker																															
AE1-026	10.96	32.77%	\$1,802,254	DVP_P4-2: 201262'	breaker																															

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN																														
3308346	25	6ELIZ CT - 6SHAWBRO 230 kV	Rebuild 10.28 miles of 230 kV Line 2021 from Elizabeth City to Shawboro with 2-636 ACSR. Project Type : FAC Cost : \$ 15.4 million Time Estimate : 30-36 Months New Ratings: 1047/1047/1204 The cost allocation table is below: <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost (\$15,420,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD1-076</td><td>143.87</td><td>59.97%</td><td>\$9,247,501</td><td>'DVP_P4-2: 246T247'</td><td>breaker</td></tr><tr><td>AD2-051</td><td>15.59</td><td>6.50%</td><td>\$1,002,075</td><td>'DVP_P4-2: 246T247'</td><td>breaker</td></tr><tr><td>AD2-089</td><td>8.98</td><td>3.74%</td><td>\$577,206</td><td>'DVP_P4-2: 246T247'</td><td>breaker</td></tr><tr><td>AE1-026</td><td>71.46</td><td>29.79%</td><td>\$4,593,219</td><td>'DVP_P4-2: 246T247'</td><td>breaker</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost (\$15,420,000)	Contingency Name	Contingency Type	AD1-076	143.87	59.97%	\$9,247,501	'DVP_P4-2: 246T247'	breaker	AD2-051	15.59	6.50%	\$1,002,075	'DVP_P4-2: 246T247'	breaker	AD2-089	8.98	3.74%	\$577,206	'DVP_P4-2: 246T247'	breaker	AE1-026	71.46	29.79%	\$4,593,219	'DVP_P4-2: 246T247'	breaker	\$15,400,000	\$4,593,219	N6218
Queue	MW contribution	Percentage of Cost	Cost (\$15,420,000)	Contingency Name	Contingency Type																															
AD1-076	143.87	59.97%	\$9,247,501	'DVP_P4-2: 246T247'	breaker																															
AD2-051	15.59	6.50%	\$1,002,075	'DVP_P4-2: 246T247'	breaker																															
AD2-089	8.98	3.74%	\$577,206	'DVP_P4-2: 246T247'	breaker																															
AE1-026	71.46	29.79%	\$4,593,219	'DVP_P4-2: 246T247'	breaker																															
3308395, 3308396, 3308887, 3308888	8	8CHCKAHM - 8ELMONT 500 kV	Rebuild 27.72 miles of 500 kV Line 557 from Chickahominy to Elmont with 3-1351.5 113C ACSR. Project Type : FAC Cost : \$ 85.932 million Time Estimate : 48-60 Months New Ratings: 4453/4453/5121 MVA The cost allocation table is below: <table><tr><th>Queue</th><th>MW contribution</th><th>Percent age of Cost</th><th>Cost (\$85,932,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD2 Queue</td><td>528.74</td><td>89.38%</td><td>\$76,802,660</td><td>'DVP_P4-2: 563T576'</td><td>breaker</td></tr><tr><td>AE1-026</td><td>62.85</td><td>10.62%</td><td>\$9,129,340</td><td>'DVP_P4-2: 563T576'</td><td>breaker</td></tr></table>	Queue	MW contribution	Percent age of Cost	Cost (\$85,932,000)	Contingency Name	Contingency Type	AD2 Queue	528.74	89.38%	\$76,802,660	'DVP_P4-2: 563T576'	breaker	AE1-026	62.85	10.62%	\$9,129,340	'DVP_P4-2: 563T576'	breaker	\$85,932,000	\$9,129,340	N6207												
Queue	MW contribution	Percent age of Cost	Cost (\$85,932,000)	Contingency Name	Contingency Type																															
AD2 Queue	528.74	89.38%	\$76,802,660	'DVP_P4-2: 563T576'	breaker																															
AE1-026	62.85	10.62%	\$9,129,340	'DVP_P4-2: 563T576'	breaker																															

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN
3308329, 3308330, 3308782, 3308783	26	8ELMONT - 8LADYSMITH 500 kV	PJM baseline upgrade b3020: Rebuild 500kV Line #574 Ladysmith to Elmont - 26.2 miles long. Project Type : FAC Cost : \$ 65.5 million Expected in-service date: 12/31/2025 New Ratings: 4330/4330/4979 MVA Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study.	\$65,500,000	\$0	b3020

3308436, 3308845, 3308847	27	8MDLTHAN - 8NO ANNA 500 kV	PJM Network Upgrade n6055: Replace wave trap at North Anna Substation for Midlothian – North Anna 500 kV line #576. This will increase emergency rating by 31% to 3424 MVA. The network project had a required in-service date of 10/01/2019. Project Type : FAC Cost : \$250,000 Time Estimate : 30-36 Months New Ratings: 3424/3424/3938 Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AE1-026 could receive cost allocation. Note 2: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study. PJM Network Upgrade N5609: Rebuild 41.13 miles of 500 kV Line 576 from Midlothian to North Anna with 3-1351.5 125C ACSR. Project Type : FAC Cost : \$127.5 million Time Estimate : 48-60 Months New Ratings: 4816/4816/5539 The cost allocation table is below: <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost \$127,503,000</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AC2-141</td><td>77.46</td><td>21.11%</td><td>\$26,911,124</td><td>'557T574'</td><td>breaker</td></tr><tr><td>AD1-025</td><td>41.93</td><td>11.43%</td><td>\$14,567,305</td><td>'DVP_P4-2: 557T574'</td><td>breaker</td></tr><tr><td>AD1-076</td><td>137.77</td><td>37.54%</td><td>\$47,864,001</td><td>'DVP_P4-2: 557T574'</td><td>breaker</td></tr><tr><td>AD1-151</td><td>41.9</td><td>11.42%</td><td>\$14,556,882</td><td>'DVP_P4-2: 557T574'</td><td>breaker</td></tr><tr><td>AE1-026</td><td>67.94</td><td>18.51%</td><td>\$23,603,689</td><td>'DVP_P4-2: 557T574'</td><td>breaker</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost \$127,503,000	Contingency Name	Contingency Type	AC2-141	77.46	21.11%	\$26,911,124	'557T574'	breaker	AD1-025	41.93	11.43%	\$14,567,305	'DVP_P4-2: 557T574'	breaker	AD1-076	137.77	37.54%	\$47,864,001	'DVP_P4-2: 557T574'	breaker	AD1-151	41.9	11.42%	\$14,556,882	'DVP_P4-2: 557T574'	breaker	AE1-026	67.94	18.51%	\$23,603,689	'DVP_P4-2: 557T574'	breaker	\$127,7500,00	\$23,603,689	N6055 & N5609
Queue	MW contribution	Percentage of Cost	Cost \$127,503,000	Contingency Name	Contingency Type																																					
AC2-141	77.46	21.11%	\$26,911,124	'557T574'	breaker																																					
AD1-025	41.93	11.43%	\$14,567,305	'DVP_P4-2: 557T574'	breaker																																					
AD1-076	137.77	37.54%	\$47,864,001	'DVP_P4-2: 557T574'	breaker																																					
AD1-151	41.9	11.42%	\$14,556,882	'DVP_P4-2: 557T574'	breaker																																					
AE1-026	67.94	18.51%	\$23,603,689	'DVP_P4-2: 557T574'	breaker																																					

3309494	22	3BTLEBRO - 3ROCKYMT115 T 115 kV	DOMINION: Upgrade no. 1: PJM Network Upgrade n6118: Upgrading the breaker leads at DVP's terminal will bring the rating to 239/239/239 MVA (Limited by terminal equipment at Rocky Mount) Dominion End Ratings: 438/478/581 MVA Project Type : FAC Cost : \$100,000 Time Estimate : 30-36 Months New Ratings: 239/239/239 MVA Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AE1-026 could receive cost allocation. Note 2: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study. Upgrade no. 2: PJM Network Upgrade n6618: 1. Split the 155 kV Bus at Hathaway into two separate buses with a 115 kV Line on each bus. 2. Rebuild Line #55 (Tarboro – Anaconda) and close the tie switch between Line 55 & 80. 3. Line #1001 is opened at Battleboro thus making Line #1001 radial from Chestnut Substation Project Type : CON Cost : \$6,900,000 Time Estimate : 20 Months The upgrade will need to be tested during the Facilities Study phase to ensure it does not cause any additional violations. Queue Project AE1-026 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AE1-026 could receive cost allocation. Note 2: Although Queue Project AE1-026 may not have cost responsibility for this upgrade, Queue Project AE1-026 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the upgrade, Queue Project AE1-026 will need an interim study. Duke Energy/Progress (CPLE):	\$7,000,000	\$0	n6118, n6618
---------	----	---------------------------------------	--	-------------	-----	-----------------

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-026	NUN
			Duke Energy/Progress Portion: A potential constraint was identified by PJM on the Duke Energy/Progress (DEP) portion of the Battleboro- Rocky Mountain 115 kV line. The Queue Project AE1-026 may be subject to operational restriction if real-time system reliability issues occur. Additionally, if a baseline Network Upgrade project is identified on the Battleboro- Rocky Mountain 115 kV line prior to the execution of the Queue Project's final agreements, the Queue Project may require this upgrade to be in-service to be deliverable to the PJM system. If Queue Project AE1-026 comes into service prior to completion of the baseline Network Upgrade, Queue Project AE1-026 will need an interim deliverability study Note: There are no mitigations currently planned for this overload. Though, the following upgrade has been scoped to increase the DUKE end rating to 301/301/301. Scope: Reconductor the circuit to 1590 ACSR will bring the rating to 185/205/214 MVA. Upgrading the terminal equipment at Rocky Mount will bring the rating to 301/301/301 MVA (Limited by 1590 ACSR conductor). Project Type: FAC Cost: TBD Time Estimate: 36 Months			
			TOTAL COST	\$ 514,807,000	\$ 160,687,353	

12.6 Flow Gate Details

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

12.6.1 Contingency Descriptions

Contingency Name	Contingency Definition
DVP_P1-2: LN 6005_FSA	CONTINGENCY 'DVP_P1-2: LN 6005_FSA' OPEN BRANCH FROM BUS 314575 TO BUS 919140 CKT 1 /* 6NUCO TP 230.00 - AA1-138 TAP 230.00 END
DVP_P1-2: LN 563	CONTINGENCY 'DVP_P1-2: LN 563' OPEN BRANCH FROM BUS 314902 TO BUS 314914 CKT 1 /* 8CARSON 500.00 - 8MDLTHAN 500.00 END
DVP_P7-1: LN 198-581	CONTINGENCY 'DVP_P7-1: LN 198-581' OPEN BRANCH FROM BUS 314135 TO BUS 314367 CKT 1 /* 3CHANCE 115.00 - 3CHANC_1 115.00 OPEN BRANCH FROM BUS 314135 TO BUS 314775 CKT 1 /* 3CHANCE 115.00 - 3NI RVER 115.00 OPEN BRANCH FROM BUS 314755 TO BUS 314779 CKT 1 /* 3SPOTSYL 115.00 - 3TDTAVRN 115.00 OPEN BRANCH FROM BUS 314775 TO BUS 314779 CKT 1 /* 3NI RVER 115.00 - 3TDTAVRN 115.00 OPEN BRANCH FROM BUS 314815 TO BUS 314877 CKT 1 /* 3OAK GRE 115.00 - 3OAK G_1 115.00 OPEN BUS 314367 /* ISLAND: 3CHANC_1 115.00 OPEN BUS 314775 /* ISLAND: 3NI RVER 115.00 OPEN BUS 314779 /* ISLAND: 3TDTAVRN 115.00 OPEN BUS 314877 /* ISLAND: 3OAK G_1 115.00 OPEN BRANCH FROM BUS 314135 TO BUS 314905 CKT 2 /* 3CHANCE 115.00 - 8CHANCE 500.00 OPEN BRANCH FROM BUS 314905 TO BUS 314911 CKT 1 /* 8CHANCE 500.00 - 8LADYSMITH 500.00 END
DVP_P4-2: 57302	CONTINGENCY 'DVP_P4-2: 57302' /* NORTH ANNA 500 KV OPEN BRANCH FROM BUS 314918 TO BUS 314934 CKT 1 /* 8NO ANNA 500.00 - 8SPOTSYL 500.00 OPEN BRANCH FROM BUS 314232 TO BUS 314918 CKT 1 /* 6NO ANNA 230.00 - 8NO ANNA 500.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 246	CONTINGENCY 'DVP_P1-2: LN 246' OPEN BRANCH FROM BUS 314537 TO BUS 919140 CKT 1 /* 6SUFFOLK 230.00 - AA1-138 TAP 230.00 OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1 /* 6EARLEYS 230.00 - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1 /* 6NUCO TP 230.00 - 6NUCOR 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 919140 CKT 1 /* 6NUCO TP 230.00 - AA1-138 TAP 230.00 OPEN BUS 314575 /* ISLAND: 6NUCO TP 230.00 OPEN BUS 314590 /* ISLAND: 6NUCOR 230.00 OPEN BUS 919140 /* ISLAND: AA1-138 TAP 230.00 END
DVP_P1-2: LN 2056-A	CONTINGENCY 'DVP_P1-2: LN 2056-A' OPEN BRANCH FROM BUS 313845 TO BUS 934330 CKT 1 /* 6HATHAWAY 230.00 - AD1-057 TAP 230.00 END
DVP_P1-2: LN 568	CONTINGENCY 'DVP_P1-2: LN 568' OPEN BRANCH FROM BUS 314911 TO BUS 314922 CKT 1 /* 8LADYSMITH 500.00 - 8POSSUM 500.00 END
DVP_P4-2: 562T563	CONTINGENCY 'DVP_P4-2: 562T563' /* CARSON 500 KV OPEN BRANCH FROM BUS 314902 TO BUS 314923 CKT 1 /* 8CARSON 500.00 - 8SEPTA 500.00 OPEN BRANCH FROM BUS 314902 TO BUS 314914 CKT 1 /* 8CARSON 500.00 - 8MDLTHAN 500.00 END
DVP_P1-2: AA1-138_FSA	CONTINGENCY 'DVP_P1-2: AA1-138_FSA' OPEN BRANCH FROM BUS 314537 TO BUS 919140 CKT 1 /* 6NUCO TP 230.00 - AA1-138 TAP 230.00 END
DVP_P4-2: WT2171	CONTINGENCY 'DVP_P4-2: WT2171' /* EARLEYS 230 KV OPEN BRANCH FROM BUS 313709 TO BUS 314569 CKT Z1 /* 6HOLLOMAN 230.00 - 6EARLEYS 230.00 OPEN BUS 313709 /* ISLAND: 6HOLLOMAN 230.00 OPEN BUS 918511 /* ISLAND: AA1-065 C OP230.00 OPEN BUS 918512 /* ISLAND: AA1-065 E OP230.00 OPEN BRANCH FROM BUS 314568 TO BUS 314569 CKT 1 /* 3EARLEYS 115.00 - 6EARLEYS 230.00 END
DVP_P4-2: 217162	CONTINGENCY 'DVP_P4-2: 217162' /* EARLEYS 230 KV OPEN BRANCH FROM BUS 313709 TO BUS 314569 CKT Z1 /* 6HOLLOMAN 230.00 - 6EARLEYS 230.00 OPEN BUS 313709 /* ISLAND: 6HOLLOMAN 230.00 OPEN BUS 918511 /* ISLAND: AA1-065 C OP230.00 OPEN BUS 918512 /* ISLAND: AA1-065 E OP230.00 OPEN BRANCH FROM BUS 314568 TO BUS 314569 CKT 2 /* 3EARLEYS 115.00 - 6EARLEYS 230.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 2131	CONTINGENCY 'DVP_P1-2: LN 2131' OPEN BRANCH FROM BUS 313714 TO BUS 313885 CKT 1 /* 6PERQUIMANS 230.00 - 6WALBMRL 230.00 OPEN BRANCH FROM BUS 313885 TO BUS 314637 CKT 1 /* 6WALBMRL 230.00 - 6EDENTON 230.00 OPEN BRANCH FROM BUS 314203 TO BUS 314637 CKT 1 /* 6MACKEYS 230.00 - 6EDENTON 230.00 OPEN BUS 313885 /* ISLAND: 6WALBMRL 230.00 OPEN BUS 314637 /* ISLAND: 6EDENTON 230.00 END
DVP_P7-1: LN 2058-2181	CONTINGENCY 'DVP_P7-1: LN 2058-2181' OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /* 6ROCKYMT230T230.00 - 6HATHAWAY 230.00 OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00 OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-RMOUNT#4230.00 - 6NASH 230.00 OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY 230.00 - 6NASH 230.00 OPEN BUS 314591 /* ISLAND: 6NASH 230.00 END
DVP_P1-2: LN 6001	CONTINGENCY 'DVP_P1-2: LN 6001' OPEN BRANCH FROM BUS 313713 TO BUS 313721 CKT 1 /* 6SWAMP 230.00 - 6ACORN HILL 230.00 END
DVP_P4-2: WT576	CONTINGENCY 'DVP_P4-2: WT576' /* NORTH ANNA 500 KV OPEN BRANCH FROM BUS 314914 TO BUS 314918 CKT 1 /* 8MDLTHAN 500.00 - 8NO ANNA 500.00 OPEN BRANCH FROM BUS 314232 TO BUS 314918 CKT 2 /* 6NO ANNA 230.00 - 8NO ANNA 500.00 END
DVP_P1-2: LN 594	CONTINGENCY 'DVP_P1-2: LN 594' OPEN BRANCH FROM BUS 314916 TO BUS 314934 CKT 1 /* 8MORRSVL 500.00 - 8SPOTSYL 500.00 END
DVP_P4-2: 201262	CONTINGENCY 'DVP_P4-2: 201262' /* EARLEYS 230 KV OPEN BRANCH FROM BUS 314266 TO BUS 314569 CKT 1 /* 6NORTHAMPTON230.00 - 6EARLEYS 230.00 OPEN BRANCH FROM BUS 314266 TO BUS 314599 CKT 1 /* 6NORTHAMPTON230.00 - 6ROA VAL 230.00 OPEN BUS 314266 /* ISLAND: 6NORTHAMPTON230.00 OPEN BRANCH FROM BUS 314568 TO BUS 314569 CKT 1 /* 3EARLEYS 115.00 - 6EARLEYS 230.00 END
DVP_P1-2: LN 2181	CONTINGENCY 'DVP_P1-2: LN 2181' OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00 OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-RMOUNT#4230.00 - 6NASH 230.00 OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY 230.00 - 6NASH 230.00 OPEN BUS 314591 /* ISLAND: 6NASH 230.00 END

Contingency Name	Contingency Definition
DVP_P4-2: 56372	CONTINGENCY 'DVP_P4-2: 56372' /* CARSON 500 KV OPEN BRANCH FROM BUS 314902 TO BUS 314914 CKT 1 /* 8CARSON 500.00 - 8MDLTHAN 500.00 OPEN BRANCH FROM BUS 314282 TO BUS 314902 CKT 1 /* 6CARSON 230.00 - 8CARSON 500.00 END
DVP_P1-3: 6EARLEYS-TX#4	CONTINGENCY 'DVP_P1-3: 6EARLEYS-TX#4' OPEN BRANCH FROM BUS 314568 TO BUS 314569 CKT 2 /* 3EARLEYS 115.00 - 6EARLEYS 230.00 END
DVP_P4-2: 24642	CONTINGENCY 'DVP_P4-2: 24642' /* SUFFOLK 230 KV OPEN BRANCH FROM BUS 314537 TO BUS 919140 CKT 1 /* 6SUFFOLK 230.00 - AA1-138 TAP 230.00 OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1 /* 6EARLEYS 230.00 - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1 /* 6NUCO TP 230.00 - 6NUCOR 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 919140 CKT 1 /* 6NUCO TP 230.00 - AA1-138 TAP 230.00 OPEN BUS 314575 /* ISLAND: 6NUCO TP 230.00 OPEN BUS 314590 /* ISLAND: 6NUCOR 230.00 OPEN BUS 919140 /* ISLAND: AA1-138 TAP 230.00 OPEN BRANCH FROM BUS 314536 TO BUS 314537 CKT 3 /* 3SUFFOLK 115.00 - 6SUFFOLK 230.00 OPEN BRANCH FROM BUS 314537 TO BUS 314928 CKT 2 /* 6SUFFOLK 230.00 - 8SUFFOLK 500.00 END
DVP_P4-2: 24662	CONTINGENCY 'DVP_P4-2: 24662' /* EARLEYS 230 KV OPEN BRANCH FROM BUS 314537 TO BUS 919140 CKT 1 /* 6SUFFOLK 230.00 - AA1-138 TAP 230.00 OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1 /* 6EARLEYS 230.00 - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1 /* 6NUCO TP 230.00 - 6NUCOR 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 919140 CKT 1 /* 6NUCO TP 230.00 - AA1-138 TAP 230.00 OPEN BUS 314575 /* ISLAND: 6NUCO TP 230.00 OPEN BUS 314590 /* ISLAND: 6NUCOR 230.00 OPEN BUS 919140 /* ISLAND: AA1-138 TAP 230.00 OPEN BRANCH FROM BUS 314568 TO BUS 314569 CKT 1 /* 3EARLEYS 115.00 - 6EARLEYS 230.00 END
DVP_P4-2: 246T247	CONTINGENCY 'DVP_P4-2: 246T247' /* SUFFOLK 230 KV OPEN BRANCH FROM BUS 314537 TO BUS 919140 CKT 1 /* 6SUFFOLK 230.00 - AA1-138 TAP 230.00 OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1 /* 6EARLEYS 230.00 - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1 /* 6NUCO TP 230.00 - 6NUCOR 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 919140 CKT 1 /* 6NUCO TP 230.00 - AA1-138 TAP 230.00 OPEN BUS 314575 /* ISLAND: 6NUCO TP 230.00 OPEN BUS 314590 /* ISLAND: 6NUCOR 230.00 OPEN BUS 919140 /* ISLAND: AA1-138 TAP 230.00 OPEN BRANCH FROM BUS 313713 TO BUS 313721 CKT 1 /* 6SWAMP 230.00 - 6ACORN HILL 230.00 OPEN BRANCH FROM BUS 313721 TO BUS 314648 CKT 1 /* 6ACORN HILL 230.00 - 6SUNBURY 230.00 OPEN BRANCH FROM BUS 314537 TO BUS 314648 CKT 1 /* 6SUFFOLK 230.00 - 6SUNBURY 230.00 OPEN BUS 313721 /* ISLAND: 6ACORN HILL 230.00 OPEN BUS 314648 /* ISLAND: 6SUNBURY 230.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 552	CONTINGENCY 'DVP_P1-2: LN 552' OPEN BRANCH FROM BUS 314135 TO BUS 314905 CKT 1 /* 3CHANCE 115.00 - 8CHANCE 500.00 OPEN BRANCH FROM BUS 314900 TO BUS 314905 CKT 1 /* 8BRISTER 500.00 - 8CHANCE 500.00 END
DVP_P4-2: 573T594	CONTINGENCY 'DVP_P4-2: 573T594' /* SPOTSYLVANIA 500 KV OPEN BRANCH FROM BUS 314918 TO BUS 314934 CKT 1 /* 8NO ANNA 500.00 - 8SPOTSYL 500.00 OPEN BRANCH FROM BUS 314916 TO BUS 314934 CKT 1 /* 8MORRSVL 500.00 - 8SPOTSYL 500.00 END
DVP_P1-3: 6EARLEYS-TX#3	CONTINGENCY 'DVP_P1-3: 6EARLEYS-TX#3' OPEN BRANCH FROM BUS 314568 TO BUS 314569 CKT 1 /* 3EARLEYS 115.00 - 6EARLEYS 230.00 END
DVP_P1-2: LN 557	CONTINGENCY 'DVP_P1-2: LN 557' OPEN BRANCH FROM BUS 314214 TO BUS 314903 CKT 1 /* 6CHCKAHM 230.00 - 8CHCKAHM 500.00 OPEN BRANCH FROM BUS 314903 TO BUS 314908 CKT 1 /* 8CHCKAHM 500.00 - 8ELMONT 500.00 END
DVP_P4-2: 2020T2144	CONTINGENCY 'DVP_P4-2: 2020T2144' /* WINFALL 230 KV OPEN BRANCH FROM BUS 313851 TO BUS 314638 CKT 1 /* 6ECITYDP2 230.00 - 6ELIZ CT 230.00 OPEN BRANCH FROM BUS 313851 TO BUS 314639 CKT 1 /* 6ECITYDP2 230.00 - 6TANGLEW 230.00 OPEN BRANCH FROM BUS 314639 TO BUS 314651 CKT 1 /* 6TANGLEW 230.00 - 6WINFALL 230.00 OPEN BUS 313851 /* ISLAND: 6ECITYDP2 230.00 OPEN BUS 314639 /* ISLAND: 6TANGLEW 230.00 OPEN BUS 917122 /* ISLAND: Z2-027 E 230.00 OPEN BRANCH FROM BUS 313713 TO BUS 314651 CKT 1 /* 6SWAMP 230.00 - 6WINFALL 230.00 END
DVP_P1-2: LN 2020	CONTINGENCY 'DVP_P1-2: LN 2020' OPEN BRANCH FROM BUS 313851 TO BUS 314638 CKT 1 /* 6ECITYDP2 230.00 - 6ELIZ CT 230.00 OPEN BRANCH FROM BUS 313851 TO BUS 314639 CKT 1 /* 6ECITYDP2 230.00 - 6TANGLEW 230.00 OPEN BRANCH FROM BUS 314639 TO BUS 314651 CKT 1 /* 6TANGLEW 230.00 - 6WINFALL 230.00 OPEN BUS 313851 /* ISLAND: 6ECITYDP2 230.00 OPEN BUS 314639 /* ISLAND: 6TANGLEW 230.00 OPEN BUS 917122 /* ISLAND: Z2-027 E 230.00 END
DVP_P1-2: LN 574	CONTINGENCY 'DVP_P1-2: LN 574' OPEN BRANCH FROM BUS 314908 TO BUS 314911 CKT 1 /* 8ELMONT 500.00 - 8LADYSMITH 500.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 238	CONTINGENCY 'DVP_P1-2: LN 238' OPEN BRANCH FROM BUS 314282 TO BUS 314435 CKT 1 /* 6CARSON 230.00 - 6SAPONY 230.00 OPEN BRANCH FROM BUS 314435 TO BUS 314563 CKT 1 /* 6SAPONY 230.00 - 6CLUBHSE 230.00 OPEN BRANCH FROM BUS 314562 TO BUS 314563 CKT 1 /* 3CLUBHSE 115.00 - 6CLUBHSE 230.00 230.00 OPEN BUS 314435 /* ISLAND: 6SAPONY 230.00 END
DVP_P1-2: LN 576	CONTINGENCY 'DVP_P1-2: LN 576' OPEN BRANCH FROM BUS 314914 TO BUS 314918 CKT 1 /* 8MDLTHAN 500.00 - 8NO ANNA 500.00 500.00 END
DVP_P4-2: 557T574	CONTINGENCY 'DVP_P4-2: 557T574' /* ELMONT 500 KV OPEN BRANCH FROM BUS 314214 TO BUS 314903 CKT 1 /* 6CHCKAHM 230.00 - 8CHCKAHM 500.00 500.00 OPEN BRANCH FROM BUS 314903 TO BUS 314908 CKT 1 /* 8CHCKAHM 500.00 - 8ELMONT 500.00 500.00 OPEN BRANCH FROM BUS 314908 TO BUS 314911 CKT 1 /* 8ELMONT 500.00 - 8LADYSMITH 500.00 500.00 END
DVP_P4-2: 57602	CONTINGENCY 'DVP_P4-2: 57602' /* NORTH ANNA 500 KV OPEN BRANCH FROM BUS 314914 TO BUS 314918 CKT 1 /* 8MDLTHAN 500.00 - 8NO ANNA 500.00 500.00 OPEN BRANCH FROM BUS 314232 TO BUS 314918 CKT 1 /* 6NO ANNA 230.00 - 8NO ANNA 500.00 500.00 END
Base Case	
DVP_P4-2: 563T576	CONTINGENCY 'DVP_P4-2: 563T576' /* MIDLOTHIAN 500 500 KV OPEN BRANCH FROM BUS 314902 TO BUS 314914 CKT 1 /* 8CARSON 500.00 - 8MDLTHAN 500.00 500.00 OPEN BRANCH FROM BUS 314914 TO BUS 314918 CKT 1 /* 8MDLTHAN 500.00 - 8NO ANNA 500.00 500.00 END
DVP_P4-2: H2T557	CONTINGENCY 'DVP_P4-2: H2T557' /* ELMONT 500 KV OPEN BRANCH FROM BUS 314214 TO BUS 314903 CKT 1 /* 6CHCKAHM 230.00 - 8CHCKAHM 500.00 500.00 OPEN BRANCH FROM BUS 314903 TO BUS 314908 CKT 1 /* 8CHCKAHM 500.00 - 8ELMONT 500.00 500.00 OPEN BRANCH FROM BUS 314218 TO BUS 314908 CKT 2 /* 6ELMONT 230.00 - 8ELMONT 500.00 500.00 END
DVP_P1-2: LN 581	CONTINGENCY 'DVP_P1-2: LN 581' OPEN BRANCH FROM BUS 314135 TO BUS 314905 CKT 2 /* 3CHANCE 115.00 - 8CHANCE 500.00 OPEN BRANCH FROM BUS 314905 TO BUS 314911 CKT 1 /* 8CHANCE 500.00 - 8LADYSMITH 500.00 500.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 573	CONTINGENCY 'DVP_P1-2: LN 573' OPEN BRANCH FROM BUS 314918 TO BUS 314934 CKT 1 /* 8NO ANNA 500.00 - 8SPOTSYL 500.00 END
DVP_P1-2: LN 2058	CONTINGENCY 'DVP_P1-2: LN 2058' OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /* 6ROCKYMT230T230.00 - 6HATHAWAY 230.00 END
DVP_P1-2: LN 2092	CONTINGENCY 'DVP_P1-2: LN 2092' OPEN BRANCH FROM BUS 313714 TO BUS 314662 CKT 1 /* 6PERQUIMANS 230.00 - 6S HERTFORD 230.00 OPEN BRANCH FROM BUS 314651 TO BUS 314662 CKT 1 /* 6WINFALL 230.00 - 6S HERTFORD 230.00 OPEN BUS 314662 /* ISLAND: 6S HERTFORD 230.00 END
DVP_P4-2: H1T594	CONTINGENCY 'DVP_P4-2: H1T594' /* MORRISVILLE 500 KV OPEN BRANCH FROM BUS 314916 TO BUS 314934 CKT 1 /* 8MORRSVL 500.00 - 8SPOTSYL 500.00 OPEN BRANCH FROM BUS 314063 TO BUS 314916 CKT 1 /* 6MORRSVL 230.00 - 8MORRSVL 500.00 OPEN BUS 314897 /* 8MORRS_1 500.00 KV END
DVP_P4-2: SPOTS H1T594	CONTINGENCY 'DVP_P4-2: SPOTS H1T594' /* SPOTSYLVANIA 500 KV OPEN BRANCH FROM BUS 314916 TO BUS 314934 CKT 1 /* 8MORRSVL 500.00 - 8SPOTSYL 500.00 OPEN BRANCH FROM BUS 314755 TO BUS 314934 CKT 1 /* 3SPOTSYL 115.00 - 8SPOTSYL 500.00 END

12.6.2 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308370	314282	6CARSON	DVP	314285	6CHRL249	DVP	1	DVP_P4-2: 562T563	breaker	684.0	112.62	114.47	AC	14.76

Bus #	Bus	MW Impact
314541	3WATKINS	0.23
314554	3BTLEBRO	0.4
314557	3BETHEL	0.39
314572	3EMPORIA	0.33
314574	6EVERETS	1.77
314578	3HORNRTN	1.92
314582	3KELFORD	1.62
314603	3SCOT NK	1.61
314617	3TUNIS	0.36
314620	6CASHIE	0.31
314623	3WITAKRS	0.69
314704	3LAWRENC	0.2
315102	1BRUNSWICKG1	3.87
315103	1BRUNSWICKG2	3.87
315104	1BRUNSWICKG3	3.87
315105	1BRUNSWICKS1	8.03
315131	1EDGECEMA	4.73
315132	1EDGECEMB	4.73
315136	1ROSEMG1	1.22
315137	1ROSEMS1	0.76
315138	1ROSEMG2	0.57
315139	1GASTONA	1.77
315141	1GASTONB	1.77
315150	1BUGGS 1	5.14
315151	1BUGGS 2	5.14
900672	V4-068 E	0.13
916301	Z1-086 C	23.58
916302	Z1-086 E	5.22
917332	Z2-043 E	0.46
917342	Z2-044 E	0.3
917512	Z2-088 E OP1	1.76
918492	AA1-063AE OP	2.11
918512	AA1-065 E OP	1.81
918532	AA1-067 E	0.31
918562	AA1-072 E	0.08
919692	AA2-053 E	2.07
919702	AA2-057 E	1.83
920042	AA2-088 E OP	4.27
920592	AA2-165 E	0.24
920672	AA2-174 E	0.24
923851	AB2-025 C	0.41

Bus #	Bus	MW Impact
923852	AB2-025 E	1.3
923911	AB2-031 C O1	1.6
923912	AB2-031 E O1	0.79
923991	AB2-040 C O1	0.74
923992	AB2-040 E O1	4.29
924022	AB2-043 E O1	2.32
924152	AB2-059 E O1	2.75
924162	AB2-060 E O1	1.9
924302	AB2-077 E O1	0.6
924312	AB2-078 E O1	0.6
924322	AB2-079 E O1	0.6
924401	AB2-089 C	1.07
924402	AB2-089 E	0.55
924491	AB2-098 C	0.24
924492	AB2-098 E	0.1
924501	AB2-099 C	0.25
924502	AB2-099 E	0.11
924511	AB2-100 C	10.62
924512	AB2-100 E	5.23
925122	AB2-169 E	2.2
925171	AB2-174 C O1	5.2
925172	AB2-174 E O1	4.7
925591	AC1-034 C	3.46
925592	AC1-034 E	2.61
925612	AC1-036 E	0.6
925781	AC1-054 C O1	3.69
925782	AC1-054 E O1	1.7
926071	AC1-086 C	14.05
926072	AC1-086 E	6.39
926201	AC1-098 C	3.11
926202	AC1-098 E	1.86
926211	AC1-099 C	1.04
926212	AC1-099 E	0.61
926271	AC1-105 C O1	2.36
926272	AC1-105 E O1	1.18
927021	AC1-189 C	3.85
927022	AC1-189 E	1.92
927141	AC1-208 C	4.79
927142	AC1-208 E	2.13
927251	AC1-221 C	0.93
927252	AC1-221 E	0.93
927261	AC1-222 C	1.47
927262	AC1-222 E	1.4
930402	AB1-081 E O1	1.94
930861	AB1-132 C O1	1.35
930862	AB1-132 E O1	4.09
931231	AB1-173 C	0.23
931232	AB1-173 E	0.75
931241	AB1-173AC	0.23
931242	AB1-173AE	0.75
932631	AC2-084 C	4.44
932632	AC2-084 E	2.19

Bus #	Bus	MW Impact
932761	AC2-100 C	2.13
932762	AC2-100 E	1.04
933991	AD1-023 C	4.64
933992	AD1-023 E	2.53
934201	AD1-047 C	5.72
934202	AD1-047 E	3.81
934231	AD1-050 C	2.36
934232	AD1-050 E	1.29
934311	AD1-055 C	1.02
934312	AD1-055 E	0.26
934331	AD1-057 C O1	5.86
934332	AD1-057 E O1	3.12
934341	AD1-058 C	2.33
934342	AD1-058 E	0.59
934521	AD1-076 C	18.15
934522	AD1-076 E	9.24
934611	AD1-087 C O1	3.67
934612	AD1-087 E O1	1.72
934621	AD1-088 C	4.84
934622	AD1-088 E	2.27
934991	AD1-131 C	0.76
934992	AD1-131 E	0.51
935171	AD1-152 C O1	3.65
935172	AD1-152 E O1	2.43
935212	AD1-156 E	0.63
936261	AD2-033 C	5.16
936262	AD2-033 E	3.44
936331	AD2-043 C	1.91
936332	AD2-043 E	2.26
936361	AD2-046 C O1	4.24
936362	AD2-046 E O1	1.95
936401	AD2-051 C O1	3.84
936402	AD2-051 E O1	1.65
936481	AD2-063 C O1	6.36
936482	AD2-063 E O1	4.24
936531	AD2-068 C	2.27
936532	AD2-068 E	1.17
936701	AD2-089 C	3.47
936702	AD2-089 E	2.31
936711	AD2-090 C O1	2.58
936712	AD2-090 E O1	1.72
937481	AD2-202 C O1	1.01
937482	AD2-202 E O1	0.51
937571	AD2-169 C	6.85
937572	AD2-169 E	4.56
938171	AE1-026 C1 O	10.2
938172	AE1-026 C2 O	1.57
938173	AE1-026 E O1	2.99
938221	AE1-035 C	0.92
938222	AE1-035 E	0.45
938491	AE1-068 C O1	33.31
938492	AE1-068 E O1	18.4

Bus #	Bus	MW Impact
938501	AE1-069 C O1	26.32
938502	AE1-069 E O1	15.05
938661	AE1-088	1.26
939181	AE1-148 C O1	4.17
939182	AE1-148 E O1	2.78
939371	AE1-168 C	5.23
939372	AE1-168 E	3.48
939411	AE1-173 C	37.39
939412	AE1-173 E	24.92
AA2-074	AA2-074	2.63
CARR	CARR	0.31
CBM-S1	CBM-S1	6.65
CBM-S2	CBM-S2	7.55
CBM-W1	CBM-W1	6.62
CBM-W2	CBM-W2	43.29
CIN	CIN	3.0
CPLE	CPLE	3.87
G-007	G-007	1.05
IPL	IPL	1.88
LGEE	LGEE	0.87
MEC	MEC	6.69
MECS	MECS	2.77
O-066	O-066	6.69
RENSSELAER	RENSSELAER	0.24
WEC	WEC	0.8

12.6.3 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308390	314285	6CHRL249	DVP	314316	6LOCKS	DVP	1	DVP_P4-2: 562T563	breaker	684.0	109.9	111.75	AC	14.76

Bus #	Bus	MW Impact
314541	3WATKINS	0.23
314554	3BTLEBRO	0.4
314557	3BETHEL	0.39
314572	3EMPORIA	0.33
314574	6EVERETS	1.77
314578	3HORNRTN	1.92
314582	3KELFORD	1.62
314603	3SCOT NK	1.61
314617	3TUNIS	0.36
314620	6CASHIE	0.31
314623	3WITAKRS	0.69
314704	3LAWRENC	0.2
315102	1BRUNSWICKG1	3.87
315103	1BRUNSWICKG2	3.87
315104	1BRUNSWICKG3	3.87
315105	1BRUNSWICKS1	8.03
315131	1EDGECEMA	4.73
315132	1EDGECEMB	4.73
315136	1ROSEMG1	1.22
315137	1ROSEMS1	0.76
315138	1ROSEMG2	0.57
315139	1GASTONA	1.77
315141	1GASTONB	1.77
315150	1BUGGS 1	5.14
315151	1BUGGS 2	5.14
900672	V4-068 E	0.13
916301	Z1-086 C	23.58
916302	Z1-086 E	5.22
917332	Z2-043 E	0.46
917342	Z2-044 E	0.3
917512	Z2-088 E OP1	1.76
918492	AA1-063AE OP	2.11
918512	AA1-065 E OP	1.81
918532	AA1-067 E	0.31
918562	AA1-072 E	0.08
919692	AA2-053 E	2.07
919702	AA2-057 E	1.83
920042	AA2-088 E OP	4.27
920592	AA2-165 E	0.24
920672	AA2-174 E	0.24
923851	AB2-025 C	0.41

Bus #	Bus	MW Impact
923852	AB2-025 E	1.3
923911	AB2-031 C O1	1.6
923912	AB2-031 E O1	0.79
923991	AB2-040 C O1	0.74
923992	AB2-040 E O1	4.29
924022	AB2-043 E O1	2.32
924152	AB2-059 E O1	2.75
924162	AB2-060 E O1	1.9
924302	AB2-077 E O1	0.6
924312	AB2-078 E O1	0.6
924322	AB2-079 E O1	0.6
924401	AB2-089 C	1.07
924402	AB2-089 E	0.55
924491	AB2-098 C	0.24
924492	AB2-098 E	0.1
924501	AB2-099 C	0.25
924502	AB2-099 E	0.11
924511	AB2-100 C	10.62
924512	AB2-100 E	5.23
925122	AB2-169 E	2.2
925171	AB2-174 C O1	5.2
925172	AB2-174 E O1	4.7
925591	AC1-034 C	3.46
925592	AC1-034 E	2.61
925612	AC1-036 E	0.6
925781	AC1-054 C O1	3.69
925782	AC1-054 E O1	1.7
926071	AC1-086 C	14.05
926072	AC1-086 E	6.39
926201	AC1-098 C	3.11
926202	AC1-098 E	1.86
926211	AC1-099 C	1.04
926212	AC1-099 E	0.61
926271	AC1-105 C O1	2.36
926272	AC1-105 E O1	1.18
927021	AC1-189 C	3.85
927022	AC1-189 E	1.92
927141	AC1-208 C	4.79
927142	AC1-208 E	2.13
927251	AC1-221 C	0.93
927252	AC1-221 E	0.93
927261	AC1-222 C	1.47
927262	AC1-222 E	1.4
930402	AB1-081 E O1	1.94
930861	AB1-132 C O1	1.35
930862	AB1-132 E O1	4.09
931231	AB1-173 C	0.23
931232	AB1-173 E	0.75
931241	AB1-173AC	0.23
931242	AB1-173AE	0.75
932631	AC2-084 C	4.44
932632	AC2-084 E	2.19

Bus #	Bus	MW Impact
932761	AC2-100 C	2.13
932762	AC2-100 E	1.04
933991	AD1-023 C	4.64
933992	AD1-023 E	2.53
934201	AD1-047 C	5.72
934202	AD1-047 E	3.81
934231	AD1-050 C	2.36
934232	AD1-050 E	1.29
934311	AD1-055 C	1.02
934312	AD1-055 E	0.26
934331	AD1-057 C O1	5.86
934332	AD1-057 E O1	3.12
934341	AD1-058 C	2.33
934342	AD1-058 E	0.59
934521	AD1-076 C	18.15
934522	AD1-076 E	9.24
934611	AD1-087 C O1	3.67
934612	AD1-087 E O1	1.72
934621	AD1-088 C	4.84
934622	AD1-088 E	2.27
934991	AD1-131 C	0.76
934992	AD1-131 E	0.51
935171	AD1-152 C O1	3.65
935172	AD1-152 E O1	2.43
935212	AD1-156 E	0.63
936261	AD2-033 C	5.16
936262	AD2-033 E	3.44
936331	AD2-043 C	1.91
936332	AD2-043 E	2.26
936361	AD2-046 C O1	4.24
936362	AD2-046 E O1	1.95
936401	AD2-051 C O1	3.84
936402	AD2-051 E O1	1.65
936481	AD2-063 C O1	6.36
936482	AD2-063 E O1	4.24
936531	AD2-068 C	2.27
936532	AD2-068 E	1.17
936701	AD2-089 C	3.47
936702	AD2-089 E	2.31
936711	AD2-090 C O1	2.58
936712	AD2-090 E O1	1.72
937481	AD2-202 C O1	1.01
937482	AD2-202 E O1	0.51
937571	AD2-169 C	6.85
937572	AD2-169 E	4.56
938171	AE1-026 C1 O	10.2
938172	AE1-026 C2 O	1.57
938173	AE1-026 E O1	2.99
938221	AE1-035 C	0.92
938222	AE1-035 E	0.45
938491	AE1-068 C O1	33.31
938492	AE1-068 E O1	18.4

Bus #	Bus	MW Impact
938501	AE1-069 C O1	26.32
938502	AE1-069 E O1	15.05
938661	AE1-088	1.26
939181	AE1-148 C O1	4.17
939182	AE1-148 E O1	2.78
939371	AE1-168 C	5.23
939372	AE1-168 E	3.48
939411	AE1-173 C	37.39
939412	AE1-173 E	24.92
AA2-074	AA2-074	2.63
CARR	CARR	0.31
CBM-S1	CBM-S1	6.65
CBM-S2	CBM-S2	7.55
CBM-W1	CBM-W1	6.62
CBM-W2	CBM-W2	43.29
CIN	CIN	3.0
CPLE	CPLE	3.87
G-007	G-007	1.05
IPL	IPL	1.88
LGEE	LGEE	0.87
MEC	MEC	6.69
MECS	MECS	2.77
O-066	O-066	6.69
RENSSELAER	RENSSELAER	0.24
WEC	WEC	0.8

12.6.4 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308856	314569	6EARLEYS	DVP	314575	6NUCO TP	DVP	1	DVP_P1-2: LN 2092	single	571.52	99.99	111.6	AC	67.55

Bus #	Bus	MW Impact
314589	3MURPHYS	0.08
314704	3LAWRENC	0.16
315126	1ROARAP2	1.11
315128	1ROARAP4	1.07
315131	1EDGECEMA	9.42
315132	1EDGECEMB	9.42
315136	1ROSEMG1	1.95
315137	1ROSEMS1	1.21
315138	1ROSEMG2	0.91
315139	1GASTONA	2.74
315141	1GASTONB	2.74
315159	1KERR 2	0.61
315163	1KERR 6	0.6
315164	1KERR 7	0.6
315292	1DOMTR78	2.33
315293	1DOMTR9	1.9
315294	1DOMTR10	3.45
916041	Z1-036 C	1.9
917331	Z2-043 C	0.53
917341	Z2-044 C	0.2
917511	Z2-088 C OP1	1.93
918491	AA1-063AC OP	1.56
918511	AA1-065 C OP	2.76
918531	AA1-067 C	0.42
918561	AA1-072 C	0.08
919691	AA2-053 C	1.41
919701	AA2-057 C	1.08
920041	AA2-088 C OP	0.57
920591	AA2-165 C	0.15
920671	AA2-174 C	0.06
920691	AA2-178 C	2.73
923911	AB2-031 C O1	1.74
923991	AB2-040 C O1	0.81
924151	AB2-059 C O1	1.49
924401	AB2-089 C	1.24
924491	AB2-098 C	0.99
924501	AB2-099 C	0.95
924511	AB2-100 C	7.11
925121	AB2-169 C	1.74
925171	AB2-174 C O1	5.2
925591	AC1-034 C	6.82

Bus #	Bus	MW Impact
925781	AC1-054 C O1	4.49
926071	AC1-086 C	21.84
926201	AC1-098 C	7.82
926211	AC1-099 C	2.62
927021	AC1-189 C	13.69
927141	AC1-208 C	10.08
930401	AB1-081 C O1	1.26
930861	AB1-132 C O1	2.1
931231	AB1-173 C	0.25
931241	AB1-173AC	0.25
932631	AC2-084 C	11.14
933991	AD1-023 C	27.2
934201	AD1-047 C	6.24
934231	AD1-050 C	2.73
934331	AD1-057 C O1	10.32
934521	AD1-076 C	110.7
936401	AD2-051 C O1	15.91
936531	AD2-068 C	6.55
936701	AD2-089 C	9.29
937571	AD2-169 C	8.2
938171	AE1-026 C1 O	58.55
938172	AE1-026 C2 O	9.0
938221	AE1-035 C	4.88
938661	AE1-088	6.3
AA2-074	AA2-074	2.73
CARR	CARR	0.19
CBM-S1	CBM-S1	5.96
CBM-S2	CBM-S2	7.05
CBM-W1	CBM-W1	6.18
CBM-W2	CBM-W2	39.17
CIN	CIN	2.77
CPLE	CPLE	4.01
IPL	IPL	1.74
LGEE	LGEE	0.8
MEC	MEC	6.12
MECS	MECS	2.7
RENSSELAER	RENSSELAER	0.15
WEC	WEC	0.74

12.6.5 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309544	314574	6EVERETS	DVP	936530	AD2-068 TAP	DVP	1	DVP_P7-1: LN 2058-2181	tower	485.0	148.13	157.18	AC	46.54

Bus #	Bus	MW Impact
314539	3UNCAMP	1.27
314541	3WATKINS	0.39
314554	3BTLEBRO	0.38
314557	3BETHEL	1.06
314566	3CRESWEL	2.23
314572	3EMPORIA	0.22
314574	6EVERETS	8.22
314578	3HORNRTN	2.14
314582	3KELFORD	3.33
314589	3MURPHYS	0.06
314594	6PLYMOTH	0.9
314603	3SCOT NK	2.69
314617	3TUNIS	0.78
314620	6CASHIE	0.97
314623	3WITAKRS	0.71
314648	6SUNBURY	0.41
314651	6WINFALL	0.99
315131	1EDGECEMA	8.55
315132	1EDGECEMB	8.55
315136	1ROSEMG1	1.46
315137	1ROSEMS1	0.9
315138	1ROSEMG2	0.68
315292	1DOMTR78	1.52
315293	1DOMTR9	1.24
315294	1DOMTR10	2.26
900672	V4-068 E	0.24
901082	W1-029 E	24.7
907092	X1-038 E	3.17
913392	Y1-086 E	1.0
916042	Z1-036 E	30.84
917122	Z2-027 E	0.52
917331	Z2-043 C	0.31
917332	Z2-043 E	0.95
917342	Z2-044 E	0.31
917511	Z2-088 C OP1	2.11
917512	Z2-088 E OP1	6.41
918491	AA1-063AC OP	0.98
918492	AA1-063AE OP	3.28
918511	AA1-065 C OP	1.58
918512	AA1-065 E OP	5.52
918531	AA1-067 C	0.48

Bus #	Bus	MW Impact
918532	AA1-067 E	1.45
918561	AA1-072 C	0.05
918562	AA1-072 E	0.16
919692	AA2-053 E	2.78
919702	AA2-057 E	2.06
920042	AA2-088 E OP	6.8
920592	AA2-165 E	0.27
920672	AA2-174 E	0.32
920691	AA2-178 C	1.26
920692	AA2-178 E	3.82
923801	AB2-015 C O1	4.71
923802	AB2-015 E O1	3.86
923832	AB2-022 E	0.57
923911	AB2-031 C O1	1.26
923912	AB2-031 E O1	0.62
923992	AB2-040 E O1	3.38
924152	AB2-059 E O1	3.09
924491	AB2-098 C	1.13
924492	AB2-098 E	0.48
924501	AB2-099 C	0.59
924502	AB2-099 E	0.25
924511	AB2-100 C	6.09
924512	AB2-100 E	3.0
925121	AB2-169 C	1.38
925122	AB2-169 E	8.74
925171	AB2-174 C O1	3.82
925172	AB2-174 E O1	3.46
925591	AC1-034 C	3.89
925592	AC1-034 E	2.93
926071	AC1-086 C	15.84
926072	AC1-086 E	7.21
926201	AC1-098 C	4.51
926202	AC1-098 E	2.69
926211	AC1-099 C	1.51
926212	AC1-099 E	0.89
927021	AC1-189 C	15.22
927022	AC1-189 E	7.58
927141	AC1-208 C	5.86
927142	AC1-208 E	2.6
930402	AB1-081 E O1	2.18
930862	AB1-132 E O1	4.61
931232	AB1-173 E	0.59
931242	AB1-173AE	0.59
932631	AC2-084 C	6.44
932632	AC2-084 E	3.17
933991	AD1-023 C	14.77
933992	AD1-023 E	8.04
934201	AD1-047 C	4.51
934202	AD1-047 E	3.01
934331	AD1-057 C O1	8.5
934332	AD1-057 E O1	4.53
934521	AD1-076 C	58.8

Bus #	Bus	MW Impact
934522	AD1-076 E	29.94
936401	AD2-051 C O1	9.44
936402	AD2-051 E O1	4.05
936701	AD2-089 C	9.53
936702	AD2-089 E	6.35
936711	AD2-090 C O1	4.25
936712	AD2-090 E O1	2.83
937571	AD2-169 C	5.83
937572	AD2-169 E	3.88
938171	AE1-026 C1 O	32.17
938172	AE1-026 C2 O	4.94
938173	AE1-026 E O1	9.42
938221	AE1-035 C	2.8
938222	AE1-035 E	1.38
938531	AE1-072 C O1	6.69
938532	AE1-072 E O1	3.49
938661	AE1-088	5.0
938771	AE1-103 C O1	1.85
938772	AE1-103 E O1	2.55
AC1-131	AC1-131	6.18
BLUEG	BLUEG	7.81
CALDERWOOD	CALDERWOOD	1.67
CANNELTON	CANNELTON	0.54
CATAWBA	CATAWBA	1.63
CBM-N	CBM-N	0.21
CHEOAH	CHEOAH	1.56
CHILHOWEE	CHILHOWEE	0.54
COFFEEN	COFFEEN	0.92
COTTONWOOD	COTTONWOOD	5.62
DUCKCREEK	DUCKCREEK	1.89
EDWARDS	EDWARDS	0.84
FARMERCITY	FARMERCITY	0.67
G-007A	G-007A	1.0
GIBSON	GIBSON	0.34
HAMLET	HAMLET	3.48
NEWTON	NEWTON	2.4
NYISO	NYISO	0.91
PRAIRIE	PRAIRIE	5.08
SANTEETLA	SANTEETLA	0.46
SMITHLAND	SMITHLAND	0.45
TATANKA	TATANKA	1.13
TILTON	TILTON	0.99
TRIMBLE	TRIMBLE	0.86
TVA	TVA	4.7
UNIONPOWER	UNIONPOWER	2.33
VFT	VFT	2.66

12.6.6 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308906	314575	6NUCO TP	DVP	919140	AA1-138 TAP	DVP	1	DVP_P1-2: LN 2092	single	571.52	93.67	105.24	AC	67.55

Bus #	Bus	MW Impact
314589	3MURPHYS	0.08
314704	3LAWRENC	0.16
315126	1ROARAP2	1.11
315128	1ROARAP4	1.07
315131	1EDGECEMA	9.42
315132	1EDGECEMB	9.42
315136	1ROSEMG1	1.95
315137	1ROSEMS1	1.21
315138	1ROSEMG2	0.91
315139	1GASTONA	2.74
315141	1GASTONB	2.74
315159	1KERR 2	0.61
315163	1KERR 6	0.6
315164	1KERR 7	0.6
315292	1DOMTR78	2.33
315293	1DOMTR9	1.9
315294	1DOMTR10	3.45
916041	Z1-036 C	1.9
917331	Z2-043 C	0.53
917341	Z2-044 C	0.2
917511	Z2-088 C OP1	1.93
918491	AA1-063AC OP	1.56
918511	AA1-065 C OP	2.76
918531	AA1-067 C	0.42
918561	AA1-072 C	0.08
919691	AA2-053 C	1.41
919701	AA2-057 C	1.08
920041	AA2-088 C OP	0.57
920591	AA2-165 C	0.15
920671	AA2-174 C	0.06
920691	AA2-178 C	2.73
923911	AB2-031 C O1	1.74
923991	AB2-040 C O1	0.81
924151	AB2-059 C O1	1.49
924401	AB2-089 C	1.24
924491	AB2-098 C	0.99
924501	AB2-099 C	0.95
924511	AB2-100 C	7.11
925121	AB2-169 C	1.74
925171	AB2-174 C O1	5.2
925591	AC1-034 C	6.82

Bus #	Bus	MW Impact
925781	AC1-054 C O1	4.49
926071	AC1-086 C	21.84
926201	AC1-098 C	7.82
926211	AC1-099 C	2.62
927021	AC1-189 C	13.69
927141	AC1-208 C	10.08
930401	AB1-081 C O1	1.26
930861	AB1-132 C O1	2.1
931231	AB1-173 C	0.25
931241	AB1-173AC	0.25
932631	AC2-084 C	11.14
933991	AD1-023 C	27.2
934201	AD1-047 C	6.24
934231	AD1-050 C	2.73
934331	AD1-057 C O1	10.32
934521	AD1-076 C	110.7
936401	AD2-051 C O1	15.91
936531	AD2-068 C	6.55
936701	AD2-089 C	9.29
937571	AD2-169 C	8.2
938171	AE1-026 C1 O	58.55
938172	AE1-026 C2 O	9.0
938221	AE1-035 C	4.88
938661	AE1-088	6.3
AA2-074	AA2-074	2.73
CARR	CARR	0.19
CBM-S1	CBM-S1	5.96
CBM-S2	CBM-S2	7.05
CBM-W1	CBM-W1	6.18
CBM-W2	CBM-W2	39.17
CIN	CIN	2.77
CPLE	CPLE	4.01
IPL	IPL	1.74
LGEE	LGEE	0.8
MEC	MEC	6.12
MECS	MECS	2.7
RENSSELAER	RENSSELAER	0.15
WEC	WEC	0.74

12.6.7 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308400	314583	6LAKEVEW	DVP	924510	AB2-100 TAP	DVP	1	DVP_P4-2: 246T247	breaker	427.0	127.66	135.54	AC	39.55

Bus #	Bus	MW Impact
314541	3WATKINS	0.3
314554	3BTLEBRO	0.85
314557	3BETHEL	0.89
314566	3CRESWEL	1.67
314574	6EVERETS	4.37
314578	3HORNRTN	3.34
314582	3KELFORD	3.8
314594	6PLYMOTH	0.69
314603	3SCOT NK	3.52
314617	3TUNIS	0.79
314620	6CASHIE	0.82
314623	3WITAKRS	1.43
315131	1EDGECEMA	10.63
315132	1EDGECEMB	10.63
315136	1ROSEMG1	3.83
315137	1ROSEMS1	2.38
315138	1ROSEMG2	1.8
315139	1GASTONA	5.7
315141	1GASTONB	5.7
900672	V4-068 E	0.24
916042	Z1-036 E	21.05
917331	Z2-043 C	0.36
917332	Z2-043 E	1.08
917341	Z2-044 C	0.2
917342	Z2-044 E	0.61
917511	Z2-088 C OP1	1.38
917512	Z2-088 E OP1	4.2
918491	AA1-063AC OP	1.25
918492	AA1-063AE OP	4.16
918511	AA1-065 C OP	1.49
918512	AA1-065 E OP	5.2
918531	AA1-067 C	0.25
918532	AA1-067 E	0.77
918561	AA1-072 C	0.05
918562	AA1-072 E	0.18
919691	AA2-053 C	1.26
919692	AA2-053 E	3.83
919701	AA2-057 C	1.06
919702	AA2-057 E	3.74
920042	AA2-088 E OP	6.85
920591	AA2-165 C	0.14

Bus #	Bus	MW Impact
920592	AA2-165 E	0.49
920671	AA2-174 C	0.06
920672	AA2-174 E	0.44
920692	AA2-178 E	2.87
924151	AB2-059 C O1	1.61
924152	AB2-059 E O1	5.86
924401	AB2-089 C	0.97
924402	AB2-089 E	0.5
924491	AB2-098 C	0.6
924492	AB2-098 E	0.26
924501	AB2-099 C	0.6
924502	AB2-099 E	0.26
925122	AB2-169 E	5.46
925591	AC1-034 C	7.36
925592	AC1-034 E	5.55
925781	AC1-054 C O1	3.7
925782	AC1-054 E O1	1.71
926071	AC1-086 C	45.38
926072	AC1-086 E	20.65
926201	AC1-098 C	6.52
926202	AC1-098 E	3.88
926211	AC1-099 C	2.18
926212	AC1-099 E	1.28
927021	AC1-189 C	9.28
927022	AC1-189 E	4.62
927141	AC1-208 C	9.44
927142	AC1-208 E	4.19
930401	AB1-081 C O1	1.36
930402	AB1-081 E O1	4.13
930861	AB1-132 C O1	4.35
930862	AB1-132 E O1	13.21
932631	AC2-084 C	9.29
932632	AC2-084 E	4.58
933991	AD1-023 C	12.17
933992	AD1-023 E	6.62
934231	AD1-050 C	2.13
934232	AD1-050 E	1.16
934331	AD1-057 C O1	16.14
934332	AD1-057 E O1	8.61
934521	AD1-076 C	45.63
934522	AD1-076 E	23.24
936401	AD2-051 C O1	9.63
936402	AD2-051 E O1	4.14
936531	AD2-068 C	4.47
936532	AD2-068 E	2.3
936701	AD2-089 C	8.05
936702	AD2-089 E	5.36
936711	AD2-090 C O1	3.68
936712	AD2-090 E O1	2.46
938171	AE1-026 C1 O	27.34
938172	AE1-026 C2 O	4.2
938173	AE1-026 E O1	8.01

Bus #	Bus	MW Impact
938221	AE1-035 C	2.64
938222	AE1-035 E	1.3
938661	AE1-088	3.12
AA2-074	AA2-074	2.26
CARR	CARR	0.19
CBM-S1	CBM-S1	4.87
CBM-S2	CBM-S2	5.86
CBM-W1	CBM-W1	4.89
CBM-W2	CBM-W2	31.81
CIN	CIN	2.2
CPLE	CPLE	3.32
G-007	G-007	0.65
IPL	IPL	1.38
LGEE	LGEE	0.64
MEC	MEC	4.91
MECS	MECS	2.08
O-066	O-066	4.11
RENSSELAER	RENSSELAER	0.15
WEC	WEC	0.59

12.6.8 Index 7

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309133	314591	6NASH	DVP	304226	6PA-RMOUNT#4	CPL	1	DVP_P1-2: LN 2058	single	470.0	98.55	102.57	AC	19.21

Bus #	Bus	MW Impact
314589	3MURPHYS	0.05
314704	3LAWRENC	0.15
315115	1S HAMPT1	0.64
315126	1ROARAP2	0.95
315128	1ROARAP4	0.92
315131	1EDGECEMA	25.25
315132	1EDGECEMB	25.25
315136	1ROSEMG1	2.49
315137	1ROSEMS1	1.54
315138	1ROSEMG2	1.17
315139	1GASTONA	2.98
315141	1GASTONB	2.98
315292	1DOMTR78	0.69
315293	1DOMTR9	0.56
315294	1DOMTR10	1.02
901081	W1-029 C	0.29
916041	Z1-036 C	0.35
917331	Z2-043 C	0.27
917341	Z2-044 C	0.24
917511	Z2-088 C OP1	1.73
918491	AA1-063AC OP	0.93
918511	AA1-065 C OP	0.86
918531	AA1-067 C	0.18
918561	AA1-072 C	0.04
919691	AA2-053 C	0.98
919701	AA2-057 C	1.17
920041	AA2-088 C OP	0.54
920591	AA2-165 C	0.16
920671	AA2-174 C	0.04
920691	AA2-178 C	0.64
923801	AB2-015 C O1	4.03
923851	AB2-025 C	0.14
923911	AB2-031 C O1	1.64
923991	AB2-040 C O1	0.76
924151	AB2-059 C O1	2.43
924401	AB2-089 C	0.74
924491	AB2-098 C	0.41
924501	AB2-099 C	0.41
924511	AB2-100 C	8.56
925121	AB2-169 C	0.57
925171	AB2-174 C O1	5.03

Bus #	Bus	MW Impact
925591	AC1-034 C	11.15
925781	AC1-054 C O1	2.89
926071	AC1-086 C	23.71
926201	AC1-098 C	6.04
926211	AC1-099 C	2.02
927021	AC1-189 C	9.92
927141	AC1-208 C	9.24
930401	AB1-081 C O1	2.06
930861	AB1-132 C O1	2.27
931231	AB1-173 C	0.23
931241	AB1-173AC	0.23
932631	AC2-084 C	8.61
933991	AD1-023 C	7.55
934201	AD1-047 C	5.89
934231	AD1-050 C	1.64
934331	AD1-057 C O1	20.2
934521	AD1-076 C	29.35
936401	AD2-051 C O1	6.26
936701	AD2-089 C	14.33
936711	AD2-090 C O1	3.77
937571	AD2-169 C	7.54
938171	AE1-026 C1 O	16.65
938172	AE1-026 C2 O	2.56
938221	AE1-035 C	1.51
938661	AE1-088	2.09
938771	AE1-103 C O1	1.52
BLUEG	BLUEG	6.03
CALDERWOOD	CALDERWOOD	1.22
CANNELTON	CANNELTON	0.41
CATAWBA	CATAWBA	1.19
CBM-N	CBM-N	0.13
CHEOAH	CHEOAH	1.14
CHILHOWEE	CHILHOWEE	0.4
COFFEEN	COFFEEN	0.7
COTTONWOOD	COTTONWOOD	4.08
DUCKCREEK	DUCKCREEK	1.44
EDWARDS	EDWARDS	0.64
FARMERCITY	FARMERCITY	0.5
G-007A	G-007A	0.66
GIBSON	GIBSON	0.26
HAMLET	HAMLET	2.34
NEWTON	NEWTON	1.82
NYISO	NYISO	0.55
PRAIRIE	PRAIRIE	3.8
SANTEETLA	SANTEETLA	0.34
SMITHLAND	SMITHLAND	0.34
TATANKA	TATANKA	0.86
TILTON	TILTON	0.76
TRIMBLE	TRIMBLE	0.66
TVA	TVA	3.43
UNIONPOWER	UNIONPOWER	1.7
VFT	VFT	1.75

12.6.9 Index 8

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308395	314903	8CHCKAHM	DVP	314908	8ELMONT	DVP	1	DVP_P4-2: 563T576	breaker	3144.0	114.47	116.19	AC	62.85

Bus #	Bus	MW Impact
314189	6PAPERMILL	10.93
314421	6WINCHST	0.23
314539	3UNCAMP	3.89
314541	3WATKINS	1.09
314554	3BTLEBRO	1.02
314557	3BETHEL	1.07
314566	3CRESWEL	3.91
314572	3EMPORIA	0.55
314574	6EVERETS	5.91
314578	3HORNRTN	4.48
314582	3KELFORD	5.02
314594	6PLYMOTH	1.35
314603	3SCOT NK	4.59
314617	3TUNIS	1.25
314620	6CASHIE	1.31
314623	3WITAKRS	1.75
314648	6SUNBURY	1.55
314651	6WINFALL	3.05
315073	1STONECA	8.7
315074	1HOPCGN1	10.49
315075	1HOPCGN2	10.36
315090	1YORKTN1	53.43
315091	1YORKTN2	55.45
315092	1YORKTN3	37.57
315110	1ELIZAR3	5.11
315131	1EDGECEMA	11.89
315132	1EDGECEMB	11.89
315233	1SURRY 2	45.3
900672	V4-068 E	0.45
901082	W1-029 E	79.92
907092	X1-038 E	9.74
913392	Y1-086 E	3.88
916042	Z1-036 E	77.51
916192	Z1-068 E	3.42
916302	Z1-086 E	13.62
917122	Z2-027 E	1.86
917332	Z2-043 E	1.43
917342	Z2-044 E	0.75
917512	Z2-088 E OP1	5.36
918492	AA1-063AE OP	5.6
918512	AA1-065 E OP	6.65

Bus #	Bus	MW Impact
918532	AA1-067 E	1.04
918562	AA1-072 E	0.24
919152	AA1-139 E	11.55
919692	AA2-053 E	5.17
919702	AA2-057 E	4.69
920042	AA2-088 E OP	15.95
920592	AA2-165 E	0.62
920672	AA2-174 E	0.6
920692	AA2-178 E	6.7
923801	AB2-015 C O1	13.71
923802	AB2-015 E O1	11.24
923832	AB2-022 E	2.19
923842	AB2-024 E	1.84
923852	AB2-025 E	1.44
923911	AB2-031 C O1	2.98
923912	AB2-031 E O1	1.47
923992	AB2-040 E O1	8.01
924152	AB2-059 E O1	6.96
924241	AB2-068 O1	619.19
924491	AB2-098 C	0.81
924492	AB2-098 E	0.35
924501	AB2-099 C	0.87
924502	AB2-099 E	0.37
924511	AB2-100 C	15.32
924512	AB2-100 E	7.54
924812	AB2-134 E O1	18.1
925051	AB2-160 C O1	6.26
925052	AB2-160 E O1	10.21
925061	AB2-161 C O1	5.12
925062	AB2-161 E O1	8.35
925122	AB2-169 E	8.79
925171	AB2-174 C O1	9.32
925172	AB2-174 E O1	8.43
925331	AB2-190 C	29.15
925332	AB2-190 E	12.49
925522	AC1-027 E	2.08
925591	AC1-034 C	8.75
925592	AC1-034 E	6.6
925781	AC1-054 C O1	8.64
925782	AC1-054 E O1	3.98
925861	AC1-065 C	5.36
925862	AC1-065 E	8.75
926071	AC1-086 C	28.11
926072	AC1-086 E	12.79
926201	AC1-098 C	8.43
926202	AC1-098 E	5.02
926211	AC1-099 C	2.83
926212	AC1-099 E	1.66
926291	AC1-107 O1	934.62
926662	AC1-147 E	2.41
926751	AC1-161 C O1	59.45
926752	AC1-161 E O1	25.38

Bus #	Bus	MW Impact
926781	AC1-164 C	67.9
926782	AC1-164 E	30.51
927021	AC1-189 C	12.08
927022	AC1-189 E	6.02
927141	AC1-208 C	12.17
927142	AC1-208 E	5.4
927221	AC1-216 C O1	14.54
927222	AC1-216 E O1	11.44
930402	AB1-081 E O1	4.91
930862	AB1-132 E O1	8.18
931232	AB1-173 E	1.4
931242	AB1-173AE	1.4
932041	AC2-012 C	18.53
932042	AC2-012 E	30.24
932532	AC2-073 E	1.95
932581	AC2-078 C O1	5.51
932582	AC2-078 E O1	8.99
932591	AC2-079 C O1	9.25
932592	AC2-079 E O1	15.09
932631	AC2-084 C	12.02
932632	AC2-084 E	5.92
932831	AC2-110 C	2.14
932832	AC2-110 E	3.5
933262	AC2-137 E	1.87
933272	AC2-138 E	1.18
933291	AC2-141 C	59.45
933292	AC2-141 E	25.38
933732	AC2-196 E	2.17
933991	AD1-023 C	20.55
933992	AD1-023 E	11.19
934011	AD1-025 C	25.0
934012	AD1-025 E	14.81
934061	AD1-033 C	13.68
934062	AD1-033 E	9.12
934141	AD1-041 C	8.48
934142	AD1-041 E	5.65
934201	AD1-047 C	10.68
934202	AD1-047 E	7.12
934212	AD1-048 E	1.38
934331	AD1-057 C O1	13.16
934332	AD1-057 E O1	7.02
934392	AD1-063 E	1.75
934521	AD1-076 C	86.27
934522	AD1-076 E	43.93
934571	AD1-082 C	11.66
934572	AD1-082 E	6.65
934611	AD1-087 C O1	9.62
934612	AD1-087 E O1	4.52
935112	AD1-144 E	1.67
935161	AD1-151 C O1	23.43
935162	AD1-151 E O1	15.62
935171	AD1-152 C O1	9.56

Bus #	Bus	MW Impact
935172	AD1-152 E O1	6.38
935212	AD1-156 E	1.72
936041	AD2-007	2.65
936051	AD2-008 C	4.35
936052	AD2-008 E	9.47
936151	AD2-021	0.33
936242	AD2-030 E	1.9
936301	AD2-039 C	2.14
936302	AD2-039 E	3.5
936341	AD2-044 C	0.29
936342	AD2-044 E	0.33
936391	AD2-049 C	3.25
936392	AD2-049 E	3.25
936401	AD2-051 C O1	12.91
936402	AD2-051 E O1	5.54
936531	AD2-068 C	6.82
936532	AD2-068 E	3.51
936591	AD2-074 C	7.44
936592	AD2-074 E	12.15
936661	AD2-085 C	5.53
936662	AD2-085 E	9.02
936701	AD2-089 C	9.67
936702	AD2-089 E	6.45
936711	AD2-090 C O1	11.19
936712	AD2-090 E O1	7.46
937221	AD2-160 C O1	10.57
937222	AD2-160 E O1	5.54
937251	AD2-164	8.86
937481	AD2-202 C O1	2.65
937482	AD2-202 E O1	1.33
937541	AD2-215 C	3.08
937542	AD2-215 E	1.64
937571	AD2-169 C	13.38
937572	AD2-169 E	8.92
938031	AE1-004 C	2.14
938032	AE1-004 E	3.5
938171	AE1-026 C1 O	43.45
938172	AE1-026 C2 O	6.68
938173	AE1-026 E O1	12.73
938181	AE1-027 C	4.25
938182	AE1-027 E	2.24
938191	AE1-028 C	2.47
938192	AE1-028 E	1.43
938221	AE1-035 C	3.37
938222	AE1-035 E	1.66
938491	AE1-068 C O1	86.89
938492	AE1-068 E O1	47.99
938501	AE1-069 C O1	68.65
938502	AE1-069 E O1	39.25
938531	AE1-072 C O1	31.5
938532	AE1-072 E O1	16.42
938631	AE1-085 C O1	12.25

Bus #	Bus	MW Impact
938632	AE1-085 E O1	6.13
938661	AE1-088	5.02
938771	AE1-103 C O1	5.9
938772	AE1-103 E O1	8.15
939191	AE1-149 C O1	14.16
939192	AE1-149 E O1	9.44
939241	AE1-155 C	16.81
939242	AE1-155 E	11.21
939311	AE1-162 C	3.67
939312	AE1-162 E	2.45
939411	AE1-173 C	146.82
939412	AE1-173 E	97.88
939421	AE1-174 C	0.66
939422	AE1-174 E	1.0
939431	AE1-175 C	4.42
939432	AE1-175 E	2.19
939611	AE1-191 C	16.96
939612	AE1-191 E	11.31
940061	AE1-248 C O1	24.23
940062	AE1-248 E O1	16.15
940071	AE1-249 C	8.25
940072	AE1-249 E	6.15
AA2-074	AA2-074	6.78
CARR	CARR	1.33
CBM-S1	CBM-S1	13.68
CBM-S2	CBM-S2	18.76
CBM-W1	CBM-W1	9.73
CBM-W2	CBM-W2	85.19
CIN	CIN	4.65
CPLE	CPLE	9.96
G-007	G-007	4.26
IPL	IPL	2.85
LGEE	LGEE	1.36
MEC	MEC	11.66
MECS	MECS	2.46
O-066	O-066	27.14
RENSSELAER	RENSSELAER	1.06
WEC	WEC	1.23

12.6.10 Index 9

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308916	919140	AA1-138 TAP	DVP	314537	6SUFFOLK	DVP	1	DVP_P1-2: LN 2092	single	571.52	93.54	105.09	AC	67.55

Bus #	Bus	MW Impact
314589	3MURPHYS	0.08
314704	3LAWRENC	0.16
315126	1ROARAP2	1.11
315128	1ROARAP4	1.07
315131	1EDGECEMA	9.42
315132	1EDGECEMB	9.42
315136	1ROSEMG1	1.95
315137	1ROSEMS1	1.21
315138	1ROSEMG2	0.91
315139	1GASTONA	2.74
315141	1GASTONB	2.74
315159	1KERR 2	0.61
315163	1KERR 6	0.6
315164	1KERR 7	0.6
315292	1DOMTR78	2.33
315293	1DOMTR9	1.9
315294	1DOMTR10	3.45
916041	Z1-036 C	1.9
917331	Z2-043 C	0.53
917341	Z2-044 C	0.2
917511	Z2-088 C OP1	1.93
918491	AA1-063AC OP	1.56
918511	AA1-065 C OP	2.76
918531	AA1-067 C	0.42
918561	AA1-072 C	0.08
919691	AA2-053 C	1.41
919701	AA2-057 C	1.08
920041	AA2-088 C OP	0.57
920591	AA2-165 C	0.15
920671	AA2-174 C	0.06
920691	AA2-178 C	2.73
923911	AB2-031 C O1	1.74
923991	AB2-040 C O1	0.81
924151	AB2-059 C O1	1.49
924401	AB2-089 C	1.24
924491	AB2-098 C	0.99
924501	AB2-099 C	0.95
924511	AB2-100 C	7.11
925121	AB2-169 C	1.74
925171	AB2-174 C O1	5.2
925591	AC1-034 C	6.82

Bus #	Bus	MW Impact
925781	AC1-054 C O1	4.49
926071	AC1-086 C	21.84
926201	AC1-098 C	7.82
926211	AC1-099 C	2.62
927021	AC1-189 C	13.69
927141	AC1-208 C	10.08
930401	AB1-081 C O1	1.26
930861	AB1-132 C O1	2.1
931231	AB1-173 C	0.25
931241	AB1-173AC	0.25
932631	AC2-084 C	11.14
933991	AD1-023 C	27.2
934201	AD1-047 C	6.24
934231	AD1-050 C	2.73
934331	AD1-057 C O1	10.32
934521	AD1-076 C	110.7
936401	AD2-051 C O1	15.91
936531	AD2-068 C	6.55
936701	AD2-089 C	9.29
937571	AD2-169 C	8.2
938171	AE1-026 C1 O	58.55
938172	AE1-026 C2 O	9.0
938221	AE1-035 C	4.88
938661	AE1-088	6.3
AA2-074	AA2-074	2.73
CARR	CARR	0.19
CBM-S1	CBM-S1	5.96
CBM-S2	CBM-S2	7.05
CBM-W1	CBM-W1	6.18
CBM-W2	CBM-W2	39.17
CIN	CIN	2.77
CPLE	CPLE	4.01
IPL	IPL	1.74
LGEE	LGEE	0.8
MEC	MEC	6.12
MECS	MECS	2.7
RENSSELAER	RENSSELAER	0.15
WEC	WEC	0.74

12.6.11 Index 10

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309232	934330	AD1-057 TAP	DVP	313845	6HATHAWAY	DVP	1	DVP_P1-2: LN 238	single	441.8	97.15	100.18	AC	12.95

Bus #	Bus	MW Impact
314589	3MURPHYS	0.05
314704	3LAWRENC	0.2
315115	1S HAMPT1	0.67
315126	1ROARAP2	1.31
315128	1ROARAP4	1.26
315136	1ROSEMG1	4.86
315137	1ROSEMS1	3.01
315138	1ROSEMG2	2.28
315139	1GASTONA	5.9
315141	1GASTONB	5.9
315158	1KERR 1	0.13
315159	1KERR 2	0.52
315160	1KERR 3	0.51
315161	1KERR 4	0.51
315162	1KERR 5	0.51
315163	1KERR 6	0.51
315164	1KERR 7	0.51
917331	Z2-043 C	0.14
918491	AA1-063AC OP	1.07
918511	AA1-065 C OP	0.65
918561	AA1-072 C	0.02
919691	AA2-053 C	1.23
920041	AA2-088 C OP	0.6
920671	AA2-174 C	0.06
923801	AB2-015 C O1	4.13
923911	AB2-031 C O1	2.15
923991	AB2-040 C O1	1.0
924021	AB2-043 C O1	0.14
924161	AB2-060 C O1	0.41
924301	AB2-077 C O1	0.09
924311	AB2-078 C O1	0.09
924321	AB2-079 C O1	0.09
924401	AB2-089 C	1.16
924501	AB2-099 C	0.34
924511	AB2-100 C	25.42
925171	AB2-174 C O1	6.51
925781	AC1-054 C O1	4.39
926071	AC1-086 C	46.93
930861	AB1-132 C O1	4.5
931231	AB1-173 C	0.31
931241	AB1-173AC	0.31

Bus #	Bus	MW Impact
933991	AD1-023 C	4.9
934201	AD1-047 C	7.71
934231	AD1-050 C	2.56
934331	AD1-057 C O1	38.45
934521	AD1-076 C	17.69
936361	AD2-046 C O1	3.68
936401	AD2-051 C O1	4.87
936711	AD2-090 C O1	4.03
937571	AD2-169 C	10.0
938171	AE1-026 C1 O	11.23
938172	AE1-026 C2 O	1.73
938221	AE1-035 C	1.15
938771	AE1-103 C O1	1.49
939181	AE1-148 C O1	3.58
BLUEG	BLUEG	3.38
CALDERWOOD	CALDERWOOD	0.66
CANNELTON	CANNELTON	0.23
CARR	CARR	0.0
CATAWBA	CATAWBA	0.64
CHEOAH	CHEOAH	0.62
CHILHOWEE	CHILHOWEE	0.22
COFFEEN	COFFEEN	0.39
COTTONWOOD	COTTONWOOD	2.24
DUCKCREEK	DUCKCREEK	0.81
EDWARDS	EDWARDS	0.36
FARMERCITY	FARMERCITY	0.28
G-007A	G-007A	0.12
GIBSON	GIBSON	0.15
HAMLET	HAMLET	1.28
NEWTON	NEWTON	1.02
PRAIRIE	PRAIRIE	2.11
RENSSELAER	RENSSELAER	0.0
SANTEETLA	SANTEETLA	0.18
SMITHLAND	SMITHLAND	0.19
TATANKA	TATANKA	0.48
TILTON	TILTON	0.43
TRIMBLE	TRIMBLE	0.37
TVA	TVA	1.88
UNIONPOWER	UNIONPOWER	0.92
VFT	VFT	0.3

12.6.12 Index 11

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309535	936530	AD2-068 TAP	DVP	304451	6GREENVILLE T	CPL	1	DVP_P7-1: LN 2058-2181	tower	478.0	156.39	165.41	AC	46.54

Bus #	Bus	MW Impact
314539	3UNCAMP	1.27
314541	3WATKINS	0.39
314554	3BTLEBRO	0.38
314557	3BETHEL	1.06
314566	3CRESWEL	2.23
314572	3EMPORIA	0.22
314574	6EVERETS	8.22
314578	3HORNRTN	2.14
314582	3KELFORD	3.33
314589	3MURPHYS	0.06
314594	6PLYMOTH	0.9
314603	3SCOT NK	2.69
314617	3TUNIS	0.78
314620	6CASHIE	0.97
314623	3WITAKRS	0.71
314648	6SUNBURY	0.41
314651	6WINFALL	0.99
315131	1EDGECEMA	8.55
315132	1EDGECEMB	8.55
315136	1ROSEMG1	1.46
315137	1ROSEMS1	0.9
315138	1ROSEMG2	0.68
315292	1DOMTR78	1.52
315293	1DOMTR9	1.24
315294	1DOMTR10	2.26
900672	V4-068 E	0.24
901082	W1-029 E	24.7
907092	X1-038 E	3.17
913392	Y1-086 E	1.0
916042	Z1-036 E	30.84
917122	Z2-027 E	0.52
917331	Z2-043 C	0.31
917332	Z2-043 E	0.95
917342	Z2-044 E	0.31
917511	Z2-088 C OP1	2.11
917512	Z2-088 E OP1	6.41
918491	AA1-063AC OP	0.98
918492	AA1-063AE OP	3.28
918511	AA1-065 C OP	1.58
918512	AA1-065 E OP	5.52
918531	AA1-067 C	0.48

Bus #	Bus	MW Impact
918532	AA1-067 E	1.45
918561	AA1-072 C	0.05
918562	AA1-072 E	0.16
919692	AA2-053 E	2.78
919702	AA2-057 E	2.06
920042	AA2-088 E OP	6.8
920592	AA2-165 E	0.27
920672	AA2-174 E	0.32
920691	AA2-178 C	1.26
920692	AA2-178 E	3.82
923801	AB2-015 C O1	4.71
923802	AB2-015 E O1	3.86
923832	AB2-022 E	0.57
923911	AB2-031 C O1	1.26
923912	AB2-031 E O1	0.62
923992	AB2-040 E O1	3.38
924152	AB2-059 E O1	3.09
924491	AB2-098 C	1.13
924492	AB2-098 E	0.48
924501	AB2-099 C	0.59
924502	AB2-099 E	0.25
924511	AB2-100 C	6.09
924512	AB2-100 E	3.0
925121	AB2-169 C	1.38
925122	AB2-169 E	8.74
925171	AB2-174 C O1	3.82
925172	AB2-174 E O1	3.46
925591	AC1-034 C	3.89
925592	AC1-034 E	2.93
926071	AC1-086 C	15.84
926072	AC1-086 E	7.21
926201	AC1-098 C	4.51
926202	AC1-098 E	2.69
926211	AC1-099 C	1.51
926212	AC1-099 E	0.89
927021	AC1-189 C	15.22
927022	AC1-189 E	7.58
927141	AC1-208 C	5.86
927142	AC1-208 E	2.6
930402	AB1-081 E O1	2.18
930862	AB1-132 E O1	4.61
931232	AB1-173 E	0.59
931242	AB1-173AE	0.59
932631	AC2-084 C	6.44
932632	AC2-084 E	3.17
933991	AD1-023 C	14.77
933992	AD1-023 E	8.04
934201	AD1-047 C	4.51
934202	AD1-047 E	3.01
934331	AD1-057 C O1	8.5
934332	AD1-057 E O1	4.53
934521	AD1-076 C	58.8

Bus #	Bus	MW Impact
934522	AD1-076 E	29.94
936401	AD2-051 C O1	9.44
936402	AD2-051 E O1	4.05
936531	AD2-068 C	19.17
936532	AD2-068 E	9.88
936701	AD2-089 C	9.53
936702	AD2-089 E	6.35
936711	AD2-090 C O1	4.25
936712	AD2-090 E O1	2.83
937571	AD2-169 C	5.83
937572	AD2-169 E	3.88
938171	AE1-026 C1 O	32.17
938172	AE1-026 C2 O	4.94
938173	AE1-026 E O1	9.42
938221	AE1-035 C	2.8
938222	AE1-035 E	1.38
938661	AE1-088	5.0
938771	AE1-103 C O1	1.85
938772	AE1-103 E O1	2.55
AC1-131	AC1-131	6.18
BLUEG	BLUEG	7.81
CALDERWOOD	CALDERWOOD	1.67
CANNELTON	CANNELTON	0.54
CATAWBA	CATAWBA	1.63
CBM-N	CBM-N	0.21
CHEOAH	CHEOAH	1.56
CHILHOWEE	CHILHOWEE	0.54
COFFEEN	COFFEEN	0.92
COTTONWOOD	COTTONWOOD	5.62
DUCKCREEK	DUCKCREEK	1.89
EDWARDS	EDWARDS	0.84
FARMERCITY	FARMERCITY	0.67
G-007A	G-007A	1.0
GIBSON	GIBSON	0.34
HAMLET	HAMLET	3.48
NEWTON	NEWTON	2.4
NYISO	NYISO	0.91
PRAIRIE	PRAIRIE	5.08
SANTEETLA	SANTEETLA	0.46
SMITHLAND	SMITHLAND	0.45
TATANKA	TATANKA	1.13
TILTON	TILTON	0.99
TRIMBLE	TRIMBLE	0.86
TVA	TVA	4.7
UNIONPOWER	UNIONPOWER	2.33
VFT	VFT	2.66

12.6.13 Index 12

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308592	313714	6PERQUIMANS	DVP	314662	6S HERTFORD	DVP	1	DVP_P4-2: 24642	breaker	897.0	98.76	107.01	AC	82.05

Bus #	Bus	MW Impact
314554	3BTLEBRO	0.52
314557	3BETHEL	0.69
314566	3CRESWEL	7.34
314574	6EVERETS	4.87
314578	3HORNRTN	2.04
314582	3KELFORD	3.49
314594	6PLYMOTH	2.21
314603	3SCOT NK	2.88
314617	3TUNIS	0.73
314620	6CASHIE	1.71
314623	3WITAKRS	0.9
315131	1EDGECEMA	6.16
315132	1EDGECEMB	6.16
315141	1GASTONB	1.71
315292	1DOMTR78	2.62
315293	1DOMTR9	2.14
315294	1DOMTR10	3.88
900672	V4-068 E	0.19
916041	Z1-036 C	3.82
916042	Z1-036 E	180.93
917331	Z2-043 C	0.33
917332	Z2-043 E	0.99
917342	Z2-044 E	0.39
917511	Z2-088 C OP1	1.3
917512	Z2-088 E OP1	3.94
918491	AA1-063AC OP	0.97
918492	AA1-063AE OP	3.22
918511	AA1-065 C OP	1.68
918512	AA1-065 E OP	5.87
918531	AA1-067 C	0.28
918532	AA1-067 E	0.86
918561	AA1-072 C	0.05
918562	AA1-072 E	0.17
919691	AA2-053 C	0.88
919692	AA2-053 E	2.67
919701	AA2-057 C	0.69
919702	AA2-057 E	2.45
920591	AA2-165 C	0.09
920592	AA2-165 E	0.32
920671	AA2-174 C	0.04
920672	AA2-174 E	0.31
920691	AA2-178 C	4.15

Bus #	Bus	MW Impact
920692	AA2-178 E	12.59
923911	AB2-031 C O1	1.09
923912	AB2-031 E O1	0.54
923992	AB2-040 E O1	2.92
924152	AB2-059 E O1	3.53
924491	AB2-098 C	0.67
924492	AB2-098 E	0.29
924501	AB2-099 C	0.59
924502	AB2-099 E	0.25
925121	AB2-169 C	1.73
925122	AB2-169 E	10.97
925591	AC1-034 C	4.44
925592	AC1-034 E	3.35
926071	AC1-086 C	13.63
926072	AC1-086 E	6.21
926201	AC1-098 C	4.9
926202	AC1-098 E	2.92
926211	AC1-099 C	1.64
926212	AC1-099 E	0.96
927021	AC1-189 C	9.24
927022	AC1-189 E	4.6
927141	AC1-208 C	6.37
927142	AC1-208 E	2.83
930402	AB1-081 E O1	2.49
930862	AB1-132 E O1	3.97
931232	AB1-173 E	0.51
931242	AB1-173AE	0.51
932631	AC2-084 C	6.98
932632	AC2-084 E	3.44
933991	AD1-023 C	28.94
933992	AD1-023 E	15.76
934201	AD1-047 C	3.9
934202	AD1-047 E	2.6
934331	AD1-057 C O1	6.61
934332	AD1-057 E O1	3.52
934521	AD1-076 C	136.51
934522	AD1-076 E	69.51
936401	AD2-051 C O1	9.76
936402	AD2-051 E O1	4.19
936531	AD2-068 C	4.42
936532	AD2-068 E	2.28
936701	AD2-089 C	6.19
936702	AD2-089 E	4.12
937571	AD2-169 C	5.13
937572	AD2-169 E	3.42
938171	AE1-026 C1 O	56.72
938172	AE1-026 C2 O	8.72
938173	AE1-026 E O1	16.62
938221	AE1-035 C	2.98
938222	AE1-035 E	1.47
938661	AE1-088	6.27
AA2-074	AA2-074	1.81

Bus #	Bus	MW Impact
CARR	CARR	0.14
CBM-S1	CBM-S1	3.89
CBM-S2	CBM-S2	4.65
CBM-W1	CBM-W1	3.98
CBM-W2	CBM-W2	25.51
CIN	CIN	1.79
CPLE	CPLE	2.66
G-007	G-007	0.48
IPL	IPL	1.12
LGEE	LGEE	0.52
MEC	MEC	3.97
MECS	MECS	1.72
O-066	O-066	3.03
RENSSELAER	RENSSELAER	0.11
WEC	WEC	0.48

12.6.14 Index 13

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308951	314905	8CHANCE	DVP	314900	8BRISTER	DVP	1	DVP_P1-2: LN 594	single	2442.12	114.02	115.2	AC	33.11

Bus #	Bus	MW Impact
314229	6MT RD221	0.12
314236	6NRTHEST	0.18
314250	6ROCKVILLE	0.33
315037	1LDYSMT1	4.42
315038	1LDYSMT2	4.41
315039	1LDYSMT3	4.67
315040	1LDYSMT4	4.68
315041	1LDYSMT5	4.7
315043	1FOUR RIVERA	3.5
315044	1FOUR RIVERB	2.71
315045	1FOUR RIVERC	3.5
315046	1FOUR RIVERD	2.71
315047	1FOUR RIVERE	2.71
315048	1FOUR RIVERF	3.5
315053	1BELMED1	19.5
315054	1BELMED2	19.5
315055	1BELMED3	16.18
315058	1CHESTF3	20.46
315059	1CHESTF4	33.16
315067	1DARBY 1	2.51
315068	1DARBY 2	2.51
315069	1DARBY 3	2.52
315070	1DARBY 4	2.52
315074	1HOPCGN1	9.38
315075	1HOPCGN2	9.26
315083	1SPRUNCA	11.94
315084	1SPRUNCB	11.94
315085	1SPRUNCC	8.85
315086	1SPRUNCD	8.85
315090	1YORKTN1	31.83
315091	1YORKTN2	33.03
315225	1N ANNA1	35.96
315226	1N ANNA2	36.04
919211	AA1-145	10.13
923801	AB2-015 C O1	9.07
923911	AB2-031 C O1	2.24
924061	AB2-050	0.6
924241	AB2-068 O1	226.37
924501	AB2-099 C	0.59
924511	AB2-100 C	11.81
925051	AB2-160 C O1	6.09

Bus #	Bus	MW Impact
925061	AB2-161 C O1	3.71
925171	AB2-174 C O1	7.07
925331	AB2-190 C	22.49
925861	AC1-065 C	3.76
926001	AC1-076 C	4.92
926071	AC1-086 C	20.65
926291	AC1-107 O1	341.7
926411	AC1-112 C	0.32
926551	AC1-134	1.49
926731	AC1-158 C	93.23
926751	AC1-161 C O1	33.47
926781	AC1-164 C	45.52
927041	AC1-191 C O1	10.77
927221	AC1-216 C O1	10.99
930121	AB1-027 C	0.41
932041	AC2-012 C	11.21
932501	AC2-070 C	0.27
932581	AC2-078 C O1	4.5
932591	AC2-079 C O1	6.27
932831	AC2-110 C	1.5
933011	AC2-125	3.08
933021	AC2-126	3.1
933261	AC2-137 C	0.33
933291	AC2-141 C	33.47
933501	AC2-165 C	9.59
933991	AD1-023 C	13.51
934011	AD1-025 C	18.89
934061	AD1-033 C	8.22
934141	AD1-041 C	5.91
934201	AD1-047 C	8.04
934211	AD1-048 C	0.39
934521	AD1-076 C	56.24
934541	AD1-078 C	3.04
934571	AD1-082 C	8.45
934781	AD1-105 C	7.78
935161	AD1-151 C O1	18.07
936041	AD2-007	2.0
936051	AD2-008 C	3.29
936151	AD2-021	0.27
936301	AD2-039 C	1.5
936341	AD2-044 C	0.26
936391	AD2-049 C	1.98
936401	AD2-051 C O1	8.82
936581	AD2-073 C	2.22
936591	AD2-074 C	5.93
936661	AD2-085 C	3.76
936711	AD2-090 C O1	7.5
937221	AD2-160 C O1	6.38
937251	AD2-164	5.33
937541	AD2-215 C	1.91
937571	AD2-169 C	10.01
938031	AE1-004 C	1.5

Bus #	Bus	MW Impact
938171	AE1-026 C1 O	28.7
938172	AE1-026 C2 O	4.41
938181	AE1-027 C	2.56
938191	AE1-028 C	1.48
938221	AE1-035 C	2.27
938491	AE1-068 C O1	59.83
938501	AE1-069 C O1	47.27
938531	AE1-072 C O1	19.09
938551	AE1-074 C	2.75
938561	AE1-075 C	2.13
938631	AE1-085 C O1	9.85
938661	AE1-088	3.36
938771	AE1-103 C O1	3.86
939191	AE1-149 C O1	11.87
939241	AE1-155 C	13.69
939261	AE1-157 C O1	16.18
939271	AE1-158 C O1	16.52
939311	AE1-162 C	2.38
939411	AE1-173 C	93.89
939421	AE1-174 C	0.41
939431	AE1-175 C	2.69
939611	AE1-191 C	11.82
939751	AE1-206 C O1	38.01
940061	AE1-248 C O1	17.57
940071	AE1-249 C	8.04
AA2-074	AA2-074	5.18
CARR	CARR	1.68
CBM-S1	CBM-S1	12.6
CBM-S2	CBM-S2	15.04
CBM-W1	CBM-W1	9.19
CBM-W2	CBM-W2	78.0
CIN	CIN	4.65
CPLE	CPLE	7.62
IPL	IPL	2.87
LGEE	LGEE	1.4
MEC	MEC	10.97
MECS	MECS	2.13
RENSSELAER	RENSSELAER	1.33
WEC	WEC	1.19

12.6.15 Index 14

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309061	314911	8LADYSMITH	DVP	314905	8CHANCE	DVP	1	DVP_P1-2: LN 573	single	2738.22	103.9	104.97	AC	33.6

Bus #	Bus	MW Impact
314229	6MT RD221	0.12
314236	6NRTHEST	0.19
314250	6ROCKVILLE	0.33
315037	1LDYSMT1	4.53
315038	1LDYSMT2	4.53
315039	1LDYSMT3	4.79
315040	1LDYSMT4	4.8
315041	1LDYSMT5	4.82
315043	1FOUR RIVERA	3.58
315044	1FOUR RIVERB	2.77
315045	1FOUR RIVERC	3.58
315046	1FOUR RIVERD	2.77
315047	1FOUR RIVERE	2.77
315048	1FOUR RIVERF	3.57
315053	1BELMED1	19.82
315054	1BELMED2	19.82
315055	1BELMED3	16.45
315058	1CHESTF3	20.82
315059	1CHESTF4	33.74
315067	1DARBY 1	2.55
315068	1DARBY 2	2.56
315069	1DARBY 3	2.57
315070	1DARBY 4	2.57
315074	1HOPCGN1	9.54
315075	1HOPCGN2	9.42
315083	1SPRUNCA	12.13
315084	1SPRUNCB	12.13
315085	1SPRUNCC	9.0
315086	1SPRUNCD	9.0
315090	1YORKTN1	32.38
315091	1YORKTN2	33.6
315225	1N ANNA1	36.53
315226	1N ANNA2	36.61
919211	AA1-145	10.34
923801	AB2-015 C O1	9.2
923911	AB2-031 C O1	2.27
924061	AB2-050	0.61
924241	AB2-068 O1	230.65
924501	AB2-099 C	0.6
924511	AB2-100 C	11.98
925051	AB2-160 C O1	6.2

Bus #	Bus	MW Impact
925061	AB2-161 C O1	3.77
925171	AB2-174 C O1	7.16
925331	AB2-190 C	22.87
925861	AC1-065 C	3.83
926071	AC1-086 C	20.93
926291	AC1-107 O1	348.14
926411	AC1-112 C	0.33
926551	AC1-134	1.52
926751	AC1-161 C O1	34.02
926781	AC1-164 C	46.35
927041	AC1-191 C O1	10.98
927221	AC1-216 C O1	11.17
930121	AB1-027 C	0.42
932041	AC2-012 C	11.39
932501	AC2-070 C	0.28
932581	AC2-078 C O1	4.57
932591	AC2-079 C O1	6.37
932831	AC2-110 C	1.53
933011	AC2-125	3.16
933021	AC2-126	3.18
933261	AC2-137 C	0.34
933291	AC2-141 C	34.02
933501	AC2-165 C	9.57
933991	AD1-023 C	13.71
934011	AD1-025 C	19.21
934061	AD1-033 C	8.36
934141	AD1-041 C	6.02
934201	AD1-047 C	8.15
934211	AD1-048 C	0.4
934521	AD1-076 C	57.08
934571	AD1-082 C	8.59
934781	AD1-105 C	7.95
935161	AD1-151 C O1	18.38
936041	AD2-007	2.04
936051	AD2-008 C	3.34
936151	AD2-021	0.27
936301	AD2-039 C	1.53
936341	AD2-044 C	0.27
936391	AD2-049 C	2.02
936401	AD2-051 C O1	8.95
936581	AD2-073 C	2.27
936591	AD2-074 C	6.05
936661	AD2-085 C	3.82
936711	AD2-090 C O1	7.61
937221	AD2-160 C O1	6.48
937251	AD2-164	5.42
937541	AD2-215 C	1.94
937571	AD2-169 C	10.14
938031	AE1-004 C	1.53
938171	AE1-026 C1 O	29.12
938172	AE1-026 C2 O	4.48
938181	AE1-027 C	2.6

Bus #	Bus	MW Impact
938191	AE1-028 C	1.51
938221	AE1-035 C	2.31
938491	AE1-068 C O1	60.69
938501	AE1-069 C O1	47.95
938531	AE1-072 C O1	19.39
938551	AE1-074 C	2.79
938561	AE1-075 C	2.13
938631	AE1-085 C O1	10.01
938661	AE1-088	3.4
938771	AE1-103 C O1	3.92
939191	AE1-149 C O1	12.06
939241	AE1-155 C	13.97
939261	AE1-157 C O1	16.58
939271	AE1-158 C O1	16.92
939311	AE1-162 C	2.42
939411	AE1-173 C	95.34
939421	AE1-174 C	0.41
939431	AE1-175 C	2.74
939611	AE1-191 C	12.04
939751	AE1-206 C O1	38.77
940061	AE1-248 C O1	17.86
940071	AE1-249 C	8.17
AA2-074	AA2-074	5.24
CARR	CARR	1.68
CBM-S1	CBM-S1	12.61
CBM-S2	CBM-S2	15.17
CBM-W1	CBM-W1	9.08
CBM-W2	CBM-W2	77.91
CIN	CIN	4.6
CPLE	CPLE	7.7
IPL	IPL	2.84
LGEE	LGEE	1.38
MEC	MEC	10.91
MECS	MECS	2.04
RENSSELAER	RENSSELAER	1.33
WEC	WEC	1.18

12.6.16 Index 15

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308862	313845	6HATHAWAY	DVP	304222	6ROCKYMT230T	CPL	1	DVP_P1-2: LN 2181	single	374.0	120.44	125.39	AC	18.84

Bus #	Bus	MW Impact
314589	3MURPHYS	0.05
314704	3LAWRENC	0.15
315115	1S HAMPT1	0.63
315126	1ROARAP2	0.93
315128	1ROARAP4	0.9
315131	1EDGECEMA	24.75
315132	1EDGECEMB	24.75
315136	1ROSEMG1	2.44
315137	1ROSEMS1	1.51
315138	1ROSEMG2	1.14
315139	1GASTONA	2.92
315141	1GASTONB	2.92
315292	1DOMTR78	0.68
315293	1DOMTR9	0.55
315294	1DOMTR10	1.0
901081	W1-029 C	0.28
916041	Z1-036 C	0.34
917331	Z2-043 C	0.26
917341	Z2-044 C	0.23
917511	Z2-088 C OP1	1.69
918491	AA1-063AC OP	0.91
918511	AA1-065 C OP	0.84
918531	AA1-067 C	0.17
918561	AA1-072 C	0.04
919691	AA2-053 C	0.96
919701	AA2-057 C	1.15
920041	AA2-088 C OP	0.53
920591	AA2-165 C	0.16
920671	AA2-174 C	0.04
920691	AA2-178 C	0.63
923801	AB2-015 C O1	3.95
923851	AB2-025 C	0.14
923911	AB2-031 C O1	1.61
923991	AB2-040 C O1	0.75
924151	AB2-059 C O1	2.39
924401	AB2-089 C	0.73
924491	AB2-098 C	0.41
924501	AB2-099 C	0.4
924511	AB2-100 C	8.39
925121	AB2-169 C	0.56
925171	AB2-174 C O1	4.93

Bus #	Bus	MW Impact
925591	AC1-034 C	10.93
925781	AC1-054 C O1	2.83
926071	AC1-086 C	23.24
926201	AC1-098 C	5.92
926211	AC1-099 C	1.98
927021	AC1-189 C	9.72
927141	AC1-208 C	9.06
930401	AB1-081 C O1	2.02
930861	AB1-132 C O1	2.23
931231	AB1-173 C	0.23
931241	AB1-173AC	0.23
932631	AC2-084 C	8.44
933991	AD1-023 C	7.4
934201	AD1-047 C	5.77
934231	AD1-050 C	1.6
934331	AD1-057 C O1	19.81
934521	AD1-076 C	28.78
936401	AD2-051 C O1	6.14
936701	AD2-089 C	14.05
936711	AD2-090 C O1	3.69
937571	AD2-169 C	7.4
938171	AE1-026 C1 O	16.33
938172	AE1-026 C2 O	2.51
938221	AE1-035 C	1.48
938661	AE1-088	2.05
938771	AE1-103 C O1	1.49
BLUEG	BLUEG	5.91
CALDERWOOD	CALDERWOOD	1.2
CANNELTON	CANNELTON	0.4
CATAWBA	CATAWBA	1.17
CBM-N	CBM-N	0.12
CHEOAH	CHEOAH	1.12
CHILHOWEE	CHILHOWEE	0.39
COFFEEN	COFFEEN	0.69
COTTONWOOD	COTTONWOOD	4.01
DUCKCREEK	DUCKCREEK	1.41
EDWARDS	EDWARDS	0.63
FARMERCITY	FARMERCITY	0.49
G-007A	G-007A	0.65
GIBSON	GIBSON	0.25
HAMLET	HAMLET	2.3
NEWTON	NEWTON	1.79
NYISO	NYISO	0.54
PRAIRIE	PRAIRIE	3.73
SANTEETLA	SANTEETLA	0.33
SMITHLAND	SMITHLAND	0.33
TATANKA	TATANKA	0.84
TILTON	TILTON	0.75
TRIMBLE	TRIMBLE	0.65
TVA	TVA	3.37
UNIONPOWER	UNIONPOWER	1.66
VFT	VFT	1.72

12.6.17 Index 16

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309021	313845	6HATHAWAY	DVP	314591	6NASH	DVP	1	DVP_P1-2: LN 2058	single	449.32	108.77	112.97	AC	19.21

Bus #	Bus	MW Impact
314589	3MURPHYS	0.05
314704	3LAWRENC	0.15
315115	1S HAMPT1	0.64
315126	1ROARAP2	0.95
315128	1ROARAP4	0.92
315131	1EDGECEMA	25.25
315132	1EDGECEMB	25.25
315136	1ROSEMG1	2.49
315137	1ROSEMS1	1.54
315138	1ROSEMG2	1.17
315139	1GASTONA	2.98
315141	1GASTONB	2.98
315292	1DOMTR78	0.69
315293	1DOMTR9	0.56
315294	1DOMTR10	1.02
901081	W1-029 C	0.29
916041	Z1-036 C	0.35
917331	Z2-043 C	0.27
917341	Z2-044 C	0.24
917511	Z2-088 C OP1	1.73
918491	AA1-063AC OP	0.93
918511	AA1-065 C OP	0.86
918531	AA1-067 C	0.18
918561	AA1-072 C	0.04
919691	AA2-053 C	0.98
919701	AA2-057 C	1.17
920041	AA2-088 C OP	0.54
920591	AA2-165 C	0.16
920671	AA2-174 C	0.04
920691	AA2-178 C	0.64
923801	AB2-015 C O1	4.03
923851	AB2-025 C	0.14
923911	AB2-031 C O1	1.64
923991	AB2-040 C O1	0.76
924151	AB2-059 C O1	2.43
924401	AB2-089 C	0.74
924491	AB2-098 C	0.41
924501	AB2-099 C	0.41
924511	AB2-100 C	8.56
925121	AB2-169 C	0.57
925171	AB2-174 C O1	5.03

Bus #	Bus	MW Impact
925591	AC1-034 C	11.15
925781	AC1-054 C O1	2.89
926071	AC1-086 C	23.71
926201	AC1-098 C	6.04
926211	AC1-099 C	2.02
927021	AC1-189 C	9.92
927141	AC1-208 C	9.24
930401	AB1-081 C O1	2.06
930861	AB1-132 C O1	2.27
931231	AB1-173 C	0.23
931241	AB1-173AC	0.23
932631	AC2-084 C	8.61
933991	AD1-023 C	7.55
934201	AD1-047 C	5.89
934231	AD1-050 C	1.64
934331	AD1-057 C O1	20.2
934521	AD1-076 C	29.35
936401	AD2-051 C O1	6.26
936701	AD2-089 C	14.33
936711	AD2-090 C O1	3.77
937571	AD2-169 C	7.54
938171	AE1-026 C1 O	16.65
938172	AE1-026 C2 O	2.56
938221	AE1-035 C	1.51
938661	AE1-088	2.09
938771	AE1-103 C O1	1.52
BLUEG	BLUEG	6.03
CALDERWOOD	CALDERWOOD	1.22
CANNELTON	CANNELTON	0.41
CATAWBA	CATAWBA	1.19
CBM-N	CBM-N	0.13
CHEOAH	CHEOAH	1.14
CHILHOWEE	CHILHOWEE	0.4
COFFEEN	COFFEEN	0.7
COTTONWOOD	COTTONWOOD	4.08
DUCKCREEK	DUCKCREEK	1.44
EDWARDS	EDWARDS	0.64
FARMERCITY	FARMERCITY	0.5
G-007A	G-007A	0.66
GIBSON	GIBSON	0.26
HAMLET	HAMLET	2.34
NEWTON	NEWTON	1.82
NYISO	NYISO	0.55
PRAIRIE	PRAIRIE	3.8
SANTEETLA	SANTEETLA	0.34
SMITHLAND	SMITHLAND	0.34
TATANKA	TATANKA	0.86
TILTON	TILTON	0.76
TRIMBLE	TRIMBLE	0.66
TVA	TVA	3.43
UNIONPOWER	UNIONPOWER	1.7
VFT	VFT	1.75

12.6.18 Index 17

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309010	314209	6SKIFF CREEK	DVP	314386	6KINGS M	DVP	1	DVP_P1-2: LN 557	single	441.8	136.07	138.02	AC	10.03

Bus #	Bus	MW Impact
314421	6WINCHST	0.1
314507	3THOMPSN	0.12
315090	1YORKTN1	22.99
315091	1YORKTN2	23.86
315092	1YORKTN3	16.23
315098	1CHESPKA	0.15
315099	1CHESPKB	0.38
315108	1ELIZAR1	1.13
315109	1ELIZAR2	1.11
315110	1ELIZAR3	1.14
315233	1SURRY 2	11.54
315260	1GOSPORTA	0.12
315261	1GOSPORTB	0.16
315262	1GOSPORTC	0.13
916191	Z1-068 C	0.02
919151	AA1-139 C	0.85
923801	AB2-015 C O1	2.73
924241	AB2-068 O1	102.65
925521	AC1-027 C	0.12
926291	AC1-107 O1	154.94
926661	AC1-147 C	0.13
926751	AC1-161 C O1	13.96
932041	AC2-012 C	4.15
932591	AC2-079 C O1	1.66
933291	AC2-141 C	13.96
933731	AC2-196 C	0.1
933991	AD1-023 C	4.14
934061	AD1-033 C	3.08
934521	AD1-076 C	17.62
935111	AD1-144 C	0.1
936391	AD2-049 C	1.14
936661	AD2-085 C	0.98
936711	AD2-090 C O1	2.18
937221	AD2-160 C O1	2.37
937251	AD2-164	3.53
937541	AD2-215 C	0.75
938171	AE1-026 C1 O	8.69
938172	AE1-026 C2 O	1.34
938181	AE1-027 C	0.95
938191	AE1-028 C	0.55
938531	AE1-072 C O1	7.04

Bus #	Bus	MW Impact
938771	AE1-103 C O1	1.2
939311	AE1-162 C	0.78
939411	AE1-173 C	27.59
939421	AE1-174 C	0.16
939431	AE1-175 C	1.08
CARR	CARR	0.19
CBM-S1	CBM-S1	2.47
CBM-S2	CBM-S2	3.14
CBM-W1	CBM-W1	2.07
CBM-W2	CBM-W2	15.68
CIN	CIN	0.96
CPLE	CPLE	1.65
IPL	IPL	0.6
LGEE	LGEE	0.28
MEC	MEC	2.27
MECS	MECS	0.7
RENSSELAER	RENSSELAER	0.15
WEC	WEC	0.26

12.6.19 Index 18

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308283	314287	6CHESTFB	DVP	314276	6BASIN	DVP	1	DVP_P4-2: 562T563	breaker	549.0	136.7	138.99	AC	14.69

Bus #	Bus	MW Impact
314314	3LOCKS	0.16
314539	3UNCAMP	0.93
314541	3WATKINS	0.27
314554	3BTLEBRO	0.3
314557	3BETHEL	0.3
314572	3EMPORIA	0.21
314574	6EVERETS	1.52
314578	3HORNRTN	1.41
314582	3KELFORD	1.36
314594	6PLYMOTH	0.3
314603	3SCOT NK	1.29
314617	3TUNIS	0.33
314620	6CASHIE	0.31
314623	3WITAKRS	0.52
315065	1CHESTF6	24.31
315073	1STONECA	4.95
315074	1HOPCGN1	5.97
315075	1HOPCGN2	5.89
315076	1HOPPOLC	0.97
315077	1HOPHCF1	1.51
315078	1HOPHCF2	1.51
315079	1HOPHCF3	1.51
315080	1HOPHCF4	2.3
315131	1EDGECEMA	3.5
315132	1EDGECEMB	3.5
900672	V4-068 E	0.12
907092	X1-038 E	2.32
916302	Z1-086 E	3.65
917332	Z2-043 E	0.39
917342	Z2-044 E	0.22
917512	Z2-088 E OP1	1.44
918492	AA1-063AE OP	1.64
918512	AA1-065 E OP	1.66
918532	AA1-067 E	0.27
918562	AA1-072 E	0.06
919692	AA2-053 E	1.57
919702	AA2-057 E	1.38
920042	AA2-088 E OP	4.19
920592	AA2-165 E	0.18
920672	AA2-174 E	0.18
923801	AB2-015 C O1	3.33

Bus #	Bus	MW Impact
923802	AB2-015 E O1	2.73
923851	AB2-025 C	0.24
923852	AB2-025 E	0.77
923911	AB2-031 C O1	1.08
923912	AB2-031 E O1	0.53
923992	AB2-040 E O1	2.89
924152	AB2-059 E O1	2.04
924491	AB2-098 C	0.21
924492	AB2-098 E	0.09
924501	AB2-099 C	0.23
924502	AB2-099 E	0.1
924511	AB2-100 C	6.67
924512	AB2-100 E	3.29
924812	AB2-134 E O1	6.68
925051	AB2-160 C O1	3.53
925052	AB2-160 E O1	5.76
925061	AB2-161 C O1	1.95
925062	AB2-161 E O1	3.17
925122	AB2-169 E	2.1
925171	AB2-174 C O1	3.46
925172	AB2-174 E O1	3.13
925331	AB2-190 C	11.43
925332	AB2-190 E	4.9
925591	AC1-034 C	2.57
925592	AC1-034 E	1.94
925821	AC1-061	0.0
926071	AC1-086 C	9.76
926072	AC1-086 E	4.44
926201	AC1-098 C	2.43
926202	AC1-098 E	1.45
926211	AC1-099 C	0.81
926212	AC1-099 E	0.48
927021	AC1-189 C	3.2
927022	AC1-189 E	1.59
927141	AC1-208 C	3.62
927142	AC1-208 E	1.61
927221	AC1-216 C O1	5.37
927222	AC1-216 E O1	4.22
930402	AB1-081 E O1	1.44
930862	AB1-132 E O1	2.84
931232	AB1-173 E	0.51
931242	AB1-173AE	0.51
932581	AC2-078 C O1	3.0
932582	AC2-078 E O1	4.9
932591	AC2-079 C O1	2.72
932592	AC2-079 E O1	4.44
932631	AC2-084 C	3.46
932632	AC2-084 E	1.71
933991	AD1-023 C	4.72
933992	AD1-023 E	2.57
934011	AD1-025 C	9.23
934012	AD1-025 E	5.47

Bus #	Bus	MW Impact
934201	AD1-047 C	3.86
934202	AD1-047 E	2.57
934331	AD1-057 C O1	4.2
934332	AD1-057 E O1	2.24
934521	AD1-076 C	19.26
934522	AD1-076 E	9.81
934571	AD1-082 C	4.43
934572	AD1-082 E	2.53
935161	AD1-151 C O1	9.19
935162	AD1-151 E O1	6.12
935211	AD1-156 C	0.29
935212	AD1-156 E	1.39
936041	AD2-007	0.98
936051	AD2-008 C	1.61
936052	AD2-008 E	3.5
936401	AD2-051 C O1	3.37
936402	AD2-051 E O1	1.45
936661	AD2-085 C	1.65
936662	AD2-085 E	2.7
936701	AD2-089 C	2.71
936702	AD2-089 E	1.81
936711	AD2-090 C O1	2.82
936712	AD2-090 E O1	1.88
937571	AD2-169 C	4.69
937572	AD2-169 E	3.13
938171	AE1-026 C1 O	10.16
938172	AE1-026 C2 O	1.56
938173	AE1-026 E O1	2.98
938221	AE1-035 C	0.84
938222	AE1-035 E	0.42
938491	AE1-068 C O1	23.33
938492	AE1-068 E O1	12.89
938501	AE1-069 C O1	18.43
938502	AE1-069 E O1	10.54
938631	AE1-085 C O1	6.41
938632	AE1-085 E O1	3.2
938661	AE1-088	1.2
938771	AE1-103 C O1	1.39
938772	AE1-103 E O1	1.92
939191	AE1-149 C O1	8.26
939192	AE1-149 E O1	5.51
939311	AE1-162 C	0.81
939312	AE1-162 E	0.54
939411	AE1-173 C	31.98
939412	AE1-173 E	21.32
940061	AE1-248 C O1	9.21
940062	AE1-248 E O1	6.14
940071	AE1-249 C	4.66
940072	AE1-249 E	3.48
AA2-074	AA2-074	1.86
CARR	CARR	0.34
CBM-S1	CBM-S1	4.02

Bus #	Bus	MW Impact
CBM-S2	CBM-S2	5.2
CBM-W1	CBM-W1	3.22
CBM-W2	CBM-W2	25.4
CIN	CIN	1.51
CPLE	CPLE	2.73
G-007	G-007	1.1
IPL	IPL	0.93
LGEE	LGEE	0.44
MEC	MEC	3.62
MECS	MECS	1.01
O-066	O-066	7.02
RENSSELAER	RENSSELAER	0.27
WEC	WEC	0.4

12.6.20 Index 19

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309067	314296	6PENNINGMAN	DVP	314415	6WALR209	DVP	1	DVP_P1-2: LN 557	single	441.8	124.85	126.8	AC	10.03

Bus #	Bus	MW Impact
314421	6WINCHST	0.1
314507	3THOMPSN	0.12
315090	1YORKTN1	22.99
315091	1YORKTN2	23.86
315092	1YORKTN3	16.23
315098	1CHESPKA	0.15
315099	1CHESPKB	0.38
315108	1ELIZAR1	1.13
315109	1ELIZAR2	1.11
315110	1ELIZAR3	1.14
315233	1SURRY 2	11.54
315260	1GOSPORTA	0.12
315261	1GOSPORTB	0.16
315262	1GOSPORTC	0.13
916191	Z1-068 C	0.02
919151	AA1-139 C	0.85
923801	AB2-015 C O1	2.73
924241	AB2-068 O1	102.65
925521	AC1-027 C	0.12
926291	AC1-107 O1	154.94
926661	AC1-147 C	0.13
926751	AC1-161 C O1	13.96
932041	AC2-012 C	4.15
932591	AC2-079 C O1	1.66
933291	AC2-141 C	13.96
933731	AC2-196 C	0.1
933991	AD1-023 C	4.14
934061	AD1-033 C	3.08
934521	AD1-076 C	17.62
935111	AD1-144 C	0.1
936391	AD2-049 C	1.14
936661	AD2-085 C	0.98
936711	AD2-090 C O1	2.18
937221	AD2-160 C O1	2.37
937251	AD2-164	3.53
937541	AD2-215 C	0.75
938171	AE1-026 C1 O	8.69
938172	AE1-026 C2 O	1.34
938181	AE1-027 C	0.95
938191	AE1-028 C	0.55
938531	AE1-072 C O1	7.04

Bus #	Bus	MW Impact
938771	AE1-103 C O1	1.2
939311	AE1-162 C	0.78
939411	AE1-173 C	27.59
939421	AE1-174 C	0.16
939431	AE1-175 C	1.08
CARR	CARR	0.19
CBM-S1	CBM-S1	2.47
CBM-S2	CBM-S2	3.14
CBM-W1	CBM-W1	2.07
CBM-W2	CBM-W2	15.68
CIN	CIN	0.96
CPLE	CPLE	1.65
IPL	IPL	0.6
LGEE	LGEE	0.28
MEC	MEC	2.27
MECS	MECS	0.7
RENSSELAER	RENSSELAER	0.15
WEC	WEC	0.26

12.6.21 Index 20

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309039	314386	6KINGS M	DVP	314296	6PENNINGMAN	DVP	1	DVP_P1-2: LN 557	single	441.8	128.46	130.4	AC	10.03

Bus #	Bus	MW Impact
314421	6WINCHST	0.1
314507	3THOMPSN	0.12
315090	1YORKTN1	22.99
315091	1YORKTN2	23.86
315092	1YORKTN3	16.23
315098	1CHESPKA	0.15
315099	1CHESPKB	0.38
315108	1ELIZAR1	1.13
315109	1ELIZAR2	1.11
315110	1ELIZAR3	1.14
315233	1SURRY 2	11.54
315260	1GOSPORTA	0.12
315261	1GOSPORTB	0.16
315262	1GOSPORTC	0.13
916191	Z1-068 C	0.02
919151	AA1-139 C	0.85
923801	AB2-015 C O1	2.73
924241	AB2-068 O1	102.65
925521	AC1-027 C	0.12
926291	AC1-107 O1	154.94
926661	AC1-147 C	0.13
926751	AC1-161 C O1	13.96
932041	AC2-012 C	4.15
932591	AC2-079 C O1	1.66
933291	AC2-141 C	13.96
933731	AC2-196 C	0.1
933991	AD1-023 C	4.14
934061	AD1-033 C	3.08
934521	AD1-076 C	17.62
935111	AD1-144 C	0.1
936391	AD2-049 C	1.14
936661	AD2-085 C	0.98
936711	AD2-090 C O1	2.18
937221	AD2-160 C O1	2.37
937251	AD2-164	3.53
937541	AD2-215 C	0.75
938171	AE1-026 C1 O	8.69
938172	AE1-026 C2 O	1.34
938181	AE1-027 C	0.95
938191	AE1-028 C	0.55
938531	AE1-072 C O1	7.04

Bus #	Bus	MW Impact
938771	AE1-103 C O1	1.2
939311	AE1-162 C	0.78
939411	AE1-173 C	27.59
939421	AE1-174 C	0.16
939431	AE1-175 C	1.08
CARR	CARR	0.19
CBM-S1	CBM-S1	2.47
CBM-S2	CBM-S2	3.14
CBM-W1	CBM-W1	2.07
CBM-W2	CBM-W2	15.68
CIN	CIN	0.96
CPLE	CPLE	1.65
IPL	IPL	0.6
LGEE	LGEE	0.28
MEC	MEC	2.27
MECS	MECS	0.7
RENSSELAER	RENSSELAER	0.15
WEC	WEC	0.26

12.6.22 Index 21

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309242	314415	6WALR209	DVP	314391	6LIGH209	DVP	1	DVP_P1-2: LN 557	single	441.8	110.47	112.42	AC	10.03

Bus #	Bus	MW Impact
314421	6WINCHST	0.1
314507	3THOMPSN	0.12
315090	1YORKTN1	22.99
315091	1YORKTN2	23.86
315092	1YORKTN3	16.23
315098	1CHESPKA	0.15
315099	1CHESPKB	0.38
315108	1ELIZAR1	1.13
315109	1ELIZAR2	1.11
315110	1ELIZAR3	1.14
315233	1SURRY 2	11.54
315260	1GOSPORTA	0.12
315261	1GOSPORTB	0.16
315262	1GOSPORTC	0.13
916191	Z1-068 C	0.02
919151	AA1-139 C	0.85
923801	AB2-015 C O1	2.73
924241	AB2-068 O1	102.65
925521	AC1-027 C	0.12
926291	AC1-107 O1	154.94
926661	AC1-147 C	0.13
926751	AC1-161 C O1	13.96
932041	AC2-012 C	4.15
932591	AC2-079 C O1	1.66
933291	AC2-141 C	13.96
933731	AC2-196 C	0.1
933991	AD1-023 C	4.14
934061	AD1-033 C	3.08
934521	AD1-076 C	17.62
935111	AD1-144 C	0.1
936391	AD2-049 C	1.14
936661	AD2-085 C	0.98
936711	AD2-090 C O1	2.18
937221	AD2-160 C O1	2.37
937251	AD2-164	3.53
937541	AD2-215 C	0.75
938171	AE1-026 C1 O	8.69
938172	AE1-026 C2 O	1.34
938181	AE1-027 C	0.95
938191	AE1-028 C	0.55
938531	AE1-072 C O1	7.04

Bus #	Bus	MW Impact
938771	AE1-103 C O1	1.2
939311	AE1-162 C	0.78
939411	AE1-173 C	27.59
939421	AE1-174 C	0.16
939431	AE1-175 C	1.08
CARR	CARR	0.19
CBM-S1	CBM-S1	2.47
CBM-S2	CBM-S2	3.14
CBM-W1	CBM-W1	2.07
CBM-W2	CBM-W2	15.68
CIN	CIN	0.96
CPLE	CPLE	1.65
IPL	IPL	0.6
LGEE	LGEE	0.28
MEC	MEC	2.27
MECS	MECS	0.7
RENSSELAER	RENSSELAER	0.15
WEC	WEC	0.26

12.6.23 Index 22

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3309494	314554	3BTLEBRO	DVP	304223	3ROCKYMT115T	CPL	1	DVP_P7-1: LN 2058-2181	tower	164.0	294.06	302.25	AC	15.3

Bus #	Bus	MW Impact
314541	3WATKINS	0.25
314554	3BTLEBRO	1.96
314557	3BETHEL	0.86
314572	3EMPORIA	0.2
314574	6EVERETS	1.77
314578	3HORNRTN	2.52
314582	3KELFORD	2.87
314603	3SCOT NK	3.65
314617	3TUNIS	0.43
314620	6CASHIE	0.32
314623	3WITAKRS	2.92
315131	1EDGECEMA	13.53
315132	1EDGECEMB	13.53
315136	1ROSEMG1	1.46
315137	1ROSEMS1	0.91
315138	1ROSEMG2	0.68
315139	1GASTONA	1.8
315141	1GASTONB	1.8
900672	V4-068 E	0.15
917331	Z2-043 C	0.27
917332	Z2-043 E	0.82
917341	Z2-044 C	0.41
917342	Z2-044 E	1.25
917511	Z2-088 C OP1	0.96
917512	Z2-088 E OP1	2.91
918492	AA1-063AE OP	2.41
918512	AA1-065 E OP	1.91
918532	AA1-067 E	0.31
918561	AA1-072 C	0.04
918562	AA1-072 E	0.14
919692	AA2-053 E	2.32
919701	AA2-057 C	1.88
919702	AA2-057 E	6.66
920042	AA2-088 E OP	4.75
920591	AA2-165 C	0.26
920592	AA2-165 E	0.88
920672	AA2-174 E	0.27
923801	AB2-015 C O1	2.99
923802	AB2-015 E O1	2.45
923911	AB2-031 C O1	1.2
923912	AB2-031 E O1	0.59

Bus #	Bus	MW Impact
923992	AB2-040 E O1	3.22
924151	AB2-059 C O1	3.35
924152	AB2-059 E O1	12.21
924491	AB2-098 C	0.24
924492	AB2-098 E	0.1
924501	AB2-099 C	0.31
924502	AB2-099 E	0.13
924511	AB2-100 C	5.31
924512	AB2-100 E	2.61
925122	AB2-169 E	2.21
925171	AB2-174 C O1	3.6
925172	AB2-174 E O1	3.26
925591	AC1-034 C	15.36
925592	AC1-034 E	11.59
926071	AC1-086 C	14.33
926072	AC1-086 E	6.52
926201	AC1-098 C	7.95
926202	AC1-098 E	4.74
926211	AC1-099 C	2.66
926212	AC1-099 E	1.56
927021	AC1-189 C	5.56
927022	AC1-189 E	2.77
927141	AC1-208 C	12.51
927142	AC1-208 E	5.56
930401	AB1-081 C O1	2.84
930402	AB1-081 E O1	8.62
930861	AB1-132 C O1	1.38
930862	AB1-132 E O1	4.17
931232	AB1-173 E	0.56
931242	AB1-173AE	0.56
932631	AC2-084 C	11.33
932632	AC2-084 E	5.58
933991	AD1-023 C	4.78
933992	AD1-023 E	2.6
934201	AD1-047 C	4.29
934202	AD1-047 E	2.86
934331	AD1-057 C O1	11.18
934332	AD1-057 E O1	5.96
934521	AD1-076 C	18.49
934522	AD1-076 E	9.41
936401	AD2-051 C O1	4.79
936402	AD2-051 E O1	2.06
936701	AD2-089 C	7.78
936702	AD2-089 E	5.19
936711	AD2-090 C O1	2.83
936712	AD2-090 E O1	1.89
937571	AD2-169 C	5.61
937572	AD2-169 E	3.74
938171	AE1-026 C1 O	10.57
938172	AE1-026 C2 O	1.63
938173	AE1-026 E O1	3.1
938221	AE1-035 C	0.97

Bus #	Bus	MW Impact
938222	AE1-035 E	0.48
938661	AE1-088	1.26
BLUEG	BLUEG	4.01
CALDERWOOD	CALDERWOOD	0.81
CANNELTON	CANNELTON	0.27
CATAWBA	CATAWBA	0.79
CBM-N	CBM-N	0.07
CHEOAH	CHEOAH	0.76
CHILHOWEE	CHILHOWEE	0.26
COFFEEN	COFFEEN	0.47
COTTONWOOD	COTTONWOOD	2.72
DUCKCREEK	DUCKCREEK	0.96
EDWARDS	EDWARDS	0.43
FARMERCITY	FARMERCITY	0.34
G-007A	G-007A	0.38
GIBSON	GIBSON	0.17
HAMLET	HAMLET	1.57
NEWTON	NEWTON	1.21
NYISO	NYISO	0.29
PRAIRIE	PRAIRIE	2.53
SANTEETLA	SANTEETLA	0.22
SMITHLAND	SMITHLAND	0.22
TATANKA	TATANKA	0.57
TILTON	TILTON	0.51
TRIMBLE	TRIMBLE	0.44
TVA	TVA	2.28
UNIONPOWER	UNIONPOWER	1.13
VFT	VFT	1.01

12.6.24 Index 23

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308310	314568	3EARLEYS	DVP	314569	6EARLEYS	DVP	2	DVP_P4-2: 201262	breaker	208.0	144.76	150.48	AC	10.96

Bus #	Bus	MW Impact
314539	3UNCAMP	1.46
314541	3WATKINS	0.52
314572	3EMPORIA	0.18
314578	3HORNRTN	2.81
314582	3KELFORD	9.44
314589	3MURPHYS	0.14
314603	3SCOT NK	6.48
314617	3TUNIS	2.15
314623	3WITAKRS	0.69
315115	1S HAMPT1	1.01
315126	1ROARAP2	0.95
315128	1ROARAP4	0.92
900672	V4-068 E	0.55
907092	X1-038 E	3.65
917331	Z2-043 C	0.89
917332	Z2-043 E	2.68
917342	Z2-044 E	0.3
918491	AA1-063AC OP	2.22
918492	AA1-063AE OP	7.42
918561	AA1-072 C	0.13
918562	AA1-072 E	0.45
919691	AA2-053 C	1.71
919692	AA2-053 E	5.21
919701	AA2-057 C	0.74
919702	AA2-057 E	2.61
920041	AA2-088 C OP	1.0
920042	AA2-088 E OP	11.54
920591	AA2-165 C	0.1
920592	AA2-165 E	0.34
920671	AA2-174 C	0.08
920672	AA2-174 E	0.6
923801	AB2-015 C O1	6.0
923802	AB2-015 E O1	4.92
923911	AB2-031 C O1	1.29
923912	AB2-031 E O1	0.63
923992	AB2-040 E O1	3.46
924401	AB2-089 C	0.89
924402	AB2-089 E	0.46
924501	AB2-099 C	1.76
924502	AB2-099 E	0.76
925171	AB2-174 C O1	3.68

Bus #	Bus	MW Impact
925172	AB2-174 E O1	3.33
925781	AC1-054 C O1	3.34
925782	AC1-054 E O1	1.54
926201	AC1-098 C	9.29
926202	AC1-098 E	5.53
926211	AC1-099 C	3.11
926212	AC1-099 E	1.83
927141	AC1-208 C	9.21
927142	AC1-208 E	4.09
931232	AB1-173 E	0.61
931242	AB1-173AE	0.61
932631	AC2-084 C	13.24
932632	AC2-084 E	6.52
934201	AD1-047 C	4.61
934202	AD1-047 E	3.07
934231	AD1-050 C	1.97
934232	AD1-050 E	1.07
936401	AD2-051 C O1	29.7
936402	AD2-051 E O1	12.75
936711	AD2-090 C O1	6.31
936712	AD2-090 E O1	4.2
937571	AD2-169 C	6.32
937572	AD2-169 E	4.21
938175	AE1-026 EBAT	10.96
938662	AE1-088 E	1.45
938771	AE1-103 C O1	1.96
938772	AE1-103 E O1	2.7
CARR	CARR	0.02
CBM-S1	CBM-S1	0.06
CBM-S2	CBM-S2	0.03
CBM-W1	CBM-W1	0.01
CBM-W2	CBM-W2	0.31
CIN	CIN	0.01
G-007	G-007	0.08
IPL	IPL	0.01
LGEE	LGEE	0.0
MEC	MEC	0.04
O-066	O-066	0.5
RENSSELAER	RENSSELAER	0.02
WEC	WEC	0.0

12.6.25 Index 24

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308426	314568	3EARLEYS	DVP	314569	6EARLEYS	DVP	1	DVP_P4-2: 217162	breaker	208.0	128.41	132.2	AC	7.57

Bus #	Bus	MW Impact
314539	3UNCAMP	1.45
314541	3WATKINS	0.52
314578	3HORNRTN	2.4
314582	3KELFORD	9.59
314589	3MURPHYS	0.14
314603	3SCOT NK	6.48
314617	3TUNIS	2.18
314623	3WITAKRS	0.63
315115	1S HAMPT1	1.0
315126	1ROARAP2	0.8
315128	1ROARAP4	0.76
900672	V4-068 E	0.55
907092	X1-038 E	3.64
917331	Z2-043 C	0.9
917332	Z2-043 E	2.73
917342	Z2-044 E	0.27
918491	AA1-063AC OP	2.17
918492	AA1-063AE OP	7.23
918561	AA1-072 C	0.13
918562	AA1-072 E	0.45
919691	AA2-053 C	1.61
919692	AA2-053 E	4.89
919701	AA2-057 C	0.69
919702	AA2-057 E	2.43
920041	AA2-088 C OP	0.97
920042	AA2-088 E OP	11.24
920591	AA2-165 C	0.09
920592	AA2-165 E	0.32
920671	AA2-174 C	0.07
920672	AA2-174 E	0.56
923801	AB2-015 C O1	5.93
923802	AB2-015 E O1	4.86
923911	AB2-031 C O1	0.99
923912	AB2-031 E O1	0.49
923992	AB2-040 E O1	2.66
924501	AB2-099 C	1.8
924502	AB2-099 E	0.77
925171	AB2-174 C O1	2.75
925172	AB2-174 E O1	2.49
925781	AC1-054 C O1	2.74
925782	AC1-054 E O1	1.26

Bus #	Bus	MW Impact
926201	AC1-098 C	9.12
926202	AC1-098 E	5.44
926211	AC1-099 C	3.06
926212	AC1-099 E	1.8
927141	AC1-208 C	8.65
927142	AC1-208 E	3.84
931232	AB1-173 E	0.47
931242	AB1-173AE	0.47
932631	AC2-084 C	13.01
932632	AC2-084 E	6.41
934201	AD1-047 C	3.55
934202	AD1-047 E	2.36
936401	AD2-051 C O1	30.42
936402	AD2-051 E O1	13.06
936711	AD2-090 C O1	6.18
936712	AD2-090 E O1	4.12
937571	AD2-169 C	4.99
937572	AD2-169 E	3.33
938175	AE1-026 EBAT	7.57
938771	AE1-103 C O1	1.96
938772	AE1-103 E O1	2.71
BLUEG	BLUEG	0.01
CARR	CARR	0.02
CBM-S1	CBM-S1	0.02
CBM-S2	CBM-S2	0.01
CBM-W2	CBM-W2	0.07
COFFEEN	COFFEEN	0.0
DUCKCREEK	DUCKCREEK	0.0
EDWARDS	EDWARDS	0.0
G-007	G-007	0.07
GIBSON	GIBSON	0.0
O-066	O-066	0.42
RENSSELAER	RENSSELAER	0.02
TATANKA	TATANKA	0.0
TILTON	TILTON	0.0
TRIMBLE	TRIMBLE	0.0

12.6.26 Index 25

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308346	314638	6ELIZ CT	DVP	314647	6SHAWBRO	DVP	1	DVP_P4-2: 246T247	breaker	658.0	132.18	143.81	AC	71.46

Bus #	Bus	MW Impact
314557	3BETHEL	0.6
314566	3CRESWEL	6.39
314574	6EVERETS	4.25
314582	3KELFORD	3.05
314594	6PLYMOTH	1.92
314603	3SCOT NK	2.51
314617	3TUNIS	0.64
314620	6CASHIE	1.49
314638	6ELIZ CT	0.83
314639	6TANGLEW	1.46
314651	6WINFALL	6.71
315292	1DOMTR78	2.28
315293	1DOMTR9	1.86
315294	1DOMTR10	3.38
900672	V4-068 E	0.17
901081	W1-029 C	3.7
901082	W1-029 E	175.15
913392	Y1-086 E	9.58
916041	Z1-036 C	3.32
916042	Z1-036 E	157.45
917122	Z2-027 E	4.42
917331	Z2-043 C	0.29
917332	Z2-043 E	0.87
917511	Z2-088 C OP1	1.13
917512	Z2-088 E OP1	3.44
918491	AA1-063AC OP	0.85
918492	AA1-063AE OP	2.82
918511	AA1-065 C OP	1.47
918512	AA1-065 E OP	5.12
918531	AA1-067 C	0.25
918532	AA1-067 E	0.75
918561	AA1-072 C	0.04
918562	AA1-072 E	0.14
919691	AA2-053 C	0.77
919692	AA2-053 E	2.34
919702	AA2-057 E	2.14
920592	AA2-165 E	0.28
920671	AA2-174 C	0.04
920672	AA2-174 E	0.27
920691	AA2-178 C	3.61
920692	AA2-178 E	10.96

Bus #	Bus	MW Impact
923831	AB2-022 C	1.42
923832	AB2-022 E	5.41
924491	AB2-098 C	0.58
924492	AB2-098 E	0.25
924501	AB2-099 C	0.51
924502	AB2-099 E	0.22
925121	AB2-169 C	1.5
925122	AB2-169 E	9.55
926071	AC1-086 C	11.91
926072	AC1-086 E	5.42
926201	AC1-098 C	4.28
926202	AC1-098 E	2.55
926211	AC1-099 C	1.44
926212	AC1-099 E	0.84
927021	AC1-189 C	8.05
927022	AC1-189 E	4.01
927141	AC1-208 C	5.57
927142	AC1-208 E	2.47
930862	AB1-132 E O1	3.47
932631	AC2-084 C	6.11
932632	AC2-084 E	3.01
933991	AD1-023 C	25.2
933992	AD1-023 E	13.72
934521	AD1-076 C	118.85
934522	AD1-076 E	60.52
936401	AD2-051 C O1	8.54
936402	AD2-051 E O1	3.67
936531	AD2-068 C	3.85
936532	AD2-068 E	1.98
936701	AD2-089 C	5.39
936702	AD2-089 E	3.59
938171	AE1-026 C1 O	49.4
938172	AE1-026 C2 O	7.59
938173	AE1-026 E O1	14.47
938221	AE1-035 C	2.6
938222	AE1-035 E	1.28
938661	AE1-088	5.46
AA2-074	AA2-074	1.56
CARR	CARR	0.13
CBM-S1	CBM-S1	3.28
CBM-S2	CBM-S2	4.0
CBM-W1	CBM-W1	3.28
CBM-W2	CBM-W2	21.43
CIN	CIN	1.47
CPL	CPL	2.29
G-007	G-007	0.45
IPL	IPL	0.92
LGEE	LGEE	0.43
MEC	MEC	3.3
MECS	MECS	1.38
O-066	O-066	2.86
RENSSELAER	RENSSELAER	0.1

Bus #	Bus	MW Impact
WEC	WEC	0.4

12.6.27 Index 26

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308782	314908	8ELMONT	DVP	314911	8LADYSMITH	DVP	1	DVP_P1-2: LN 576	single	2442.12	138.87	140.79	AC	54.31

Bus #	Bus	MW Impact
314229	6MT RD221	0.2
314236	6NRTHEST	0.3
314309	6IRON208	0.71
315053	1BELMED1	31.54
315054	1BELMED2	31.54
315055	1BELMED3	26.18
315058	1CHESTF3	33.5
315059	1CHESTF4	54.31
315060	1CHESTF5	16.25
315061	1CHESTG7	6.37
315062	1CHESTS7	2.9
315063	1CHESTG8	6.25
315064	1CHESTS8	3.22
315067	1DARBY 1	4.16
315068	1DARBY 2	4.16
315069	1DARBY 3	4.18
315070	1DARBY 4	4.18
315074	1HOPCGN1	15.45
315075	1HOPCGN2	15.25
315083	1SPRUNCA	19.09
315084	1SPRUNCB	19.09
315085	1SPRUNCC	14.15
315086	1SPRUNCD	14.15
315090	1YORKTN1	53.26
315091	1YORKTN2	55.27
315092	1YORKTN3	37.36
315233	1SURRY 2	40.58
923801	AB2-015 C O1	14.85
923911	AB2-031 C O1	3.63
924241	AB2-068 O1	428.17
924501	AB2-099 C	0.97
924511	AB2-100 C	19.03
925051	AB2-160 C O1	9.89
925061	AB2-161 C O1	6.06
925171	AB2-174 C O1	11.41
925331	AB2-190 C	37.1
925861	AC1-065 C	6.03
926071	AC1-086 C	33.51
926291	AC1-107 O1	646.29
926411	AC1-112 C	0.54
926751	AC1-161 C O1	55.54

Bus #	Bus	MW Impact
926781	AC1-164 C	77.68
927041	AC1-191 C O1	17.04
927221	AC1-216 C O1	18.13
930121	AB1-027 C	0.7
932041	AC2-012 C	18.48
932501	AC2-070 C	0.46
932581	AC2-078 C O1	7.32
932591	AC2-079 C O1	10.27
932831	AC2-110 C	2.41
933061	AC2-130	0.64
933261	AC2-137 C	0.56
933291	AC2-141 C	55.54
933991	AD1-023 C	22.17
934011	AD1-025 C	31.17
934061	AD1-033 C	13.56
934141	AD1-041 C	9.41
934201	AD1-047 C	12.99
934211	AD1-048 C	0.65
934521	AD1-076 C	92.34
934571	AD1-082 C	13.81
935161	AD1-151 C O1	29.82
936041	AD2-007	3.31
936051	AD2-008 C	5.43
936151	AD2-021	0.45
936301	AD2-039 C	2.41
936391	AD2-049 C	3.3
936401	AD2-051 C O1	14.43
936661	AD2-085 C	6.16
936711	AD2-090 C O1	12.26
937221	AD2-160 C O1	10.52
937251	AD2-164	8.89
937541	AD2-215 C	3.15
937571	AD2-169 C	16.18
938031	AE1-004 C	2.41
938171	AE1-026 C1 O	47.08
938172	AE1-026 C2 O	7.24
938181	AE1-027 C	4.22
938191	AE1-028 C	2.45
938221	AE1-035 C	3.72
938491	AE1-068 C O1	93.17
938501	AE1-069 C O1	73.62
938531	AE1-072 C O1	31.46
938551	AE1-074 C	4.28
938631	AE1-085 C O1	16.04
938661	AE1-088	5.51
938771	AE1-103 C O1	6.33
939191	AE1-149 C O1	19.28
939311	AE1-162 C	3.93
939411	AE1-173 C	149.31
939421	AE1-174 C	0.68
939431	AE1-175 C	4.55
939611	AE1-191 C	18.81

Bus #	Bus	MW Impact
939751	AE1-206 C O1	50.31
940061	AE1-248 C O1	28.71
940071	AE1-249 C	13.04
AA2-074	AA2-074	8.67
CARR	CARR	2.58
CBM-S1	CBM-S1	26.83
CBM-S2	CBM-S2	26.11
CBM-W1	CBM-W1	28.21
CBM-W2	CBM-W2	175.2
CIN	CIN	13.15
CPLE	CPLE	12.75
IPL	IPL	8.28
LGEE	LGEE	3.88
MEC	MEC	27.98
MECS	MECS	11.81
RENSSELAER	RENSSELAER	2.04
WEC	WEC	3.45
Z1-043	Z1-043	13.6

12.6.28 Index 27

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308436	314914	8MDLTHAN	DVP	314918	8NO ANNA	DVP	1	DVP_P4-2: 557T574	breaker	3637.0	111.97	113.4	AC	67.94

Bus #	Bus	MW Impact
314189	6PAPERMILL	9.09
314539	3UNCAMP	4.13
314541	3WATKINS	1.17
314554	3BTLEBRO	1.24
314557	3BETHEL	1.27
314566	3CRESWEL	4.09
314572	3EMPORIA	0.69
314574	6EVERETS	6.73
314578	3HORNRTN	5.34
314582	3KELFORD	5.72
314594	6PLYMOTH	1.44
314603	3SCOT NK	5.32
314617	3TUNIS	1.39
314620	6CASHIE	1.42
314623	3WITAKRS	2.11
314648	6SUNBURY	1.55
314651	6WINFALL	3.07
315053	1BELMED1	25.09
315054	1BELMED2	25.09
315055	1BELMED3	20.83
315058	1CHESTF3	25.42
315059	1CHESTF4	41.2
315073	1STONECA	10.42
315074	1HOPCGN1	12.56
315075	1HOPCGN2	12.4
315083	1SPRUNCA	16.07
315084	1SPRUNCB	16.07
315085	1SPRUNCC	11.91
315086	1SPRUNCD	11.91
315090	1YORKTN1	47.1
315091	1YORKTN2	48.87
315102	1BRUNSWICKG1	12.69
315103	1BRUNSWICKG2	12.69
315104	1BRUNSWICKG3	12.69
315105	1BRUNSWICKS1	26.36
315108	1ELIZAR1	4.92
315109	1ELIZAR2	4.83
315110	1ELIZAR3	4.98
315131	1EDGECEMA	14.4
315132	1EDGECEMB	14.4
315233	1SURRY 2	38.3

Bus #	Bus	MW Impact
900672	V4-068 E	0.5
901082	W1-029 E	80.24
907092	X1-038 E	10.33
913392	Y1-086 E	3.86
916042	Z1-036 E	78.94
916191	Z1-068 C	0.07
916192	Z1-068 E	3.34
916301	Z1-086 C	77.23
916302	Z1-086 E	17.1
917122	Z2-027 E	1.86
917332	Z2-043 E	1.63
917342	Z2-044 E	0.91
917512	Z2-088 E OP1	6.22
918492	AA1-063AE OP	6.51
918512	AA1-065 E OP	7.34
918532	AA1-067 E	1.19
918562	AA1-072 E	0.27
919152	AA1-139 E	11.35
919692	AA2-053 E	6.07
919702	AA2-057 E	5.61
920042	AA2-088 E OP	17.61
920592	AA2-165 E	0.74
920672	AA2-174 E	0.7
920692	AA2-178 E	7.02
923801	AB2-015 C O1	14.65
923802	AB2-015 E O1	12.01
923832	AB2-022 E	2.18
923842	AB2-024 E	1.53
923852	AB2-025 E	1.85
923862	AB2-026 E	1.69
923911	AB2-031 C O1	3.65
923912	AB2-031 E O1	1.8
923992	AB2-040 E O1	9.79
924022	AB2-043 E O1	3.79
924152	AB2-059 E O1	8.43
924241	AB2-068 O1	340.41
924401	AB2-089 C	3.24
924402	AB2-089 E	1.67
924491	AB2-098 C	0.92
924492	AB2-098 E	0.4
924501	AB2-099 C	0.97
924502	AB2-099 E	0.42
924511	AB2-100 C	19.02
924512	AB2-100 E	9.37
924812	AB2-134 E O1	19.52
925051	AB2-160 C O1	8.44
925052	AB2-160 E O1	13.78
925061	AB2-161 C O1	5.69
925062	AB2-161 E O1	9.28
925122	AB2-169 E	9.65
925171	AB2-174 C O1	11.46
925172	AB2-174 E O1	10.37

Bus #	Bus	MW Impact
925331	AB2-190 C	31.84
925332	AB2-190 E	13.65
925521	AC1-027 C	0.5
925522	AC1-027 E	2.03
925591	AC1-034 C	10.6
925592	AC1-034 E	8.0
925781	AC1-054 C O1	11.02
925782	AC1-054 E O1	5.08
925861	AC1-065 C	4.62
925862	AC1-065 E	7.54
926071	AC1-086 C	33.67
926072	AC1-086 E	15.33
926201	AC1-098 C	9.89
926202	AC1-098 E	5.89
926211	AC1-099 C	3.31
926212	AC1-099 E	1.95
926291	AC1-107 O1	513.82
926662	AC1-147 E	2.35
926751	AC1-161 C O1	55.17
926752	AC1-161 E O1	23.55
926781	AC1-164 C	54.86
926782	AC1-164 E	24.65
927021	AC1-189 C	13.91
927022	AC1-189 E	6.93
927141	AC1-208 C	14.45
927142	AC1-208 E	6.42
927221	AC1-216 C O1	15.69
927222	AC1-216 E O1	12.34
930402	AB1-081 E O1	5.95
930862	AB1-132 E O1	9.8
931232	AB1-173 E	1.71
931242	AB1-173AE	1.71
932041	AC2-012 C	18.09
932042	AC2-012 E	29.52
932532	AC2-073 E	1.64
932581	AC2-078 C O1	6.66
932582	AC2-078 E O1	10.86
932591	AC2-079 C O1	9.82
932592	AC2-079 E O1	16.02
932631	AC2-084 C	14.1
932632	AC2-084 E	6.94
932831	AC2-110 C	1.85
932832	AC2-110 E	3.02
933262	AC2-137 E	1.82
933291	AC2-141 C	55.17
933292	AC2-141 E	23.55
933501	AC2-165 C	14.57
933502	AC2-165 E	10.96
933731	AC2-196 C	0.45
933732	AC2-196 E	2.12
933991	AD1-023 C	22.1
933992	AD1-023 E	12.03

Bus #	Bus	MW Impact
934011	AD1-025 C	26.97
934012	AD1-025 E	15.97
934061	AD1-033 C	13.35
934062	AD1-033 E	8.9
934141	AD1-041 C	7.26
934142	AD1-041 E	4.84
934201	AD1-047 C	13.06
934202	AD1-047 E	8.7
934212	AD1-048 E	1.63
934231	AD1-050 C	7.13
934232	AD1-050 E	3.9
934331	AD1-057 C O1	15.86
934332	AD1-057 E O1	8.46
934392	AD1-063 E	1.46
934521	AD1-076 C	91.96
934522	AD1-076 E	46.83
934571	AD1-082 C	12.96
934572	AD1-082 E	7.39
934611	AD1-087 C O1	12.81
934612	AD1-087 E O1	6.02
934621	AD1-088 C	19.52
934622	AD1-088 E	9.16
935112	AD1-144 E	1.64
935161	AD1-151 C O1	25.59
935162	AD1-151 E O1	17.06
935171	AD1-152 C O1	12.73
935172	AD1-152 E O1	8.49
935212	AD1-156 E	2.31
936041	AD2-007	2.86
936051	AD2-008 C	4.69
936052	AD2-008 E	10.22
936151	AD2-021	0.32
936242	AD2-030 E	1.61
936301	AD2-039 C	1.85
936302	AD2-039 E	3.02
936361	AD2-046 C O1	13.17
936362	AD2-046 E O1	6.06
936391	AD2-049 C	3.0
936392	AD2-049 E	3.0
936401	AD2-051 C O1	14.45
936402	AD2-051 E O1	6.2
936531	AD2-068 C	8.1
936532	AD2-068 E	4.17
936661	AD2-085 C	5.89
936662	AD2-085 E	9.6
936701	AD2-089 C	11.43
936702	AD2-089 E	7.62
936711	AD2-090 C O1	12.13
936712	AD2-090 E O1	8.09
937221	AD2-160 C O1	10.37
937222	AD2-160 E O1	5.44
937251	AD2-164	7.94

Bus #	Bus	MW Impact
937481	AD2-202 C O1	3.53
937482	AD2-202 E O1	1.78
937541	AD2-215 C	3.03
937542	AD2-215 E	1.61
937571	AD2-169 C	16.27
937572	AD2-169 E	10.85
938031	AE1-004 C	1.85
938032	AE1-004 E	3.02
938171	AE1-026 C1 O	46.96
938172	AE1-026 C2 O	7.22
938173	AE1-026 E O1	13.76
938181	AE1-027 C	4.16
938182	AE1-027 E	2.19
938191	AE1-028 C	2.41
938192	AE1-028 E	1.4
938221	AE1-035 C	3.72
938222	AE1-035 E	1.83
938491	AE1-068 C O1	108.99
938492	AE1-068 E O1	60.19
938501	AE1-069 C O1	86.11
938502	AE1-069 E O1	49.23
938531	AE1-072 C O1	31.02
938532	AE1-072 E O1	16.17
938551	AE1-074 C	4.0
938552	AE1-074 E	2.02
938561	AE1-075 C	3.31
938562	AE1-075 E	1.61
938631	AE1-085 C O1	14.64
938632	AE1-085 E O1	7.32
938661	AE1-088	5.52
938771	AE1-103 C O1	6.23
938772	AE1-103 E O1	8.6
939181	AE1-148 C O1	12.97
939182	AE1-148 E O1	8.65
939191	AE1-149 C O1	17.43
939192	AE1-149 E O1	11.62
939311	AE1-162 C	3.66
939312	AE1-162 E	2.44
939411	AE1-173 C	164.81
939412	AE1-173 E	109.87
939421	AE1-174 C	0.56
939422	AE1-174 E	0.84
939431	AE1-175 C	3.74
939432	AE1-175 E	1.85
939611	AE1-191 C	14.52
939612	AE1-191 E	9.68
940061	AE1-248 C O1	26.94
940062	AE1-248 E O1	17.96
940071	AE1-249 C	11.14
940072	AE1-249 E	8.3
AA2-074	AA2-074	8.96
CARR	CARR	2.11

Bus #	Bus	MW Impact
CBM-S1	CBM-S1	23.78
CBM-S2	CBM-S2	26.14
CBM-W1	CBM-W1	22.62
CBM-W2	CBM-W2	153.22
CIN	CIN	10.57
CPLE	CPLE	13.18
G-007	G-007	7.0
IPL	IPL	6.62
LGEE	LGEE	3.1
MEC	MEC	23.44
MECS	MECS	8.7
O-066	O-066	44.51
RENSSELAER	RENSSELAER	1.67
WEC	WEC	2.78
Z1-043	Z1-043	10.97

12.6.29 Index 28

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3308927	924510	AB2-100 TAP	DVP	314563	6CLUBHSE	DVP	1	DVP_P1-2: LN 246	single	375.06	113.03	120.39	AC	29.33

Bus #	Bus	MW Impact
314589	3MURPHYS	0.06
315115	1S HAMPT1	0.59
315126	1ROARAP2	1.15
315128	1ROARAP4	1.1
315131	1EDGECEMA	10.42
315132	1EDGECEMB	10.42
315136	1ROSEMG1	3.79
315137	1ROSEMS1	2.35
315138	1ROSEMG2	1.78
315139	1GASTONA	5.65
315141	1GASTONB	5.65
315159	1KERR 2	0.4
315164	1KERR 7	0.4
315292	1DOMTR78	0.96
315293	1DOMTR9	0.79
315294	1DOMTR10	1.43
916041	Z1-036 C	0.31
917331	Z2-043 C	0.35
917341	Z2-044 C	0.2
917511	Z2-088 C OP1	1.34
918491	AA1-063AC OP	1.22
918511	AA1-065 C OP	1.44
918531	AA1-067 C	0.24
918561	AA1-072 C	0.05
919691	AA2-053 C	1.23
919701	AA2-057 C	1.03
920041	AA2-088 C OP	0.59
920591	AA2-165 C	0.14
920671	AA2-174 C	0.06
920691	AA2-178 C	0.8
923801	AB2-015 C O1	3.46
924151	AB2-059 C O1	1.57
924401	AB2-089 C	0.94
924491	AB2-098 C	0.58
924501	AB2-099 C	0.58
924511	AB2-100 C	42.5
925121	AB2-169 C	0.8
925591	AC1-034 C	7.22
925781	AC1-054 C O1	3.62
926071	AC1-086 C	44.94
926201	AC1-098 C	6.36

Bus #	Bus	MW Impact
926211	AC1-099 C	2.13
927021	AC1-189 C	8.97
927141	AC1-208 C	9.23
930401	AB1-081 C O1	1.34
930861	AB1-132 C O1	4.31
932631	AC2-084 C	9.07
933991	AD1-023 C	11.18
934231	AD1-050 C	2.07
934331	AD1-057 C O1	15.93
934521	AD1-076 C	41.0
936401	AD2-051 C O1	9.31
936531	AD2-068 C	4.32
936701	AD2-089 C	7.84
936711	AD2-090 C O1	3.67
938171	AE1-026 C1 O	25.42
938172	AE1-026 C2 O	3.91
938221	AE1-035 C	2.54
938661	AE1-088	2.91
AA2-074	AA2-074	2.2
CARR	CARR	0.18
CBM-S1	CBM-S1	4.75
CBM-S2	CBM-S2	5.71
CBM-W1	CBM-W1	4.77
CBM-W2	CBM-W2	31.01
CIN	CIN	2.15
CPLE	CPLE	3.23
IPL	IPL	1.34
LGEE	LGEE	0.62
MEC	MEC	4.79
MECS	MECS	2.03
RENSSELAER	RENSSELAER	0.14
WEC	WEC	0.58

Affected Systems

13 Affected Systems

13.1 Duke Energy Progress

Duke Energy Progress Impacts to be determined during facility study phases.

Potential constraints were identified by PJM on the following Dominion – Duke Energy/Progress (DEP) tie lines. There are no mitigations currently planned for the DEP portions of these overloads. The Queue Project AE1-024 / AE1-025 / AE1-026 may be subject to operational restriction if real-time system reliability issues occur. The following facilities were identified in this report:

- Nash – City of Rocky Mount 230 kV line
- Battleboro – Rocky Mt. 115 kV line
- Everetts - Greenville 230 kV line
- Rocky Mt. – Hathaway 230 kV line

Short Circuit

14 Short Circuit

The following breakers are overdutied:

None

Stability Analysis

15 Stability Analysis

15.1 Steady-State Voltage Requirements

Project meets the steady-state voltage requirements

15.2 Stability and Reactive Power Requirement for Low Voltage Ride Through

No violations identified as part of this study.

Attachment 1

System Configuration

